

Deep Learning for Computer Vision

Lecture 10: Semi & Self-supervised Learning

Constantin Pape

Institute of Computer Science, Georg August Universität Göttingen



@cppape.bsky.social

<https://user.informatik.uni-goettingen.de/~pape41/>

Thesis Marketplace & Research Opportunities



- January 22nd (Thursday) from 15:00 in the geology foyer
- Research groups will present possible topics for internships and thesis related to data science
- Our research group will also be there and present our research and possible topics.

Agenda for today

Learning with few labels:

- Semi-supervised learning
- Weakly supervised learning
- Self-supervised learning
 - Pretext tasks
 - Contrastive learning

Learning with few labels

Supervised Deep Learning is data hungry!

- Supervised learning: needs a lot of labeled data for state-of-the-art results
- Generating new labeled data is very costly!

Supervised Deep Learning is data hungry!

- Supervised learning: needs a lot of labeled data for state-of-the-art results
- Generating new labeled data is very costly!

Mitigation strategies:

- **Transfer learning:** model trained on large dataset (ImageNet) + fine-tuning

Lecture 4

Supervised Deep Learning is data hungry!

- Supervised learning: needs a lot of labeled data for state-of-the-art results
- Generating new labeled data is very costly!

Mitigation strategies:

- **Transfer learning:** model trained on large dataset (ImageNet) + fine-tuning
- **Semi-supervised learning:** train model jointly on labeled and unlabeled data

This lecture

Supervised Deep Learning is data hungry!

- Supervised learning: needs a lot of labeled data for state-of-the-art results
- Generating new labeled data is very costly!

Mitigation strategies:

- **Transfer learning:** model trained on large dataset (ImageNet) + fine-tuning
- **Semi-supervised learning:** train model jointly on labeled and unlabeled data
- **Weakly-supervised learning:** use cheaper form of labels for training

This lecture: overview
Next lecture: text
supervision

Supervised Deep Learning is data hungry!

- Supervised learning: needs a lot of labeled data for state-of-the-art results
- Generating new labeled data is very costly!

Mitigation strategies:

- **Transfer learning:** model trained on large dataset (ImageNet) + fine-tuning
- **Semi-supervised learning:** train model jointly on labeled and unlabeled data
- **Weakly-supervised learning:** use cheaper form of labels for training
- **Self-supervised learning:** pre-train model unsupervised + fine-tuning

This lecture

Supervised Deep Learning is data hungry!

- Supervised learning: needs a lot of labeled data for state-of-the-art results
- Generating new labeled data is very costly!

Mitigation strategies:

- **Transfer learning:** model trained on large dataset (ImageNet) + fine-tuning
- **Semi-supervised learning:** train model jointly on labeled and unlabeled data
- **Weakly-supervised learning:** use cheaper form of labels for training
- **Self-supervised learning:** pre-train model unsupervised + fine-tuning

Basis of many vision
foundation models

Supervised Deep Learning is data hungry!

- Supervised learning: needs a lot of labeled data for state-of-the-art results
- Generating new labeled data is very costly!

Mitigation strategies:

- **Transfer learning:** model trained on large dataset, then fine-tuned on target dataset
- **Semi-supervised learning:** train model on unlabeled data, then fine-tune on labeled data
- **Weakly-supervised learning:** train model on weakly-labeled data, then fine-tune on labeled data
- **Self-supervised learning:** pre-train model unsupervised + fine-tuning

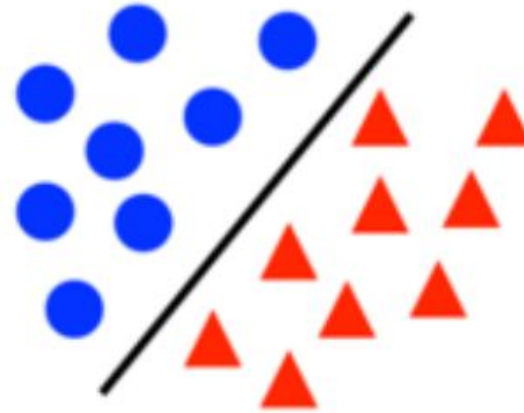
In practice: strategies can be combined

- transfer learning or self-supervised learning for pre-training
- semi-supervised or weakly-supervised learning for fine-tuning

Semi-supervised learning

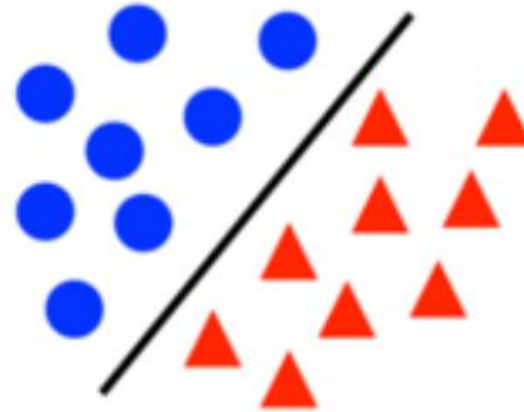
Semi-supervised learning

Supervised learning: all data points have labels

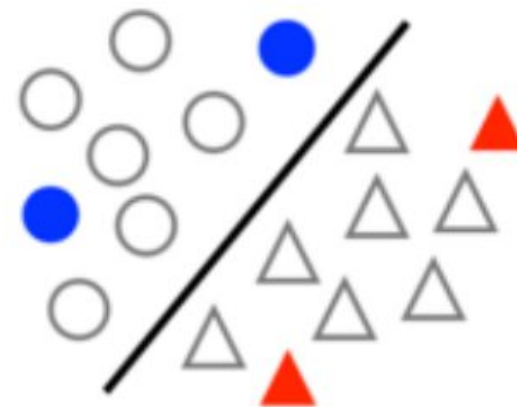


Semi-supervised learning

Supervised learning: all data points have labels



Semi-supervised learning (SSL):
only labels for subset of the data

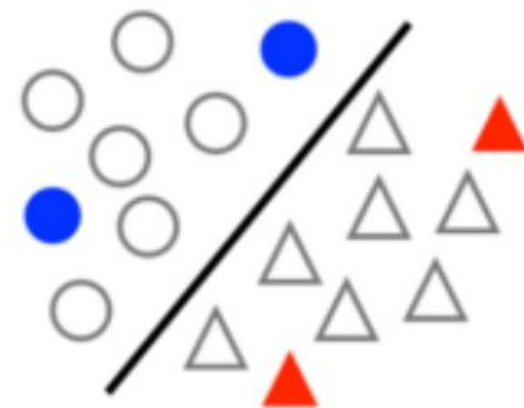
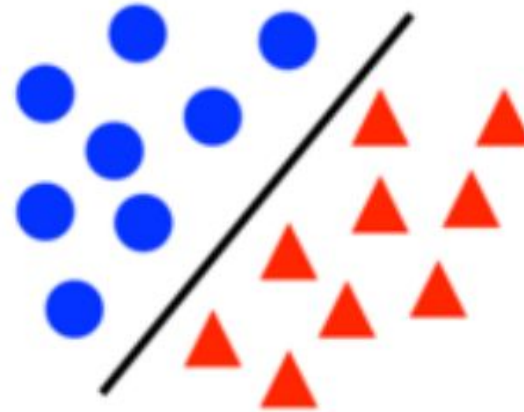


Semi-supervised learning

Supervised learning: all data points have labels

How can we use the unlabeled data for training a better model?

Semi-supervised learning (SSL):
only labels for subset of the data



Semi-supervised learning

Use supervised and unsupervised loss:

$$L = L_s + \mu(t) L_u$$

Semi-supervised learning

Use supervised and unsupervised loss:

$$L = L_s + \mu(t) L_u$$

 supervised loss for labeled samples
(discussed in prev. lectures)

Semi-supervised learning

Use supervised and unsupervised loss:

$$L = L_s + \mu(t) L_u$$

supervised loss for labeled samples
(discussed in prev. lectures)

unsupervised loss for unlabeled samples

Semi-supervised learning

Use supervised and unsupervised loss:

$$L = L_s + \mu(t) L_u$$

supervised loss for labeled samples
(discussed in prev. lectures)

unsupervised loss for unlabeled samples

time dependent weight factor: increases
importance of unsupervised loss over time

Semi-supervised learning

Use supervised and unsupervised loss:

$$L = L_s + \mu(t) L_u$$

supervised loss for labeled samples
(discussed in prev. lectures)

unsupervised loss for unlabeled samples

will focus on L_u and ignore L_s (already discussed)
and $\mu(t)$ (implementation detail)

Semi-supervised learning: approaches

- Consistency based:
 - Joint training on labeled and unlabeled samples
 - Predict unlabeled samples under different augmentations; unsupervised loss for consistent prediction
- Pseudo-label based:
 - Train teacher model on labeled samples
 - Predict teacher on unlabeled samples, derive “pseudo-labels” from predictions
 - Train student model on labeled and unlabeled samples (with pseudo-labels)
 - Make student teacher, repeat

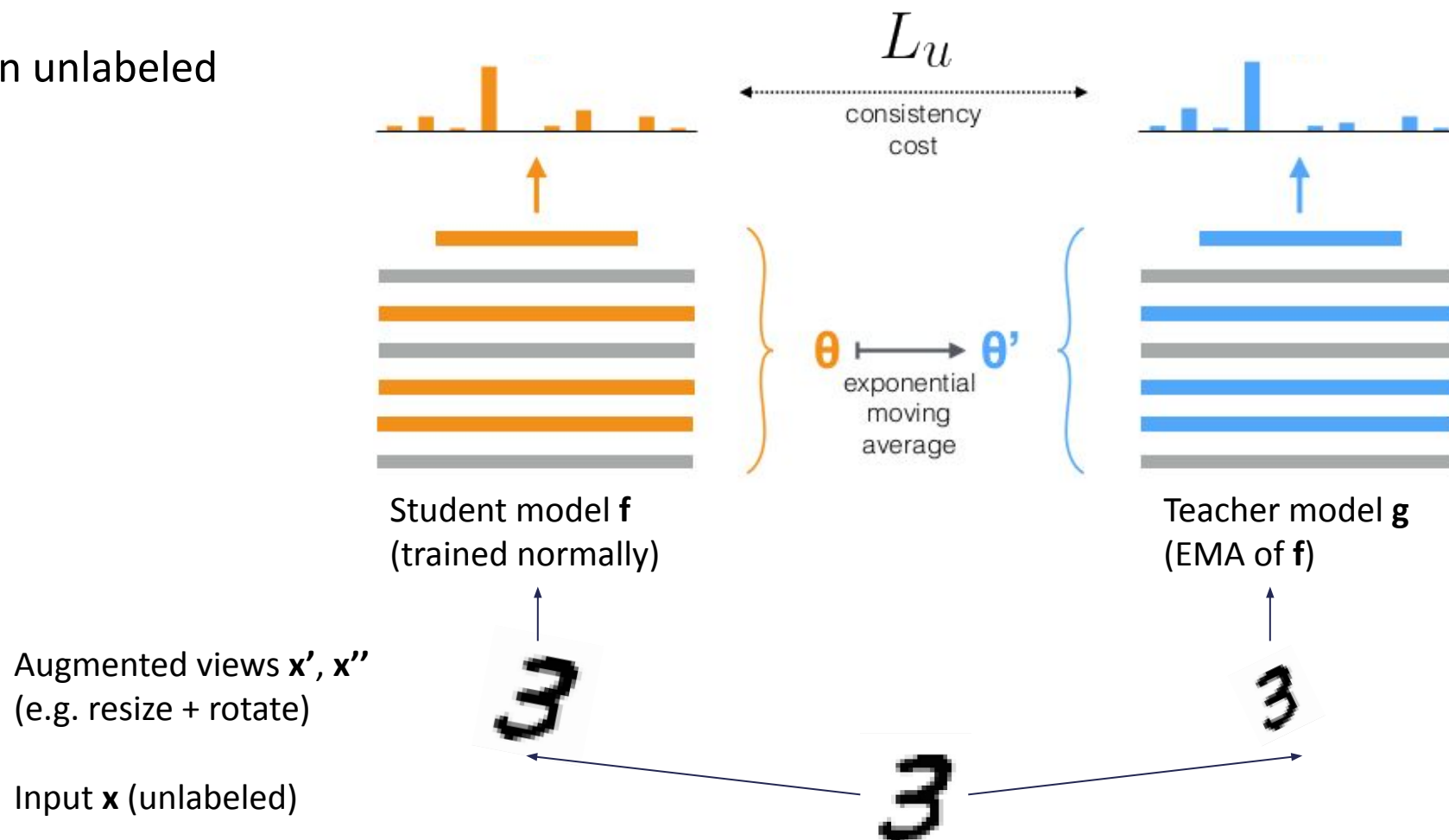
Semi-supervised learning: approaches

- Consistency based:
 - Joint training on labeled and unlabeled samples
 - Predict unlabeled samples under different augmentations; unsupervised loss for consistent prediction
- Pseudo-label based:
 - Train teacher model on labeled samples
 - Predict teacher on unlabeled samples, derive “pseudo-labels” from predictions
 - Train student model on labeled and unlabeled samples (with pseudo-labels)
 - Make student teacher, repeat

**Both approaches are quite similar;
some methods implement a mixture**

Consistency based: Mean teacher

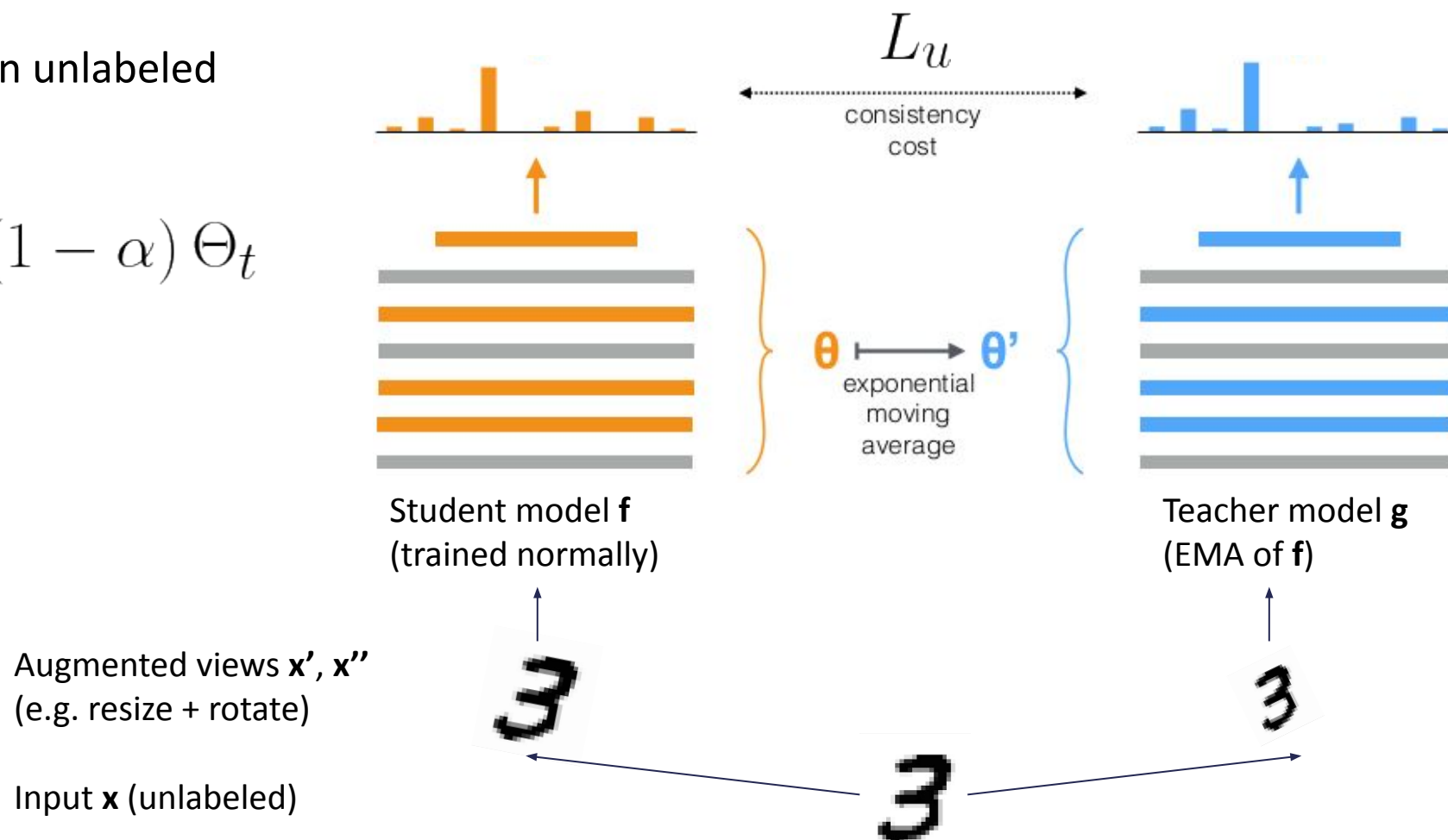
Use consistency loss on unlabeled samples



Consistency based: Mean teacher

Use consistency loss on unlabeled samples

$$\Theta'_t = \alpha \Theta'_{t-1} + (1 - \alpha) \Theta_t$$

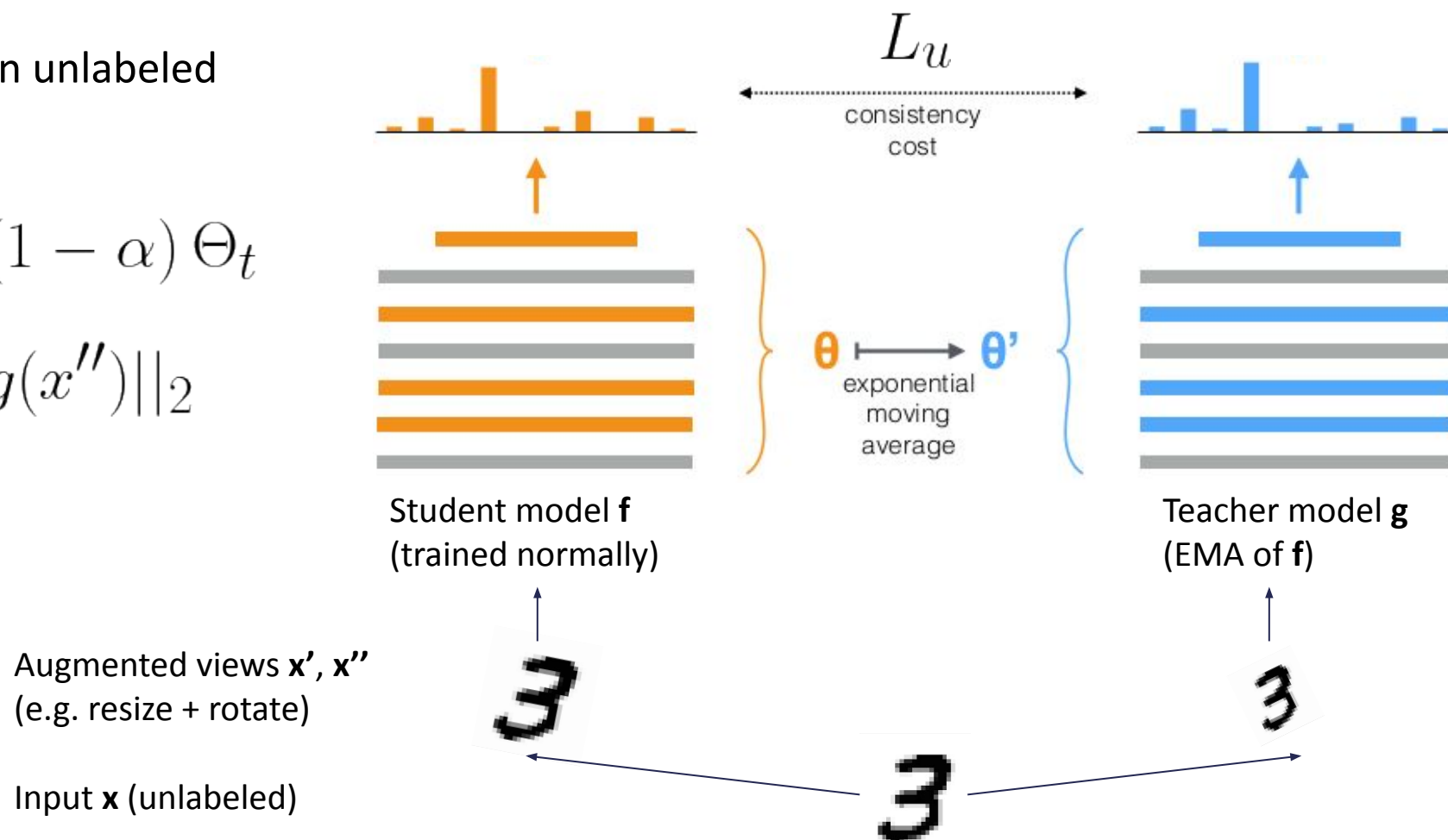


Consistency based: Mean teacher

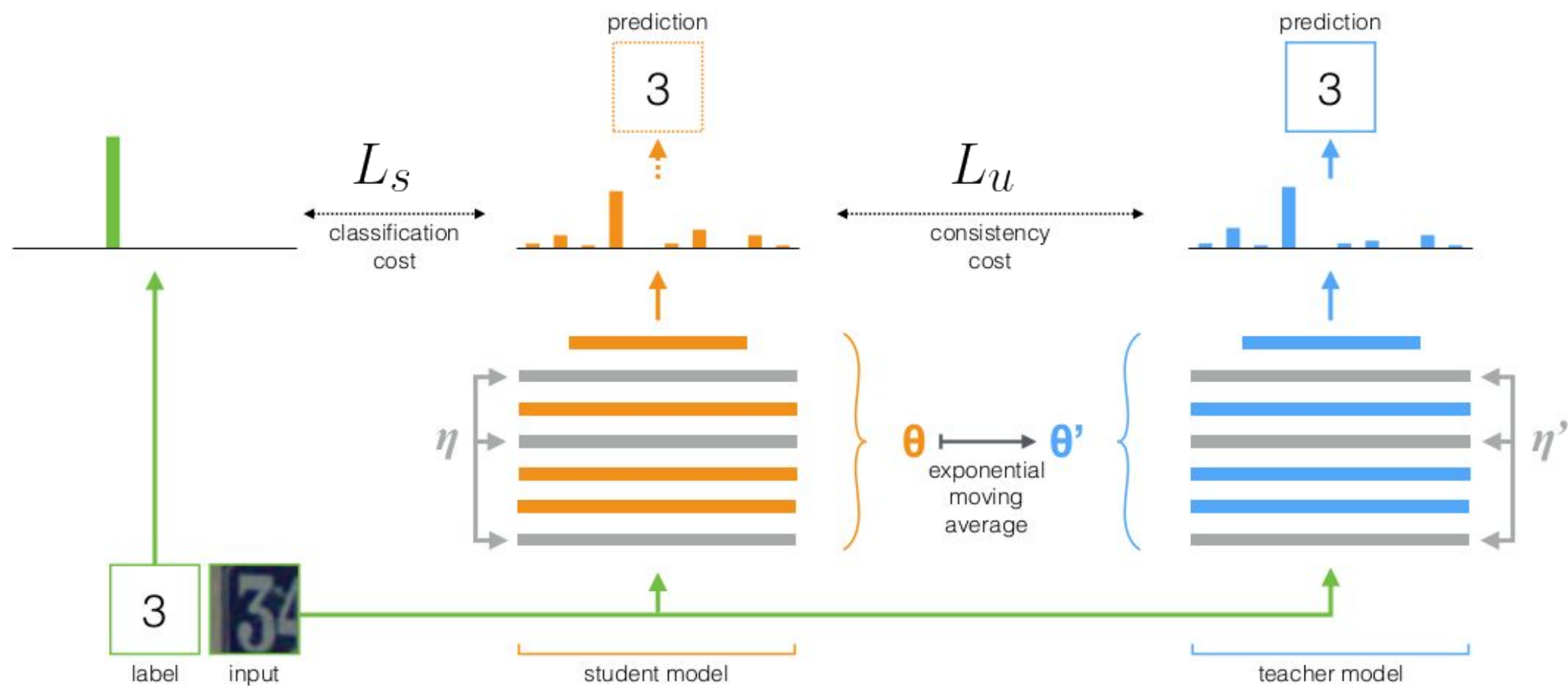
Use consistency loss on unlabeled samples

$$\Theta'_t = \alpha \Theta'_{t-1} + (1 - \alpha) \Theta_t$$

$$L_u = ||f(x') - g(x'')||_2$$



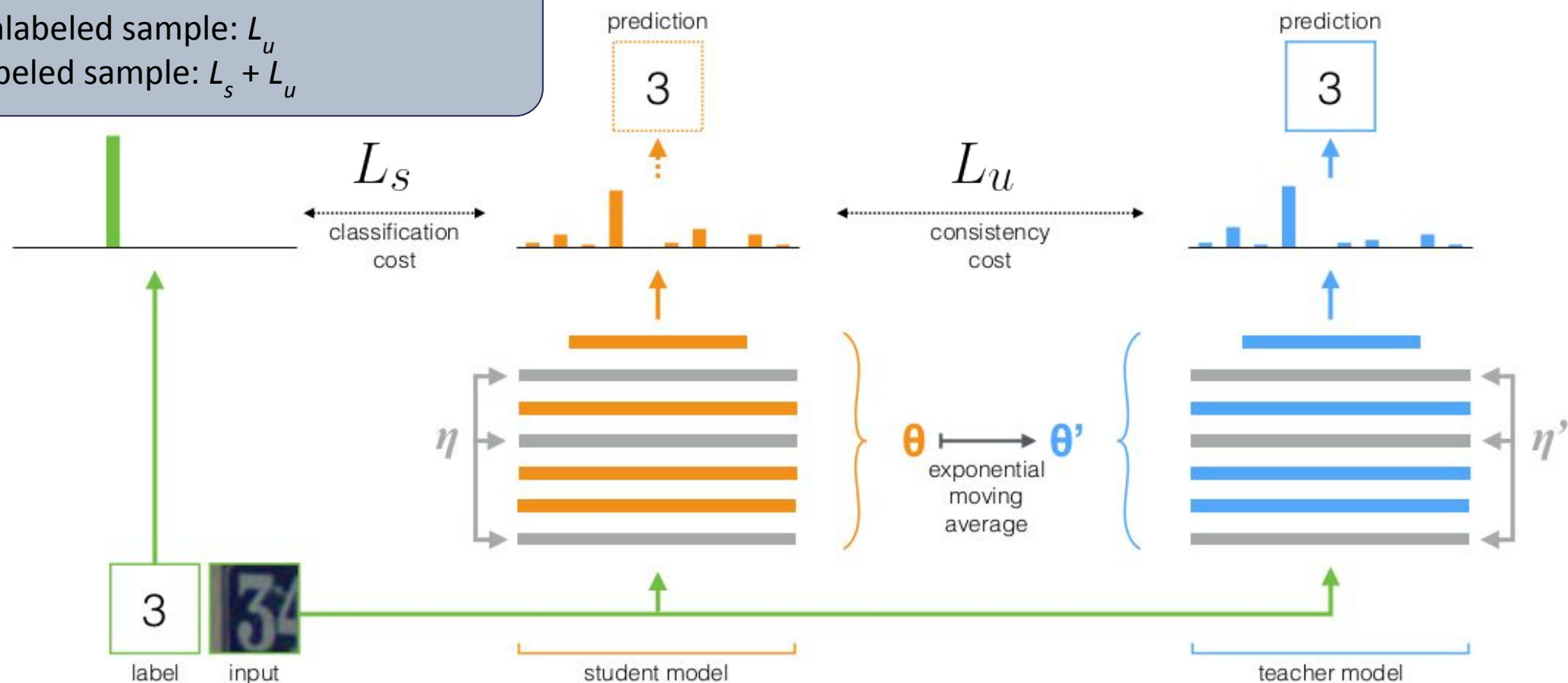
Consistency based: Mean teacher



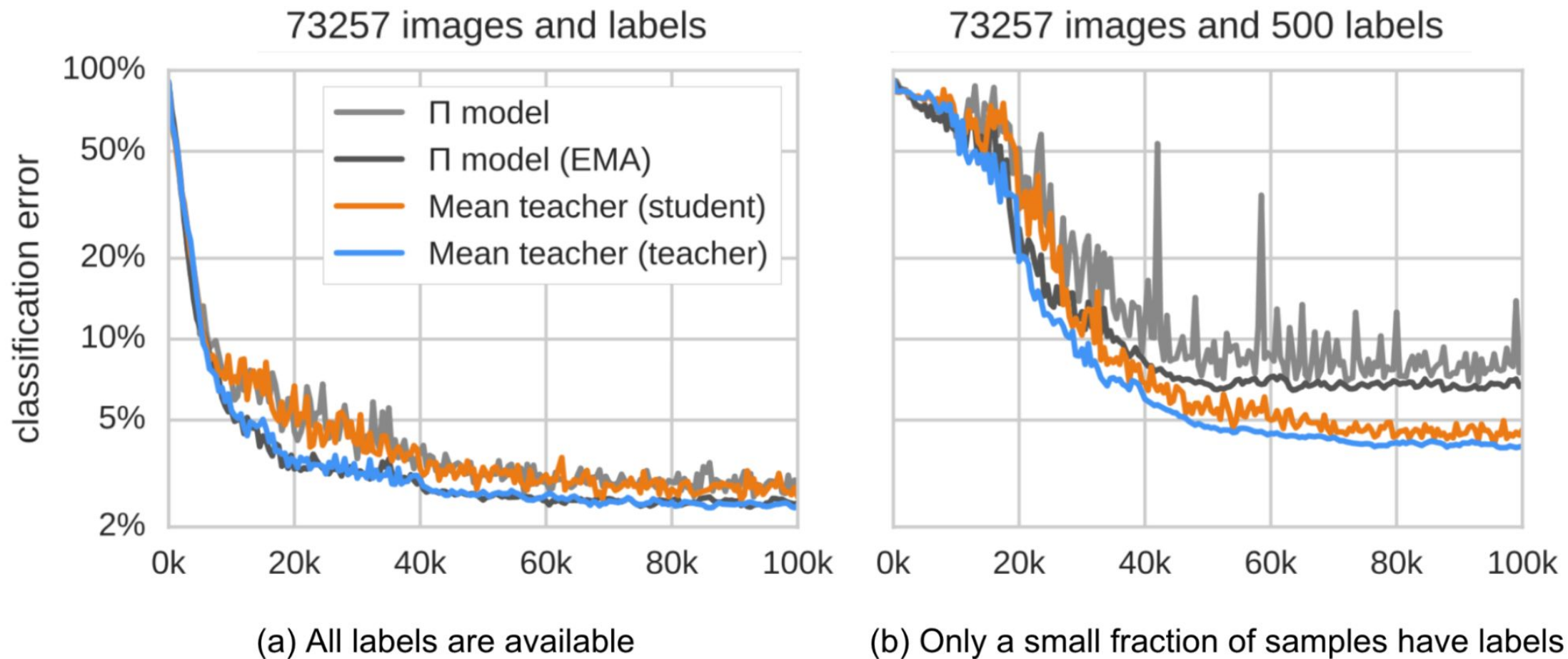
Consistency based: Mean teacher

In practice:

- unlabeled sample: L_u
- labeled sample: $L_s + L_u$



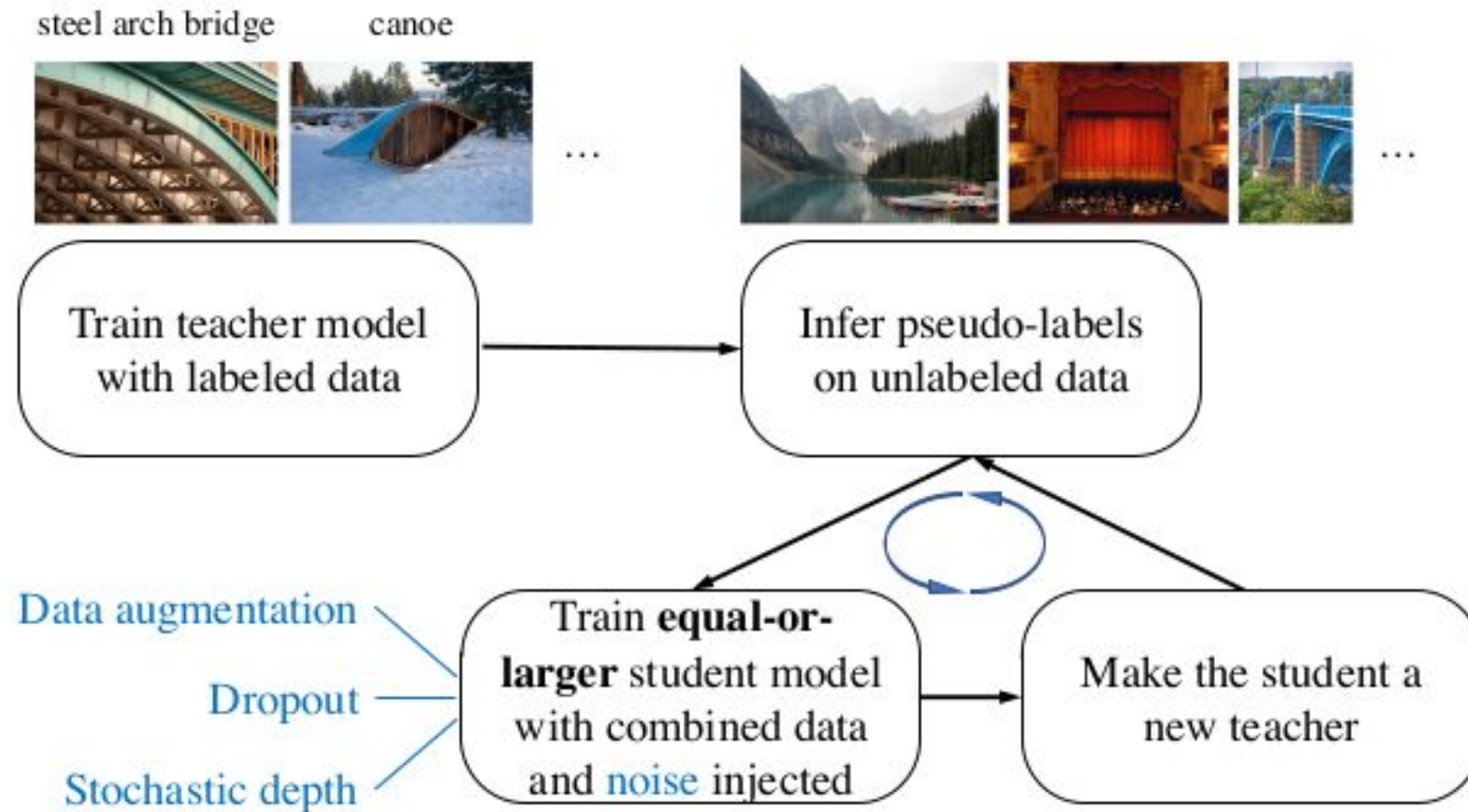
Consistency based: Mean teacher



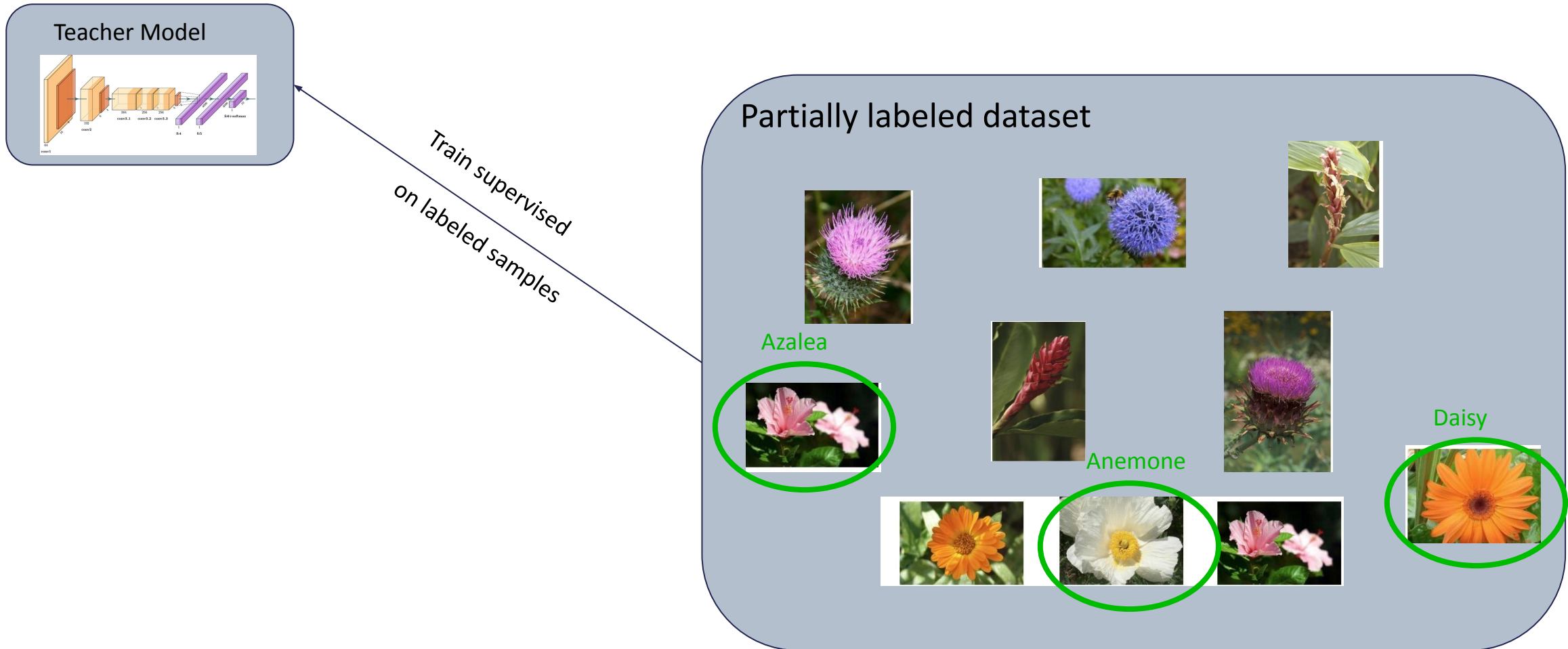
Further work on consistency based SSL

- FixMatch: <https://arxiv.org/abs/2001.07685>
 - Strong and weak augmentation; fixed teacher (stop gradient) instead of mean
- UniMatch: <https://arxiv.org/abs/2208.09910>
 - Applying semi supervised learning to segmentation (following the FixMatch design)
- UniMatch V2: <https://arxiv.org/abs/2410.10777>
 - Extension of UniMatch to stronger vision backbones (ViT-based, e.g. Dino v2)

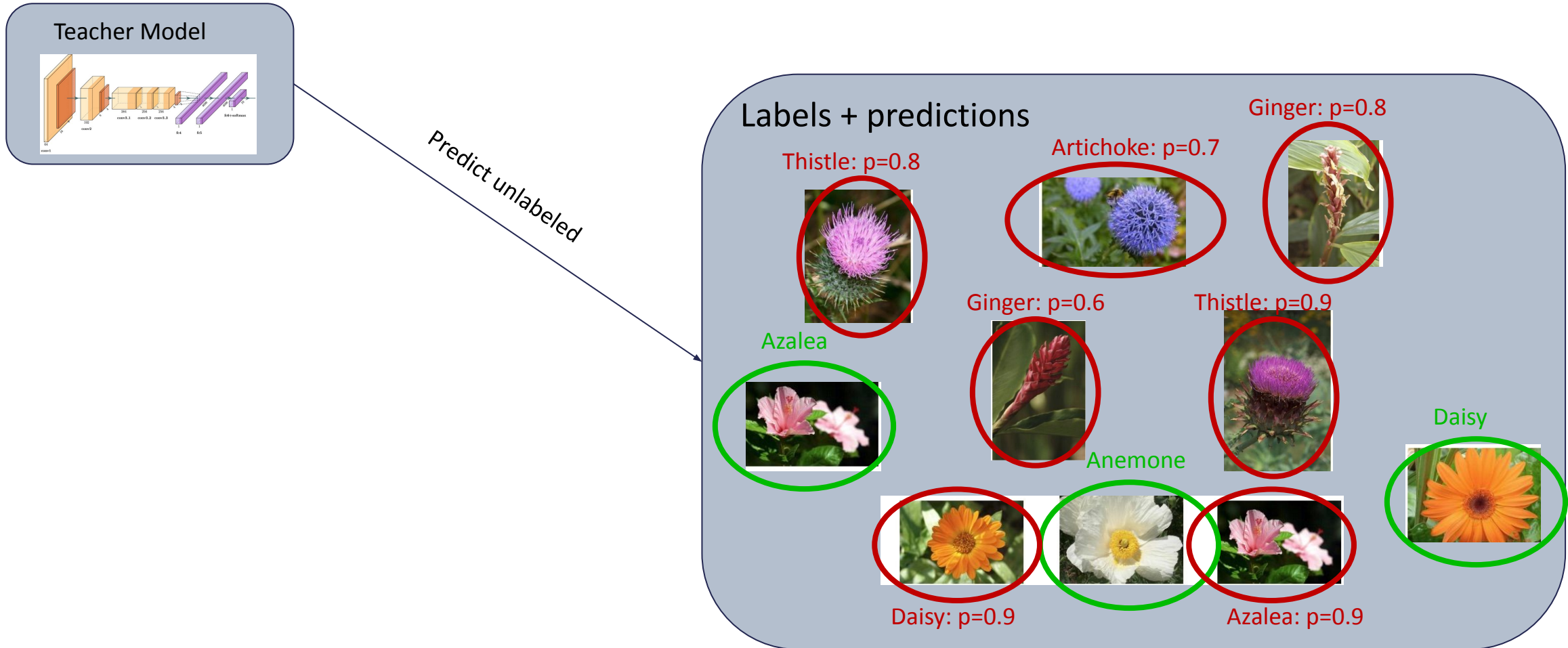
Pseudo-label based



Pseudo-label based

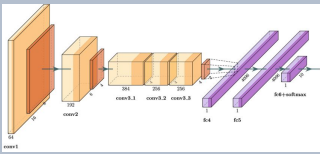


Pseudo-label based



Pseudo-label based

Teacher Model



Different strategies for predictions->pseudo-labels.
Here: keep confident predictions (e.g. $p \geq 0.8$)

Keep confident predictions

Thistle: $p=0.8$



Azalea



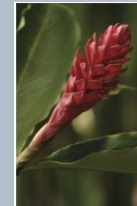
Ginger: $p=0.8$



Thistle: $p=0.9$



Anemone



Daisy: $p=0.9$



Azalea: $p=0.9$

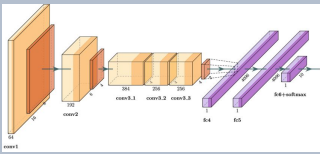


Daisy

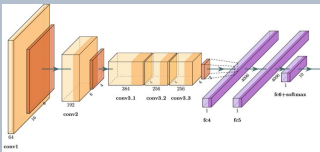


Pseudo-label based

Teacher Model



Student Model



Train on labeled samples
and pseudo-labels

Labels + pseudo-labels

Thistle: $p=0.8$



Azalea



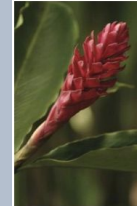
Ginger: $p=0.8$



Thistle: $p=0.9$



Anemone



Daisy

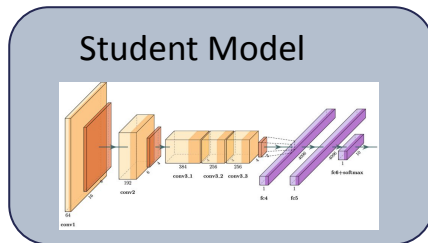
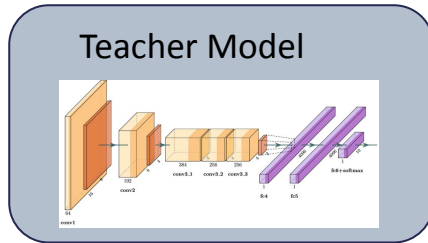


Daisy: $p=0.9$



Azalea: $p=0.9$

Pseudo-label based

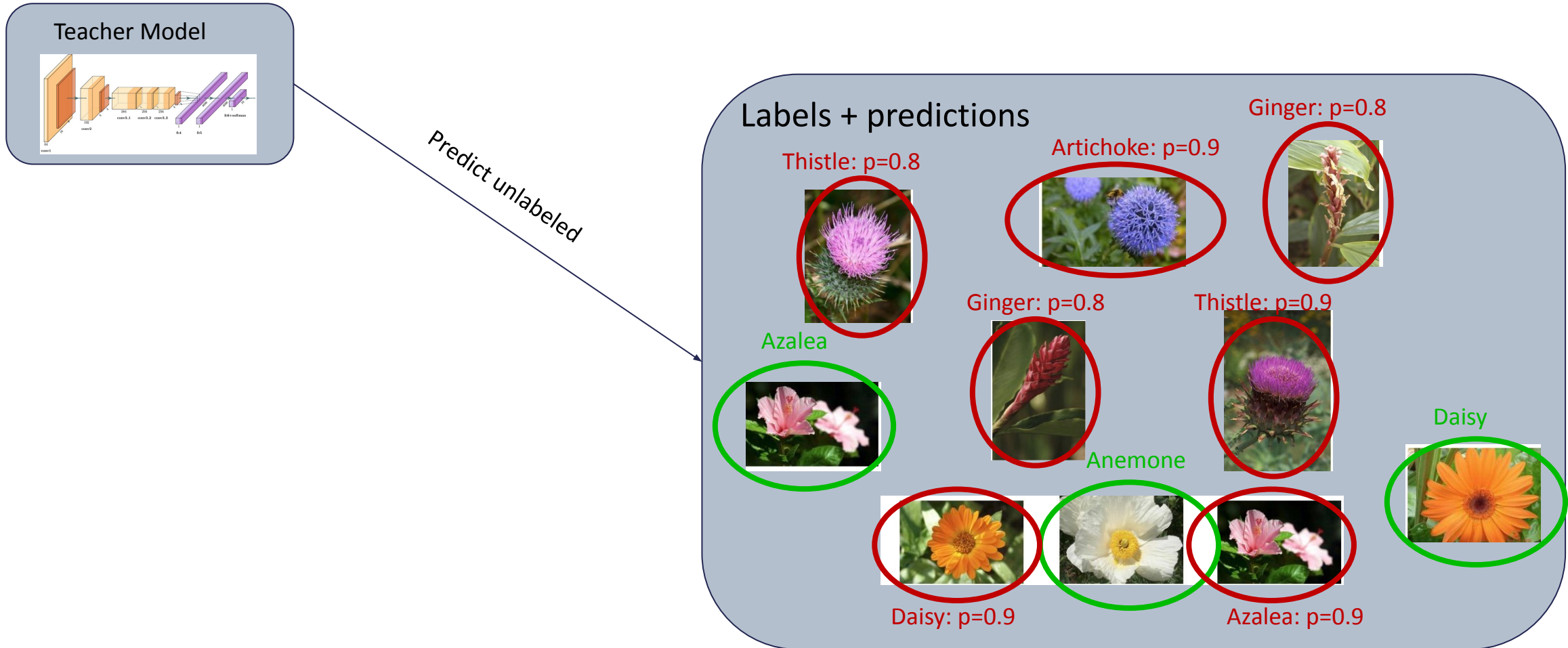


Make student the new teacher

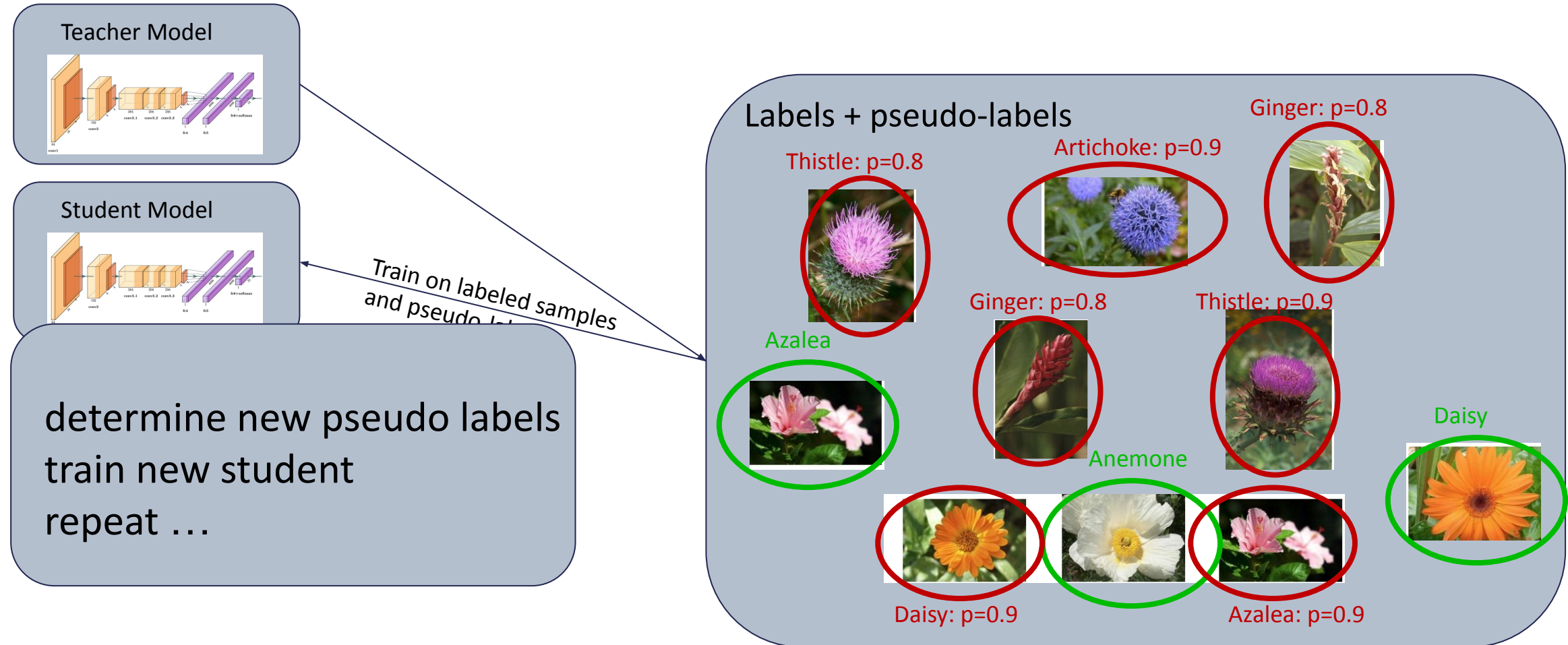
Labels + pseudo-labels



Pseudo-label based



Pseudo-label based



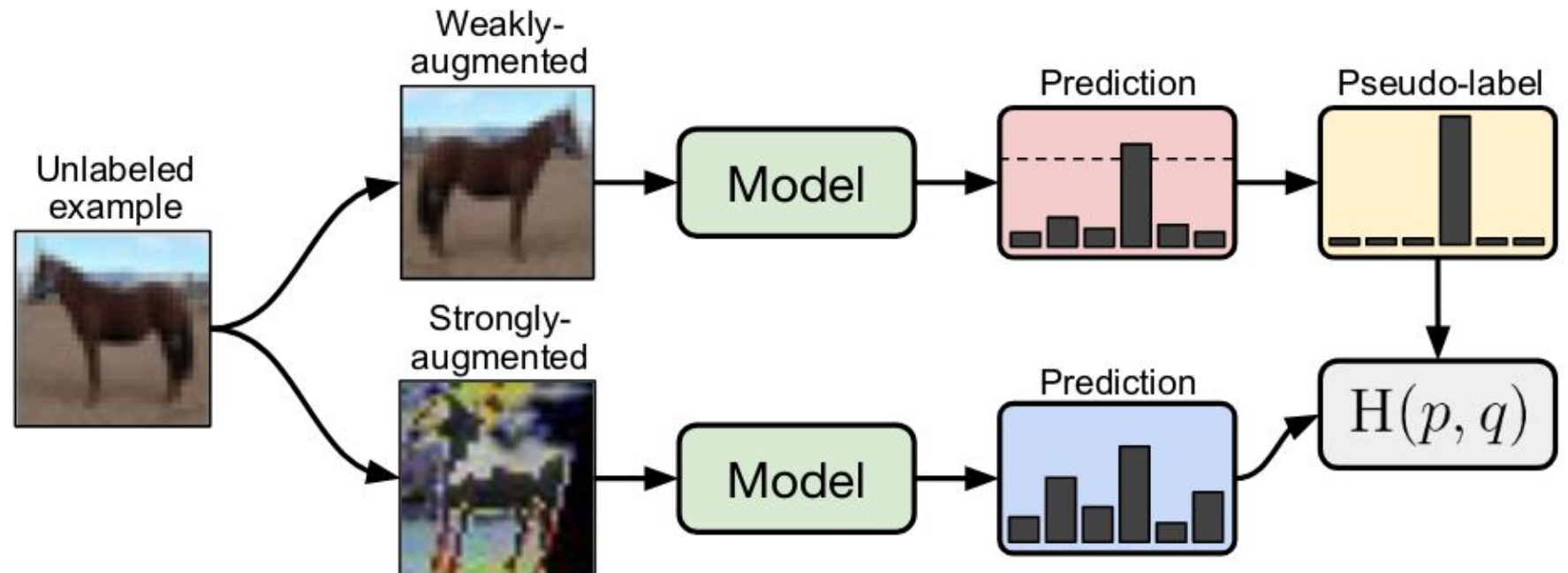
Pseudo-labeling strategies

- Our example: only keep confident pseudo-labels
- Original paper: use all predictions as pseudo labels
<http://citeseerx.ist.psu.edu/viewdoc/summary?doi=10.1.1.664.3543>
- Noisy student: use noise (augmentations, dropout, etc.) in student training
<https://arxiv.org/abs/1911.04252>
- Label propagation: create similarity graph between samples based on deep features of the network, propagate labels between similar samples
<https://arxiv.org/abs/1904.04717>

Current state-of-the-art: FixMatch

<http://arxiv.org/abs/2001.07685>

- Similar set-up to “mean teacher”, but:
 - weakly and strongly augmented version of unlabeled image
 - prediction from weakly augmented image is used as pseudo-label for strongly labeled (if probability > threshold)



Current state-of-the-art: FixMatch

<http://arxiv.org/abs/2001.07685>

Many semi-supervised learning methods can be mapped to common schema

Algorithm	Artificial label augmentation	Prediction augmentation	Artificial label post-processing	Notes
TS / Π -Model	Weak	Weak	None	
Temporal Ensembling	Weak	Weak	None	Uses model from earlier in training
Mean Teacher	Weak	Weak	None	Uses an EMA of parameters
Virtual Adversarial Training	None	Adversarial	None	
UDA	Weak	Strong	Sharpening	Ignores low-confidence artificial labels
MixMatch	Weak	Weak	Sharpening	Averages multiple artificial labels
ReMixMatch	Weak	Strong	Sharpening	Sums losses for multiple predictions
FixMatch	Weak	Strong	Pseudo-labeling	

Current state-of-the-art: FixMatch

<http://arxiv.org/abs/2001.07685>

FixMatch beats other methods on most “standard” image classification tasks.

Method	CIFAR-10		
	40 labels	250 labels	4000 labels
Π -Model	-	54.26 ± 3.97	14.01 ± 0.38
Pseudo-Labeling	-	49.78 ± 0.43	16.09 ± 0.28
Mean Teacher	-	32.32 ± 2.30	9.19 ± 0.19
MixMatch	47.54 ± 11.50	11.05 ± 0.86	6.42 ± 0.10
UDA	29.05 ± 5.93	8.82 ± 1.08	4.88 ± 0.18
ReMixMatch	19.10 ± 9.64	5.44 ± 0.05	4.72 ± 0.13
FixMatch (RA)	13.81 ± 3.37	5.07 ± 0.65	4.26 ± 0.05
FixMatch (CTA)	11.39 ± 3.35	5.07 ± 0.33	4.31 ± 0.15



FixMatch reaches 78% CIFAR-10 accuracy using only above 10 labeled images.

Weakly supervised learning

Weakly supervised learning

Use a “cheaper” version of labels for training the model:

- Classification: use images from the web with scraped text
 - images + captions
 - images & hashtags from instagram



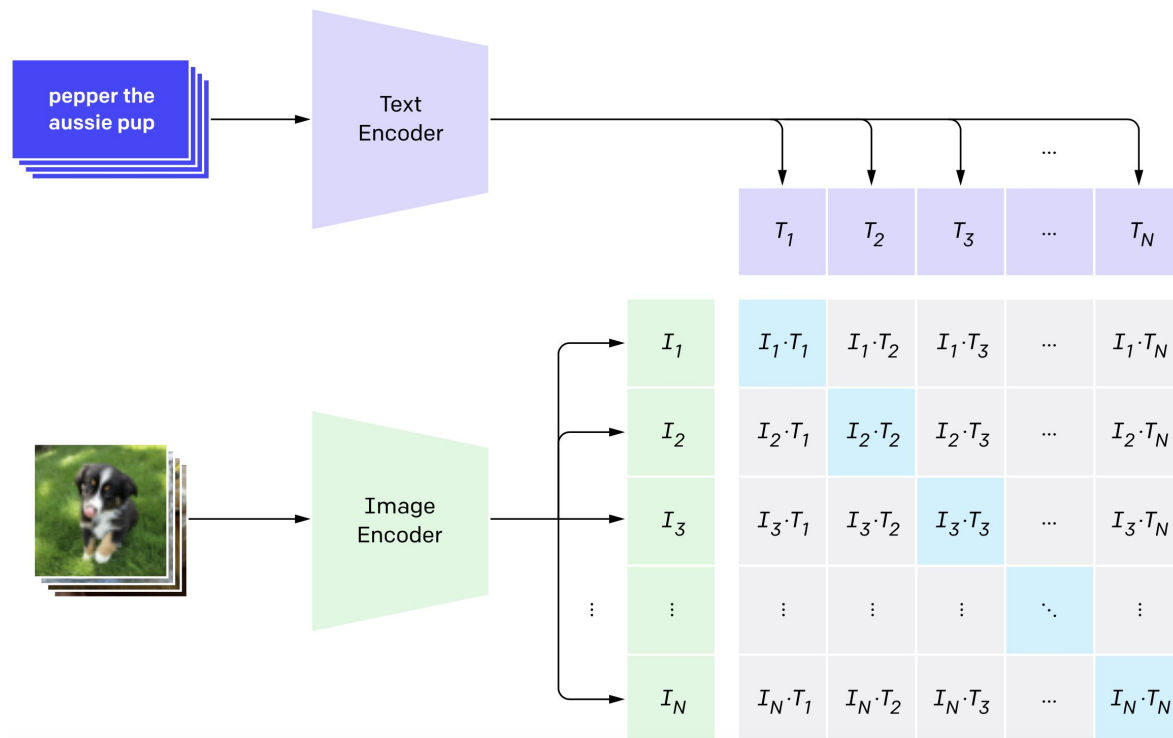
#dog #cute #fluffy #goodboy

Weakly supervised learning

Use a “cheaper” version of labels for training the model:

- Classification: use images from the web with scraped text
 - **images + captions (will cover in detail next week)**

CLIP & SigLIP: joint image and text embedding
Contrastive training of images and captions



CLIP: <https://arxiv.org/abs/2103.00020>

SigLIP: <https://arxiv.org/abs/2303.15343>

Weakly supervised learning

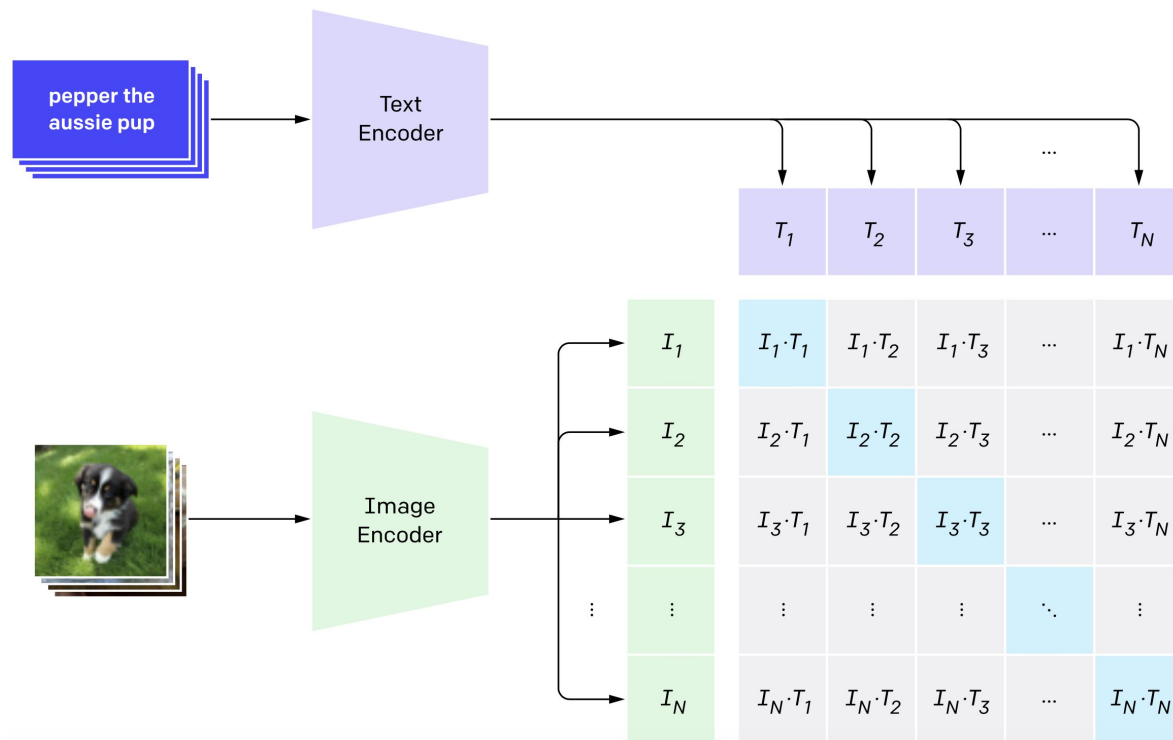
Use a “cheaper” version of labels for training the model:

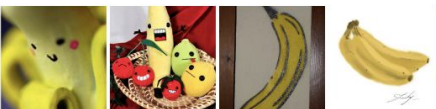
- Classification: use images from the web with scraped text
 - **images + captions (will cover in detail next week)**

CLIP & SigLIP: joint image and text embedding

Contrastive training of images and captions

Zero-shot classification and basis for many text/image tasks



DATASET	IMAGENET RESNET101	CLIP VIT-L
 ImageNet	76.2%	76.2%
 ImageNet V2	64.3%	70.1%
 ImageNet Rendition	37.7%	88.9%
 ObjectNet	32.6%	72.3%
 ImageNet Sketch	25.2%	60.2%

CLIP: <https://arxiv.org/abs/2103.00020>

SigLIP: <https://arxiv.org/abs/2303.15343>

Weakly supervised learning

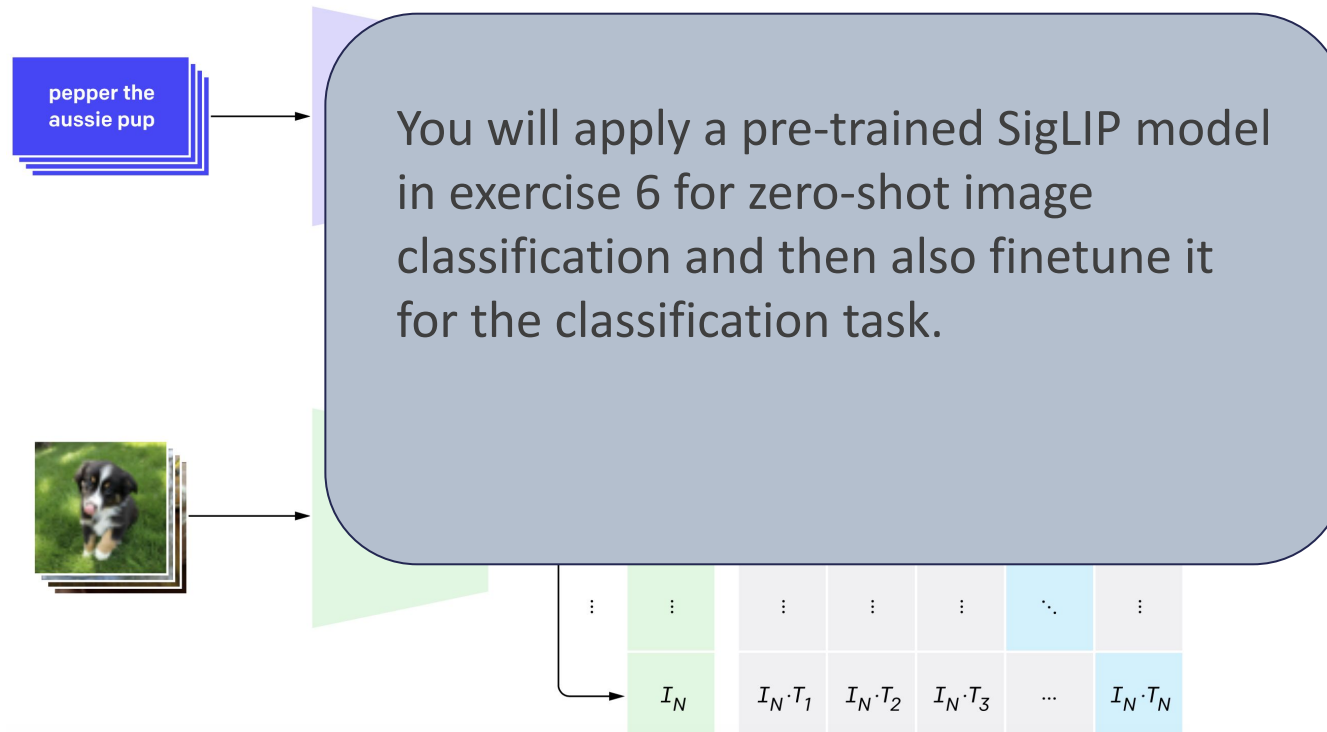
Use a “cheaper” version of labels for training the model:

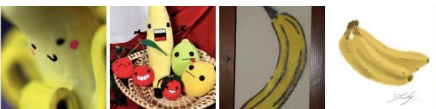
- Classification: use images from the web with scraped text
 - **images + captions (will cover in detail next week)**

CLIP & SigLIP: joint image and text embedding

Contrastive training of images and captions

Zero-shot classification and basis for many text/image tasks



DATASET	IMAGENET RESNET101	CLIP VIT-L
 ImageNet	76.2%	76.2%
 ImageNet V2	64.3%	70.1%
 ImageNet Rendition	37.7%	88.9%
 ObjectNet	32.6%	72.3%
 ImageNet Sketch	25.2%	60.2%

CLIP: <https://arxiv.org/abs/2103.00020>

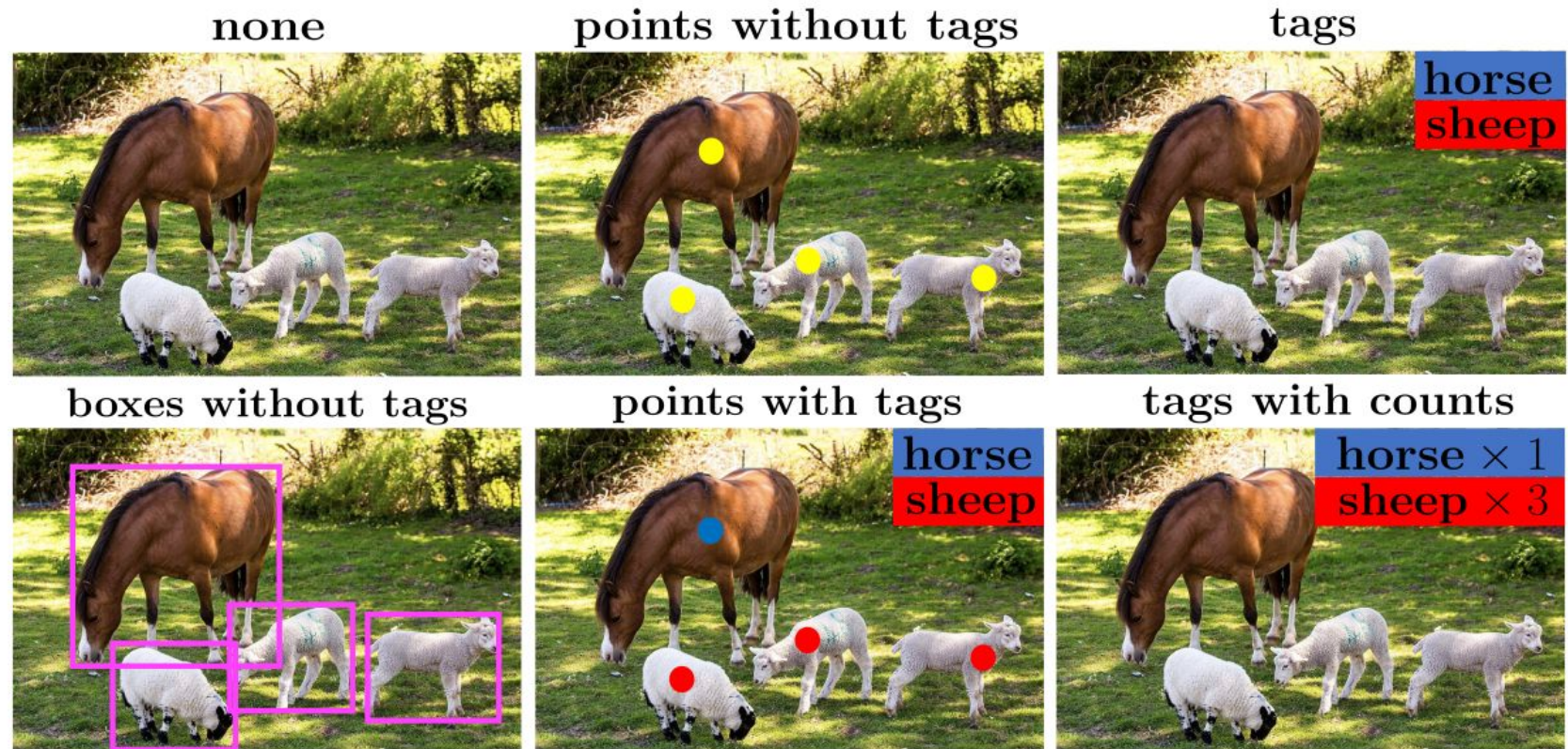
SigLIP: <https://arxiv.org/abs/2303.15343>

Weakly supervised learning

<https://arxiv.org/abs/2203.16089>

Use a “cheaper” version of labels for training the model:

- Classification: use images from the web with scraped text
 - e.g. images & hashtags from instagram
- Object detection
 - image tags, points, ...

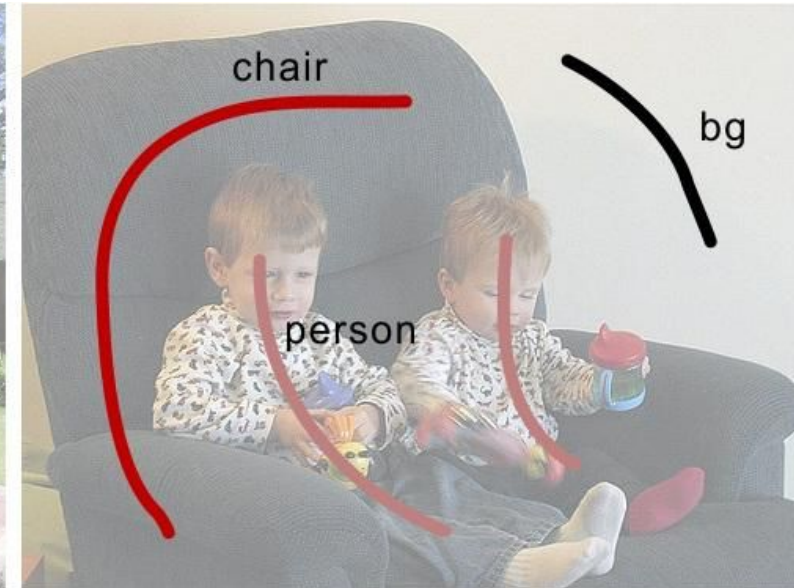
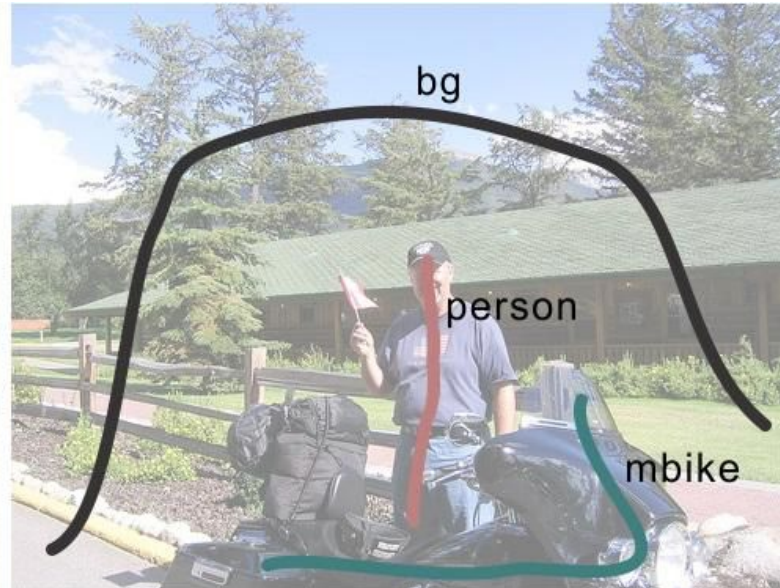
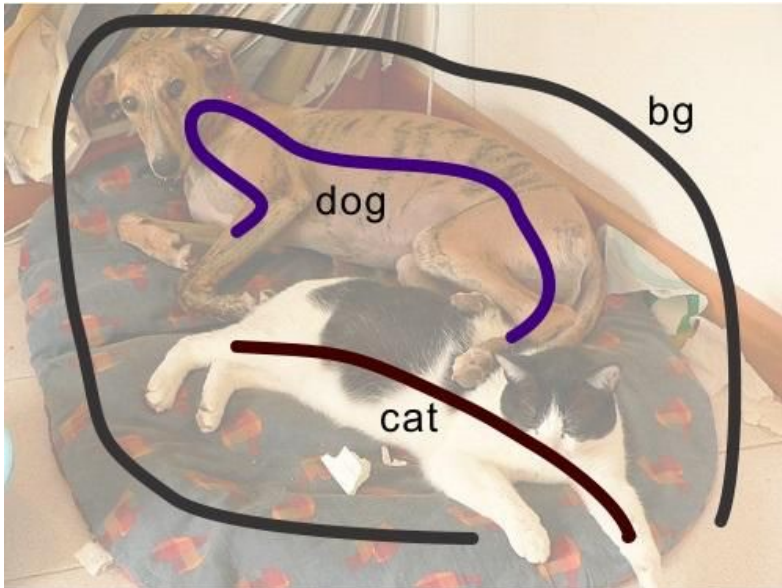


Weakly supervised learning

Use a “cheaper” version of labels for training the model:

- Classification: use images from the web with scraped text
 - e.g. images & hashtags from instagram
- Object detection
 - image tags, points, ...
- Semantic segmentation
 - “scribble” annotations*

*can also be seen as semi-supervised problem



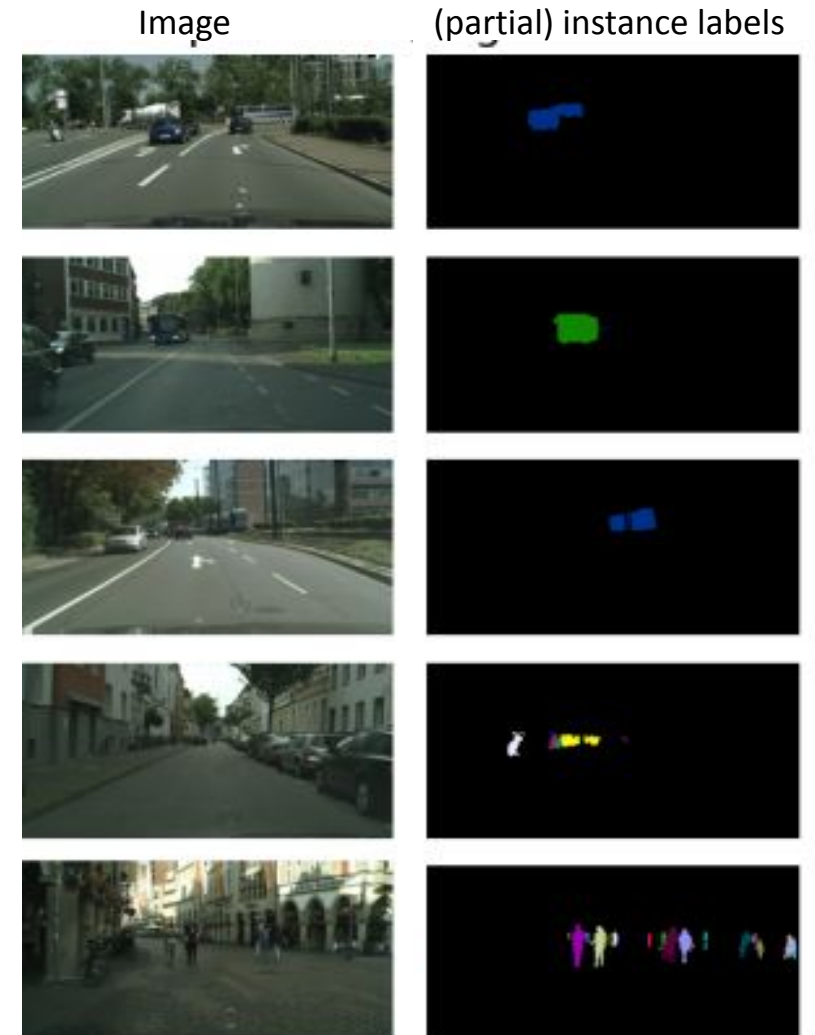
Weakly supervised learning

Use a “cheaper” version of labels for training the model:

- Classification: use images from the web with scraped text
 - e.g. images & hashtags from instagram
- Object detection
 - image tags, points, ...
- Semantic segmentation
 - “scribble” annotations
- Instance segmentation
 - image tags, points, bounding boxes, partial object annotations ...

Example: only some of the objects are labeled (positive unlabeled learning)

<https://arxiv.org/abs/2103.14572>



Weakly supervised learning

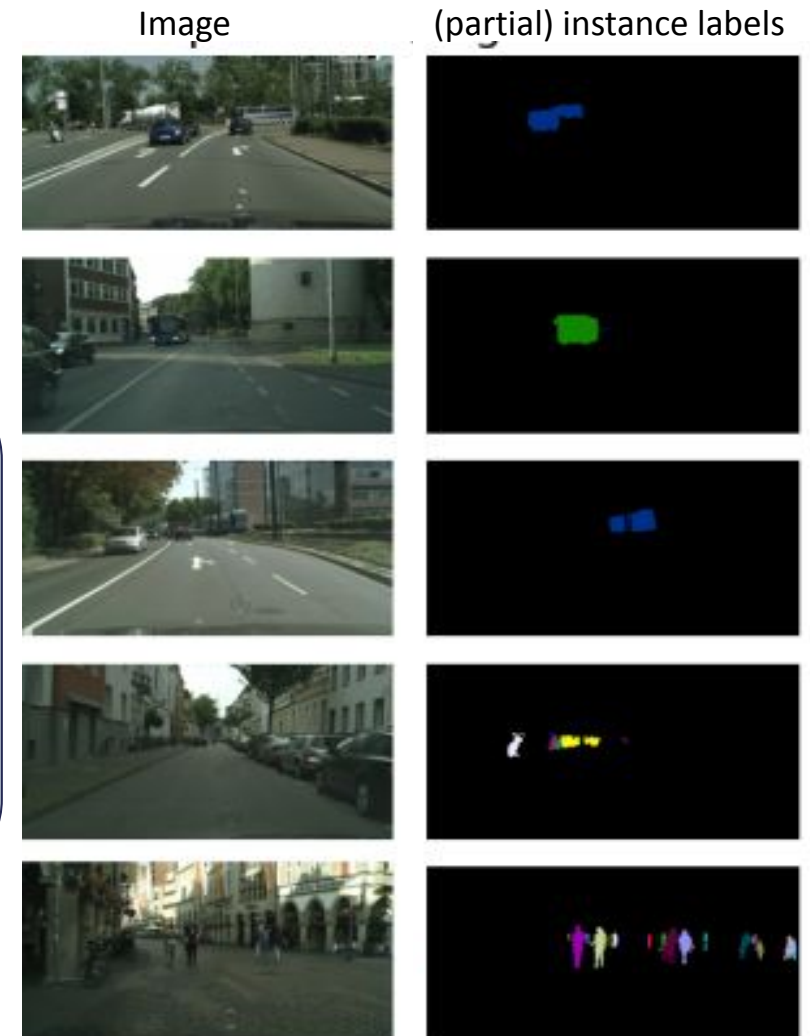
Use a “cheaper” version of labels for training the model:

- Classification: use images from the web with scraped text
 - e.g. images & hashtags from instagram
- Object detection
 - image tags, points, ...
- Semantic segmentation

Each setting requires different method: will not cover in further detail here, but will discuss text supervision next lecture

are labeled (positive unlabeled learning)

<https://arxiv.org/abs/2103.14572>



Self-supervised learning

Recap: Transfer learning

1

Pre-training

(massive data)

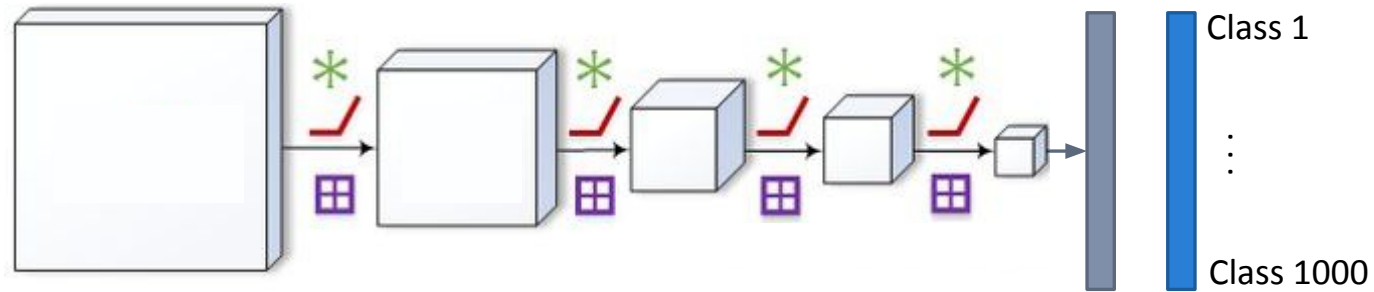
IMAGENET

1.2 million

images

1000 object

classes



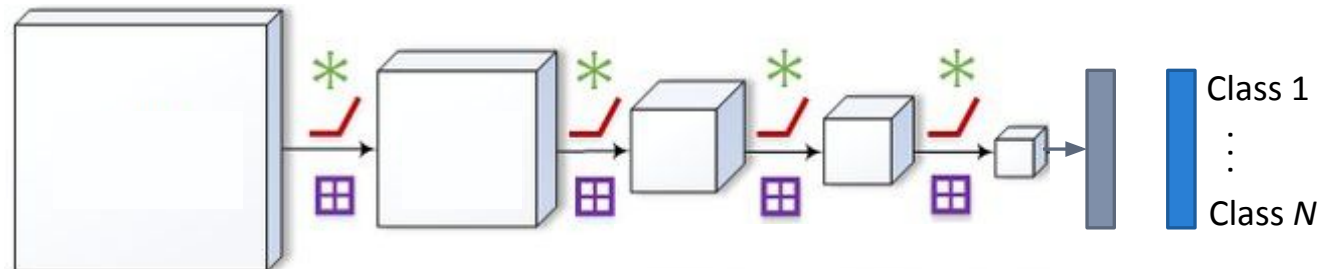
Transfer weights

Replace classifier

2

Fine-tuning

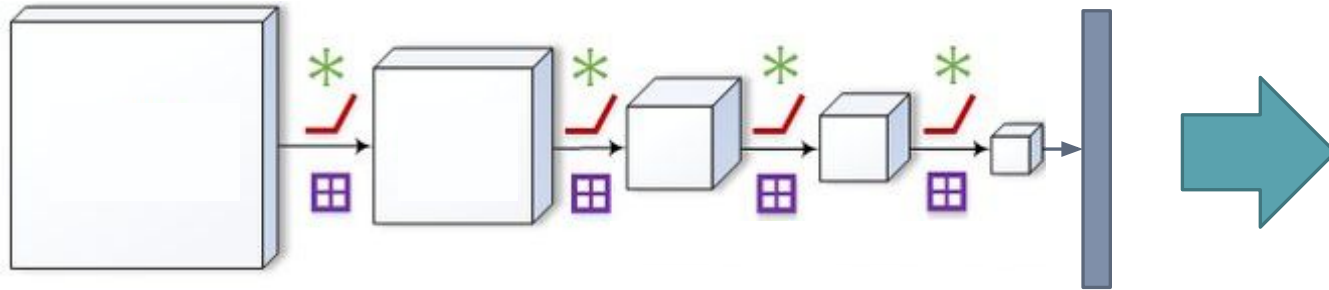
(much less data)



Recognize flowers

Self-supervised learning

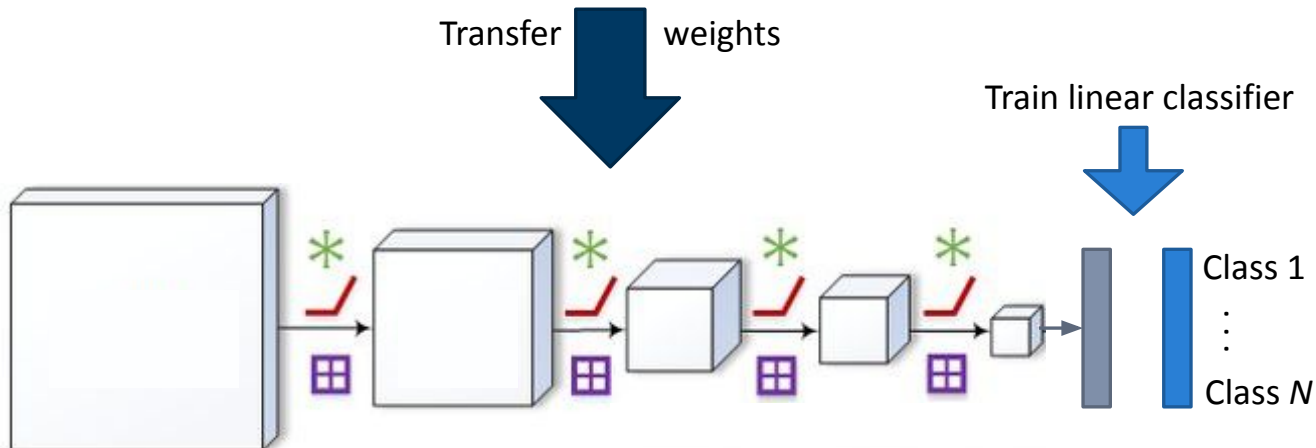
1
Pre-training
Large **unlabeled**
dataset



~~IMAGENET~~

Some other loss
function that
doesn't require
labeled data

2
Fine-tuning
small labeled
dataset

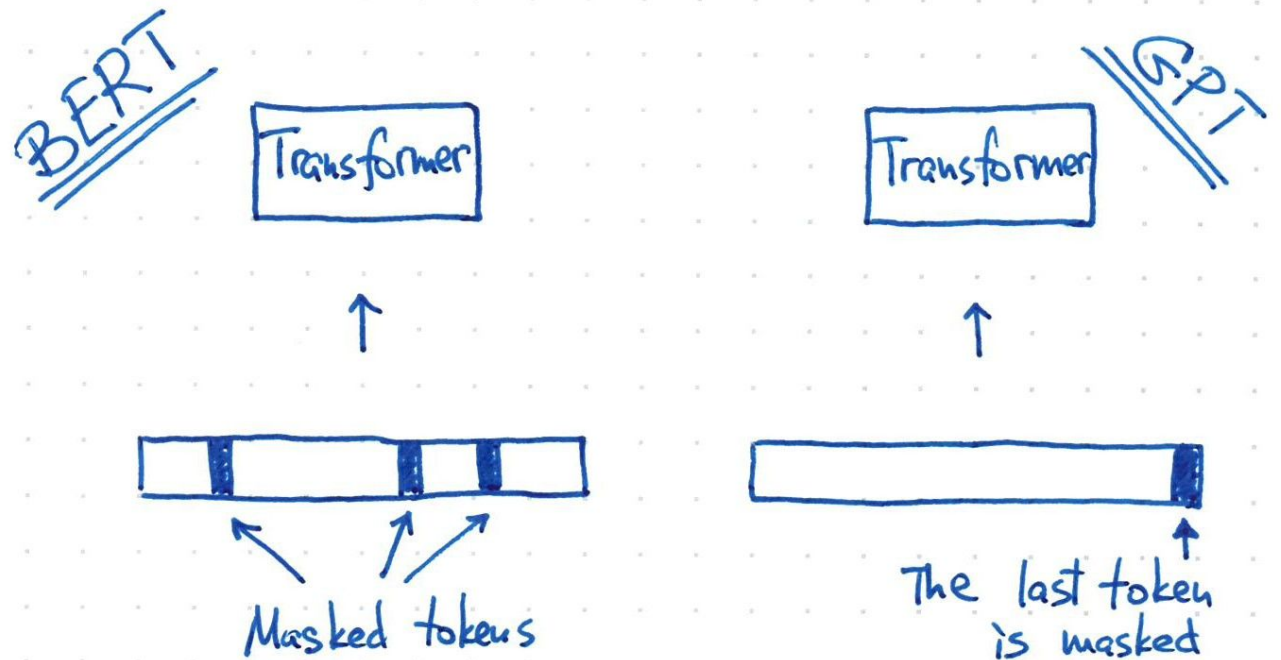


Recognize flowers

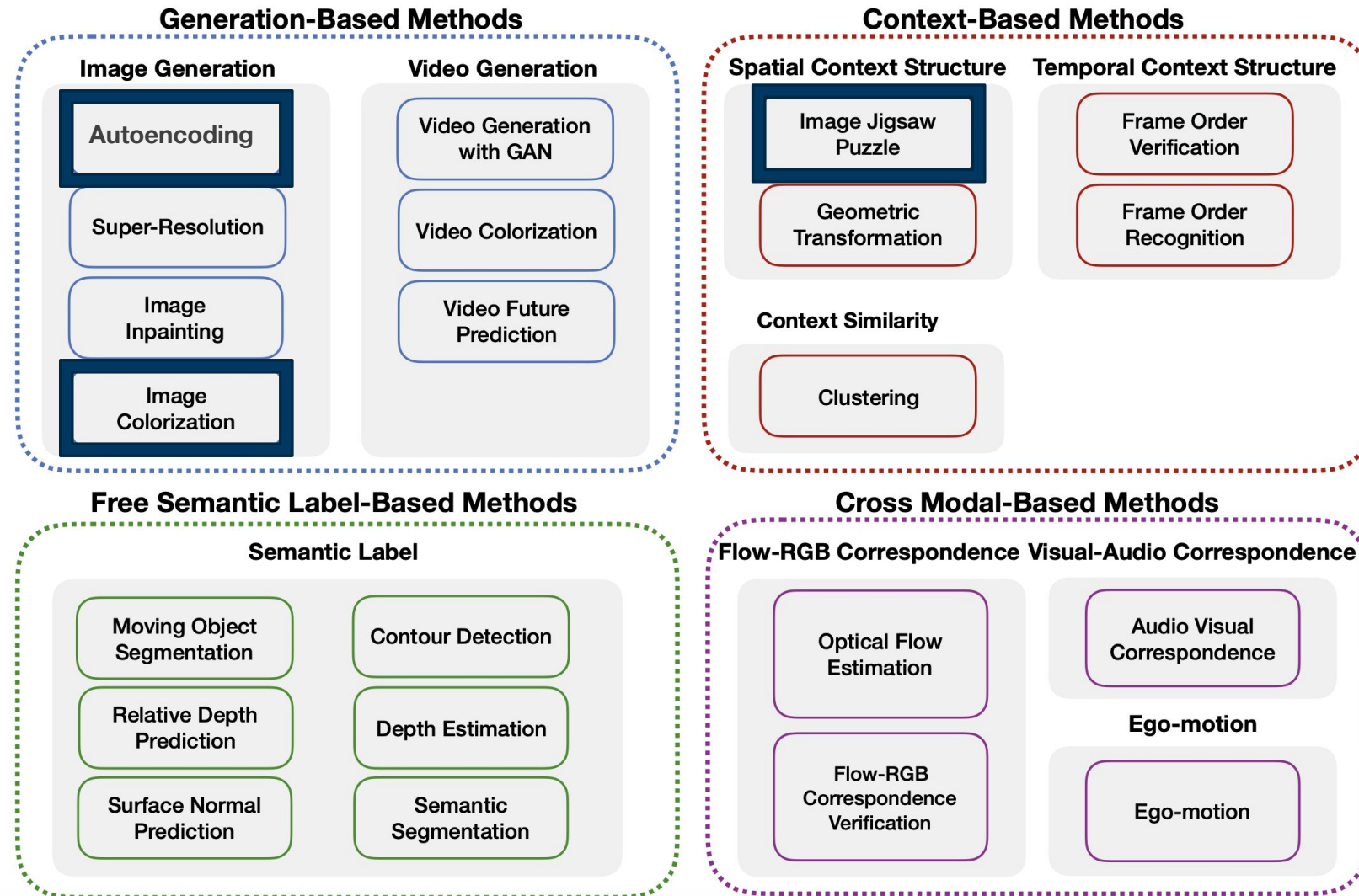
Self-supervised learning in LLMs

Remember last lecture: LLMs pre-trained on large text corpus to predict masked tokens (words)

Can we find similar tasks without explicit labels for vision?



A zoo of pretext tasks



Contrastive and
masking-based
methods
(recent)

Example pretext tasks

Image and video colorization

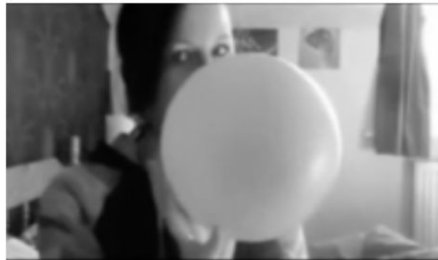


Reference Frame

Future Frame (gray)

Predicted Color

True Color



Input 1

Input 2

Prediction

Ground truth

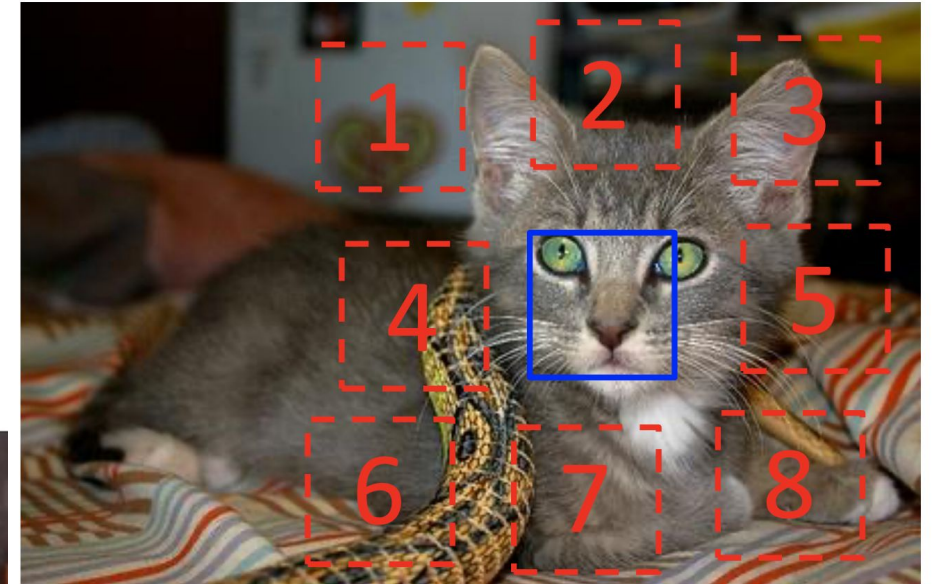
Vondrick et al. "Tracking emerges by colorizing videos", ECCV 2018

Example pretext tasks

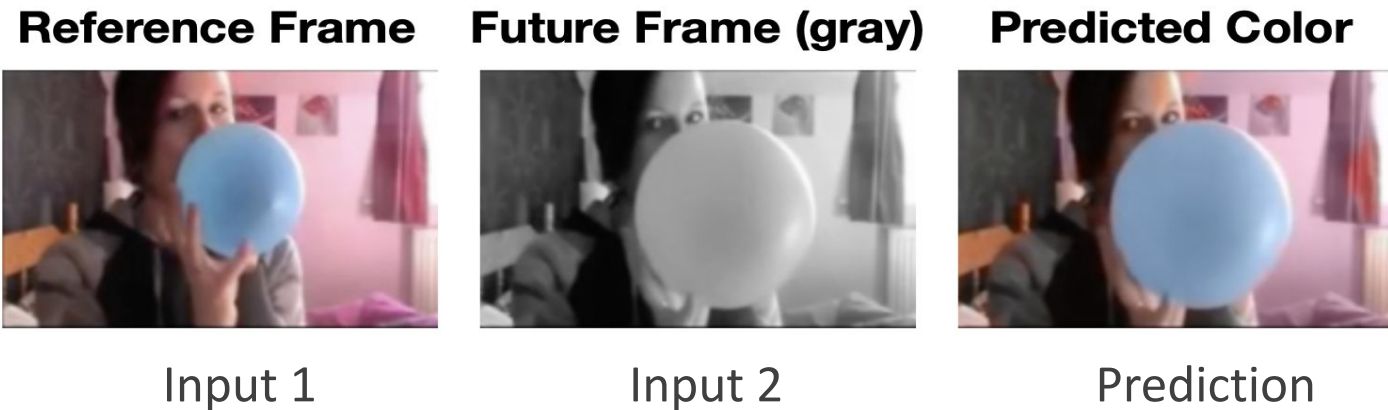
Image and video colorization



Image context prediction



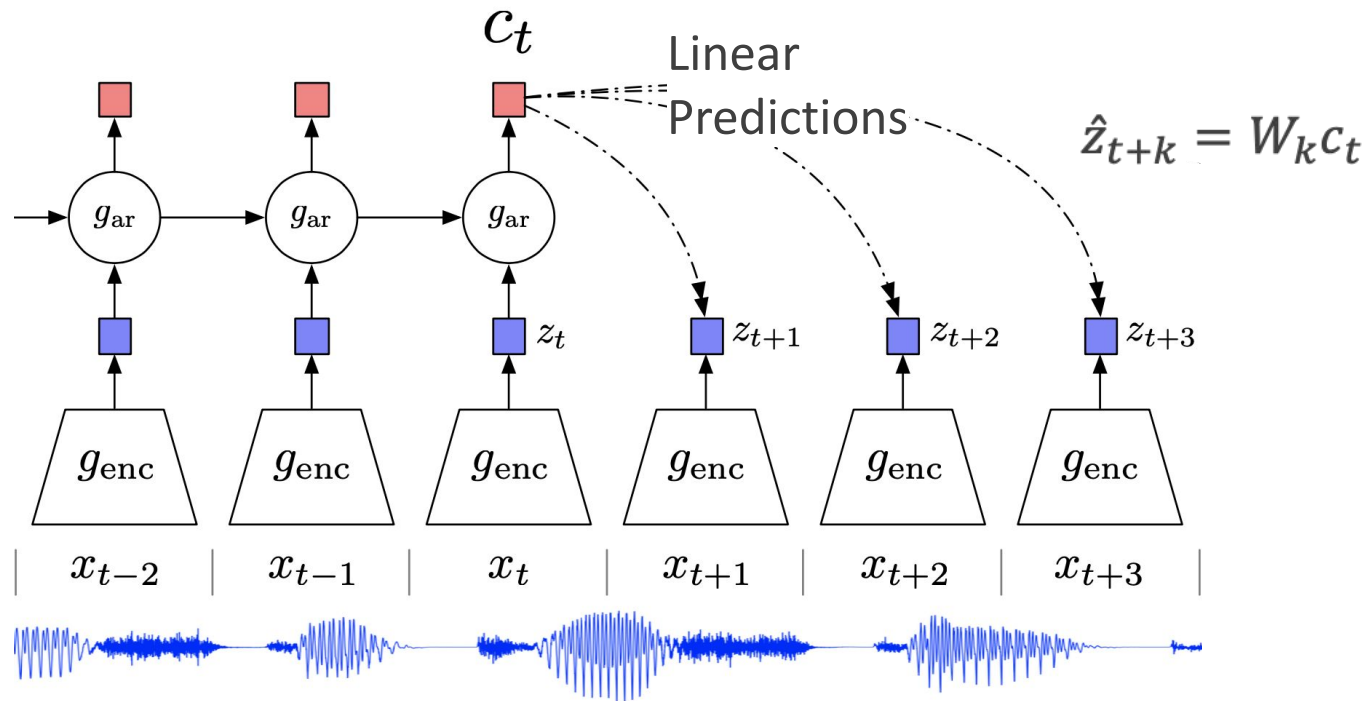
$$X = \left(\begin{array}{c} \text{cat face} \\ \text{cat ear} \end{array} \right); Y = 3$$



Vondrick et al. "Tracking emerges by colorizing videos", ECCV 2018

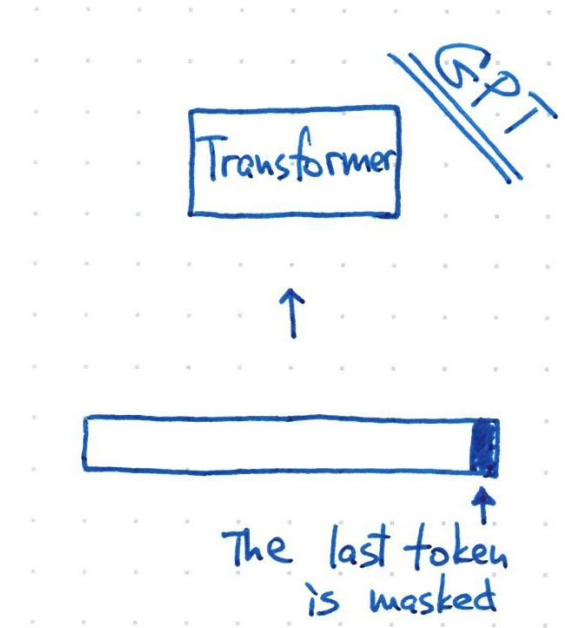
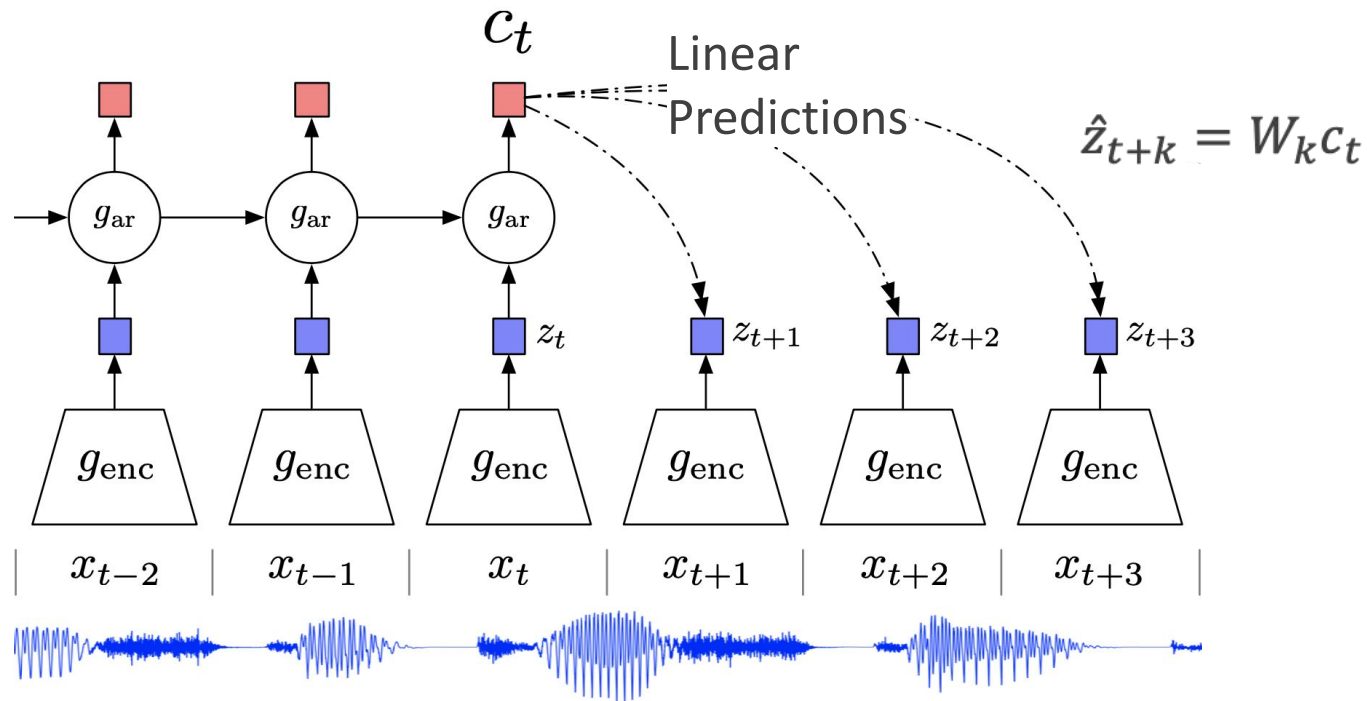
Contrastive learning

Contrastive Predictive Coding (CPC)



Idea: Learn to predict future embeddings z_{t+i} from current context c_t .

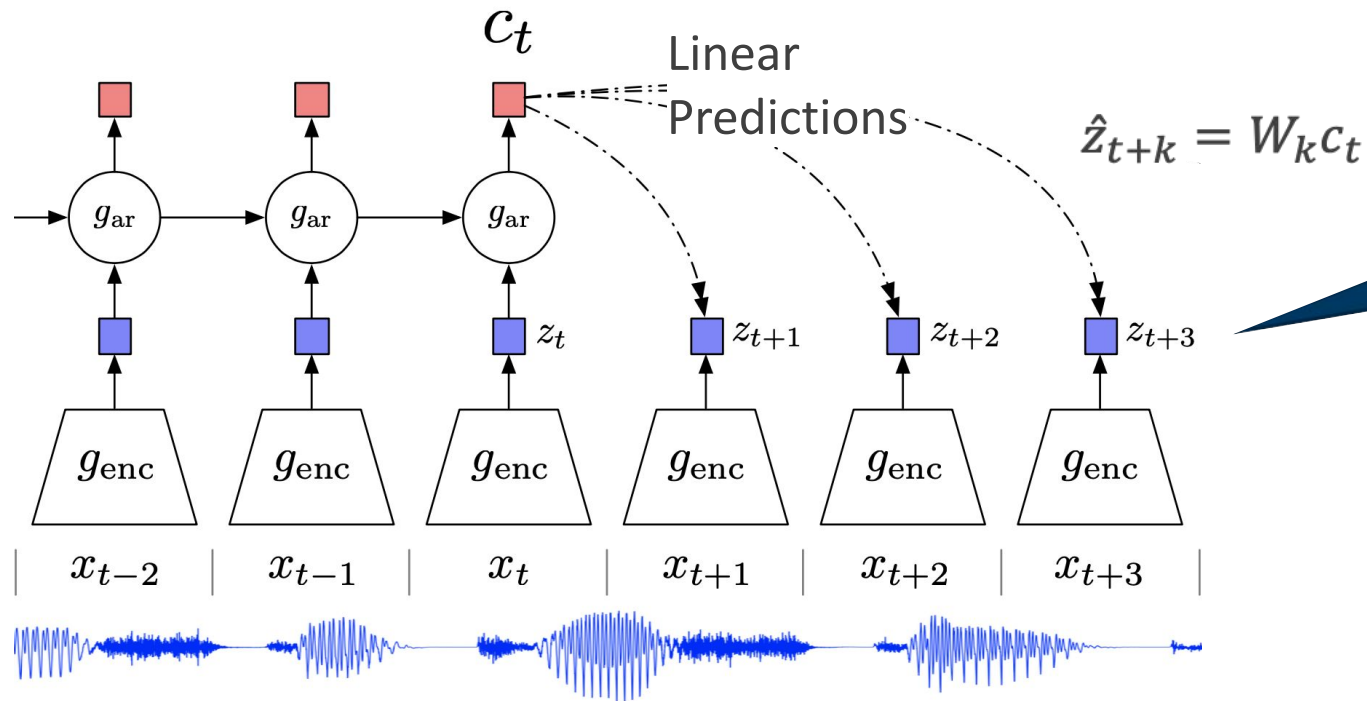
Contrastive Predictive Coding (CPC)



Similar idea to next word prediction, but predict embeddings instead of words.

Idea: Learn to predict future embeddings z_{t+i} from current context c_t .

Contrastive Predictive Coding (CPC)



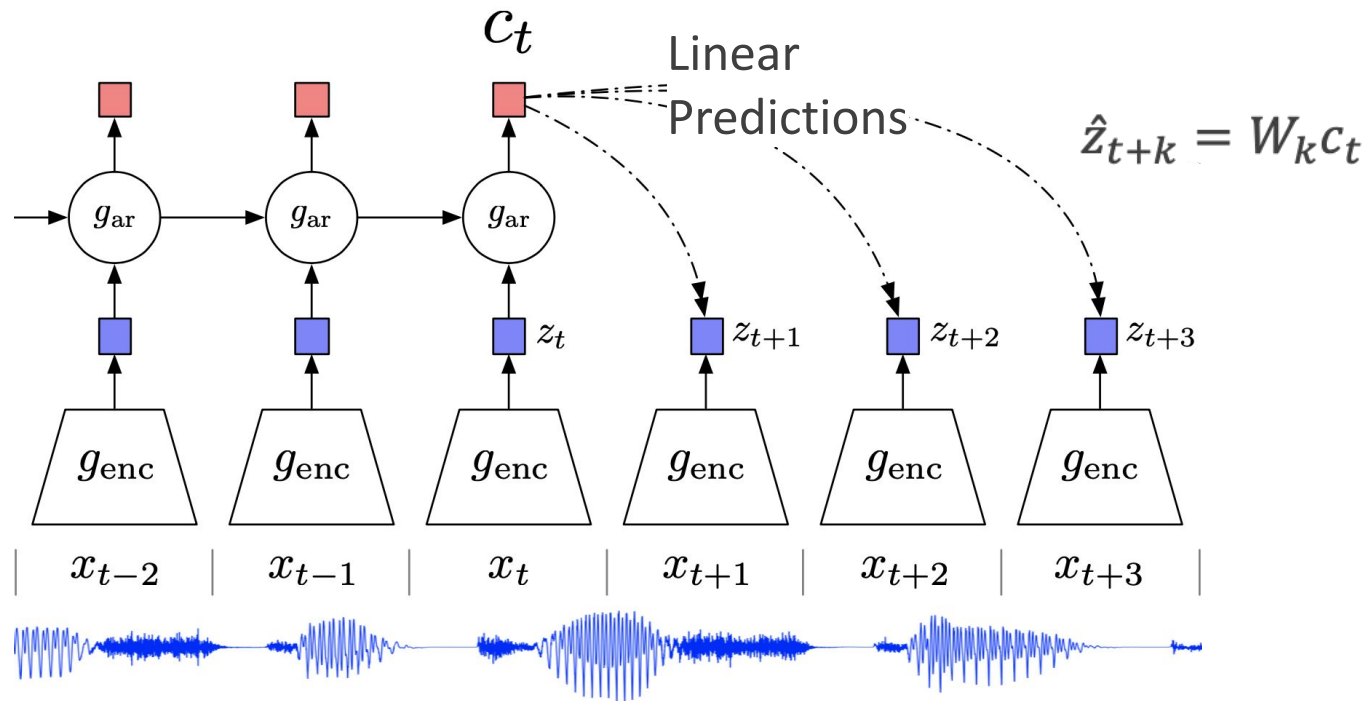
How about MSE loss?

$$\mathcal{L} = \sum (\hat{z}_{t+k} - z_{t+k})^2$$

Will learn $g_{\text{enc}}(x) = 0$!
→ Not useful

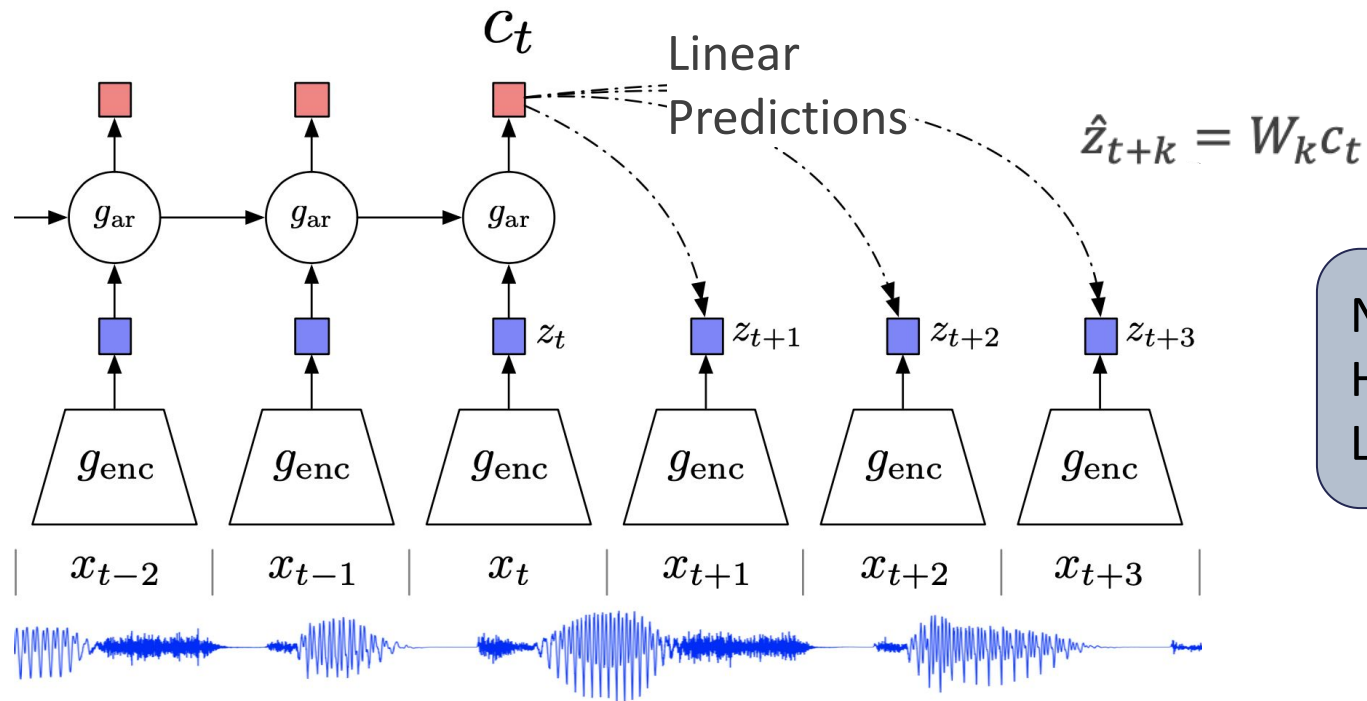
Idea: Learn to predict future embeddings z_{t+i} from current context c_t .

Contrastive Predictive Coding (CPC)



Solution: Predicted embeddings should be more similar to true embeddings than to random samples from the dataset

Contrastive Predictive Coding (CPC)

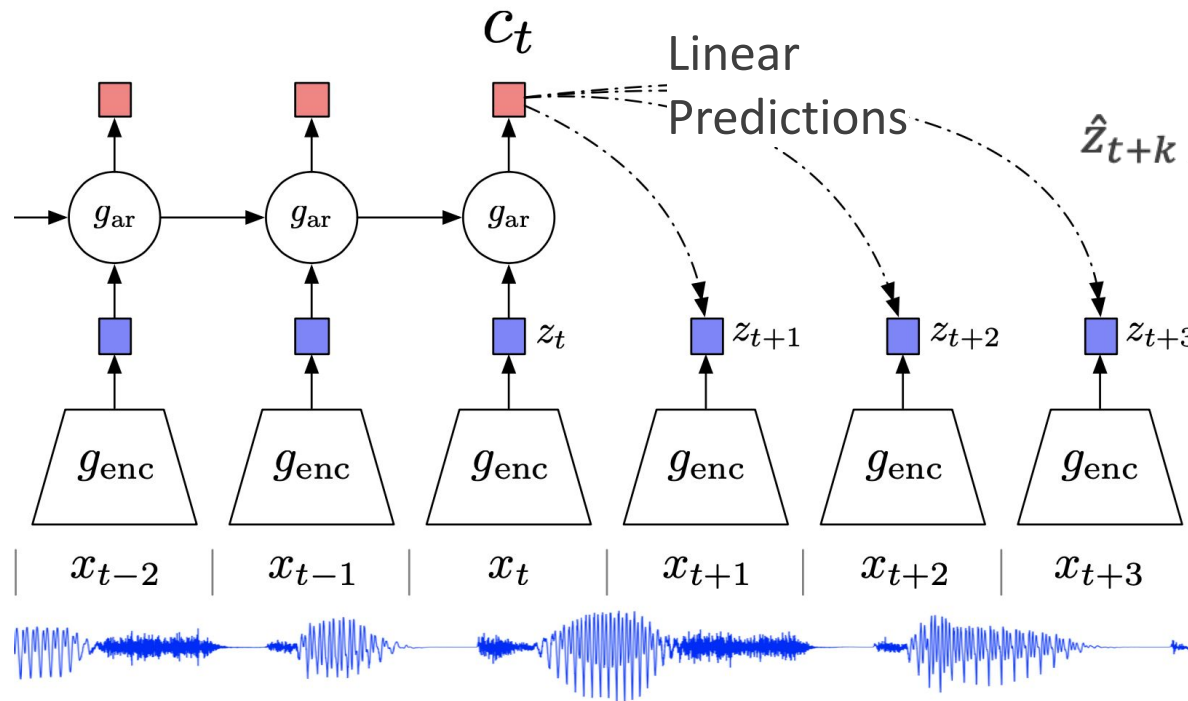


Need similarity measure for embeddings.
High similarity: large value
Low similarity: small value

$$\begin{aligned} f_k(x_{t+k}, c_t) &= \exp(z_{t+k}^T \hat{z}_{t+k}) \\ &= \exp(z_{t+k}^T W_k c_t) \end{aligned}$$

Solution: Predicted embeddings should be more similar to true embeddings than to random samples from the dataset

Contrastive Predictive Coding (CPC)



“InfoNCE” loss:

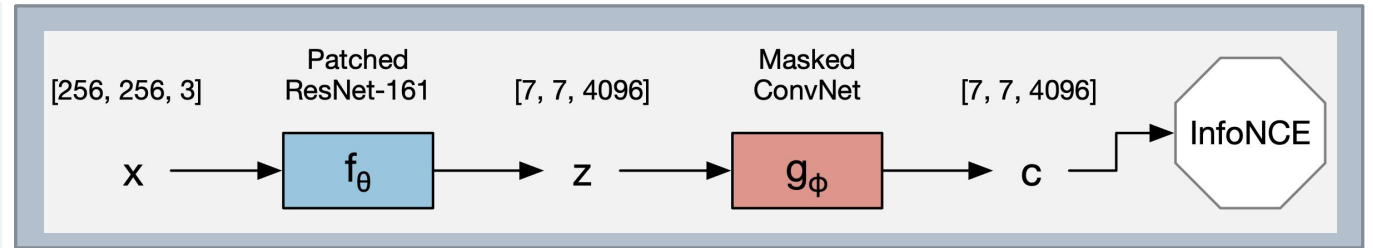
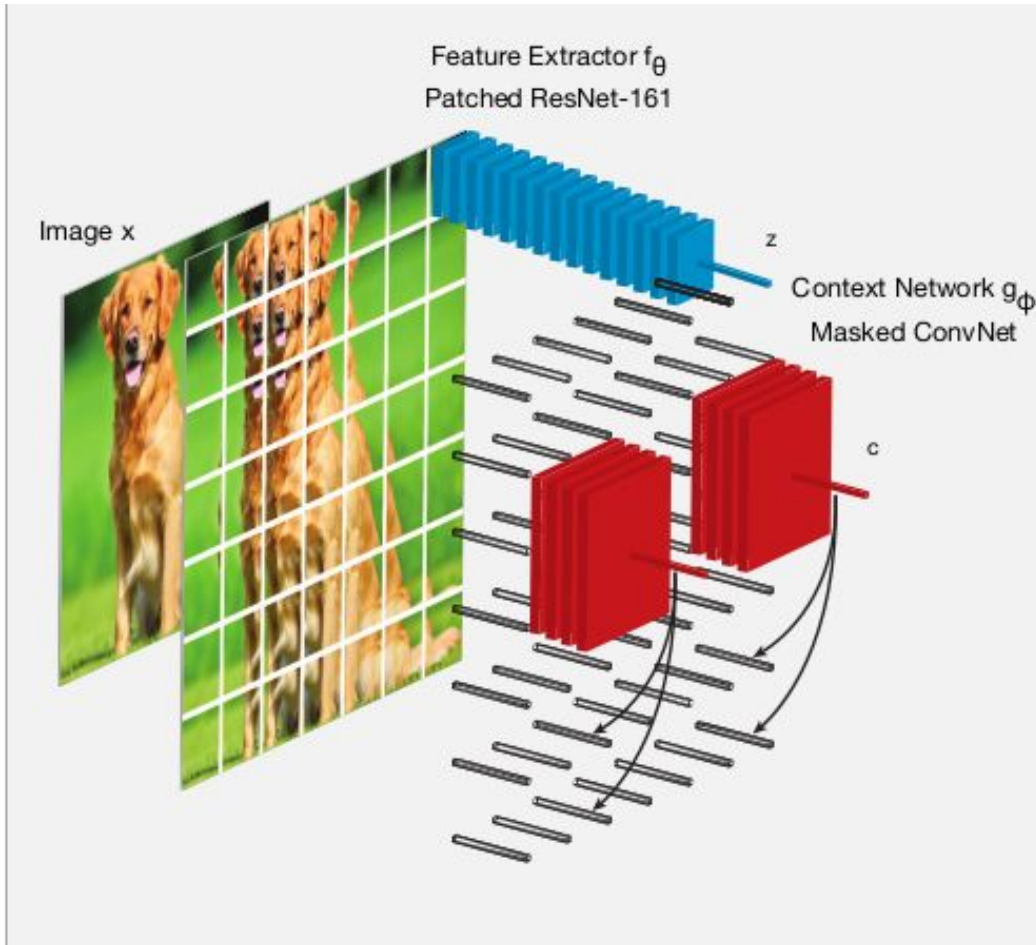
$$\mathcal{L} = -\mathbb{E}_{\mathcal{X}} \left[\log \frac{f_k(x_{t+k}, c_t)}{\sum_{x_j \in \mathcal{X}} f_k(x_j, c_t)} \right]$$

Random samples
from the dataset

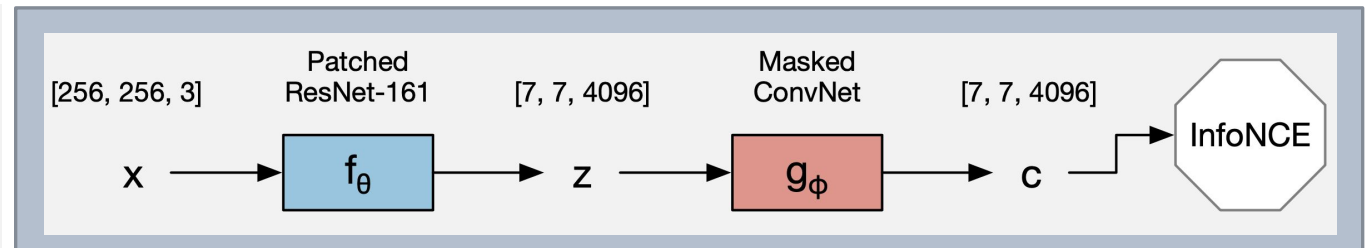
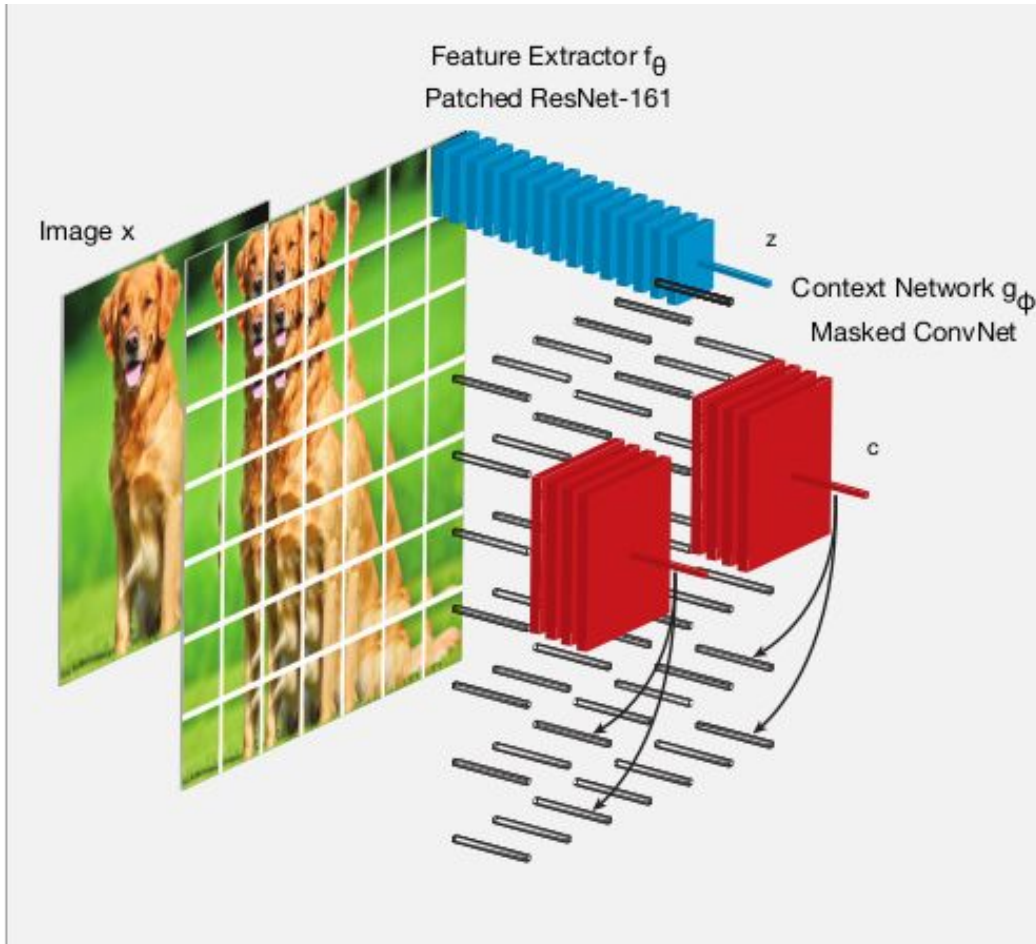
$$\begin{aligned} f_k(x_{t+k}, c_t) &= \exp(z_{t+k}^T \hat{z}_{t+k}) \\ &= \exp(z_{t+k}^T W_k c_t) \end{aligned}$$

Solution: Predicted embeddings should be more similar to true embeddings than to random samples from the dataset

Contrastive Predictive Coding V2 (CPC-v2)



Contrastive Predictive Coding V2 (CPC-v2)



CPC applied to images: predict embeddings of next patches.

- ResNet based architecture.
- Independent data augmentation for patches.

How to evaluate self-supervised representations?

One common approach: Linear evaluation protocol on ImageNet

1. Self-supervised pre-training on ImageNet without labels
2. Freeze encoder weights
3. Train linear classifier on ImageNet using all labels of training set
4. Evaluate performance on validation/test set

CPC-v2: ImageNet linear evaluation top-1 accuracy

ResNet-50: 63.8%

ResNet-161: 71.5%

How to evaluate self-supervised representations?

One common approach: Linear evaluation protocol on ImageNet

1. Self-supervised pre-training on ImageNet without labels
2. Freeze encoder weights
3. Train linear classifier on ImageNet using all labels of training set
4. Evaluate performance on validation/test set

CPC-v2: ImageNet linear evaluation top-1 accuracy

ResNet-50: 63.8%

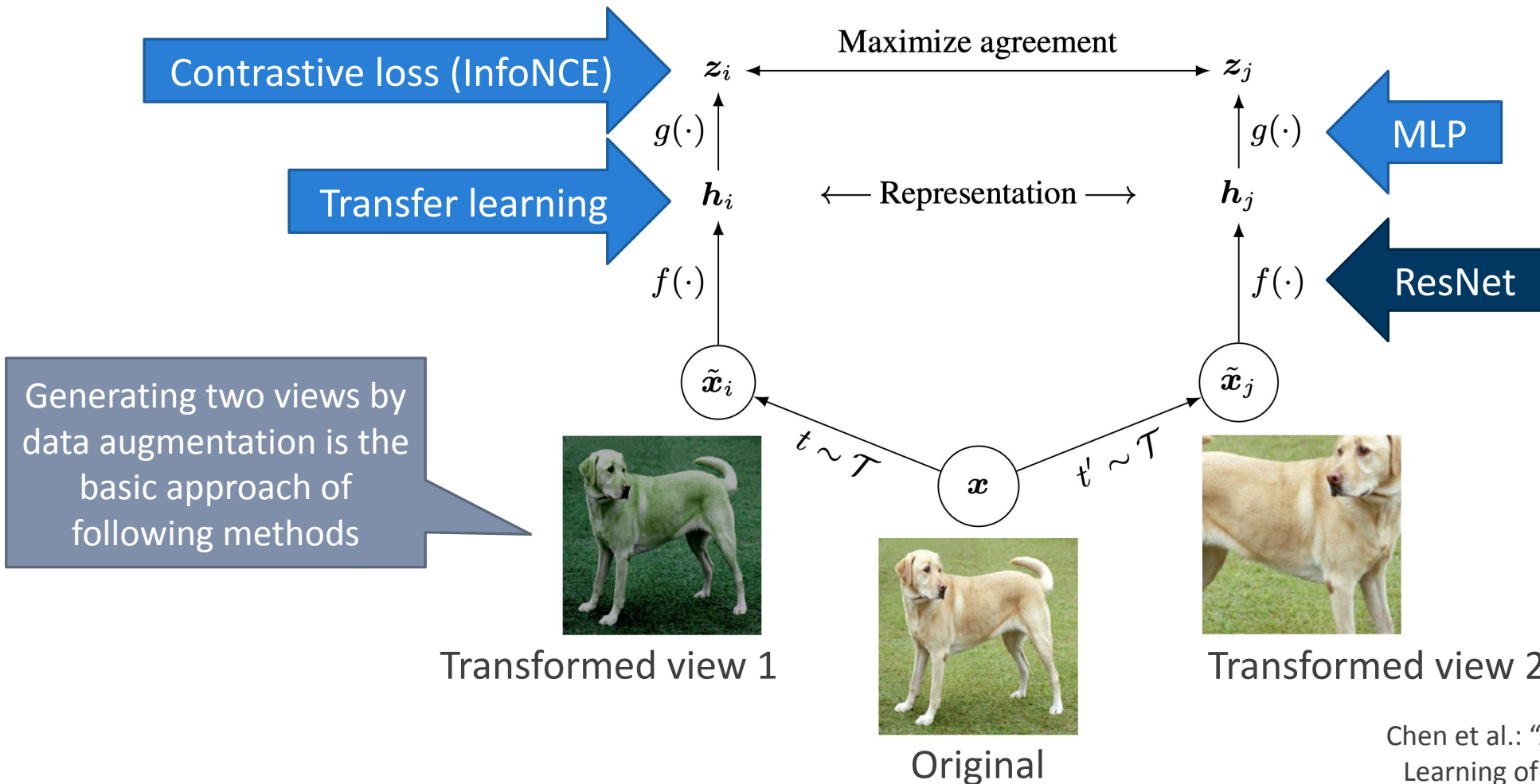
ResNet-161: 71.5%

79.3%

80.62 % (ResNet-152)

Fully supervised baseline

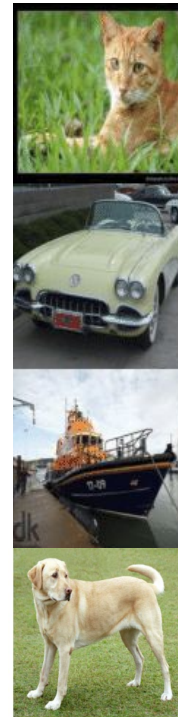
SimCLR: Maximize agreement between representations of two views



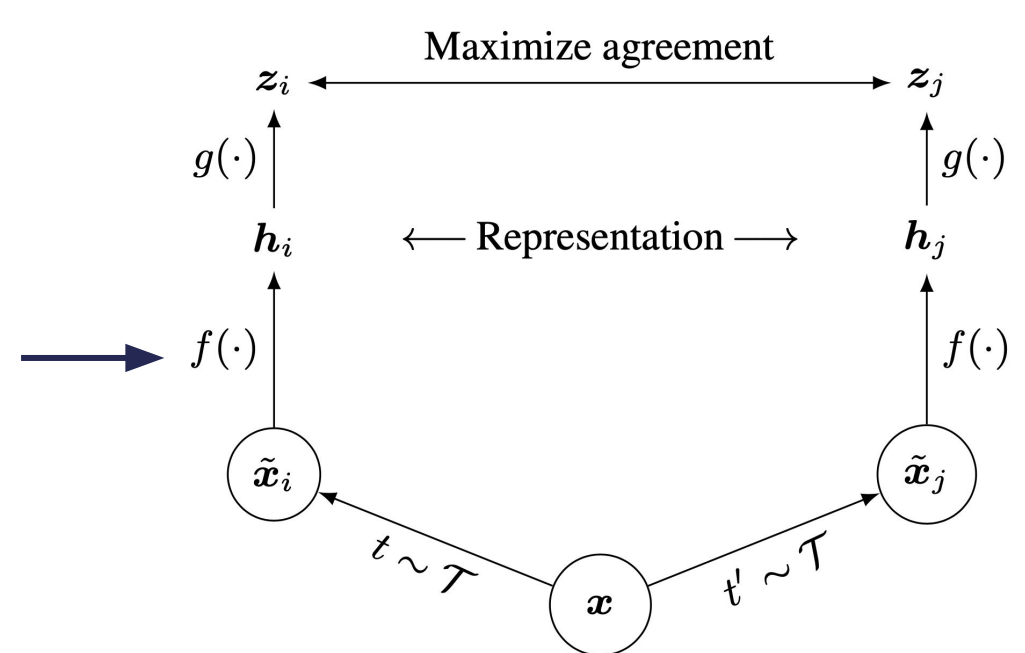
SimCLR: Training & InfoNCE Loss

Training:

- Sample batch with N images
- Apply augmentations to each image to get 2 views
- Compute InfoNCE for each view
 - Views from same image: *positive*
 - Other views: *negatives*



Batch



SimCLR: Training & InfoNCE Loss

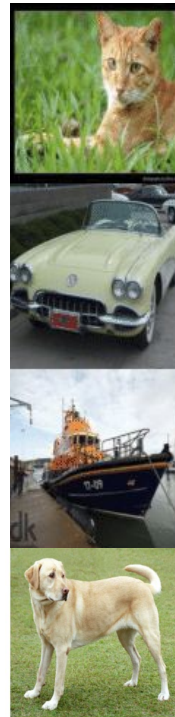
InfoNCE:

positive (views of same image)

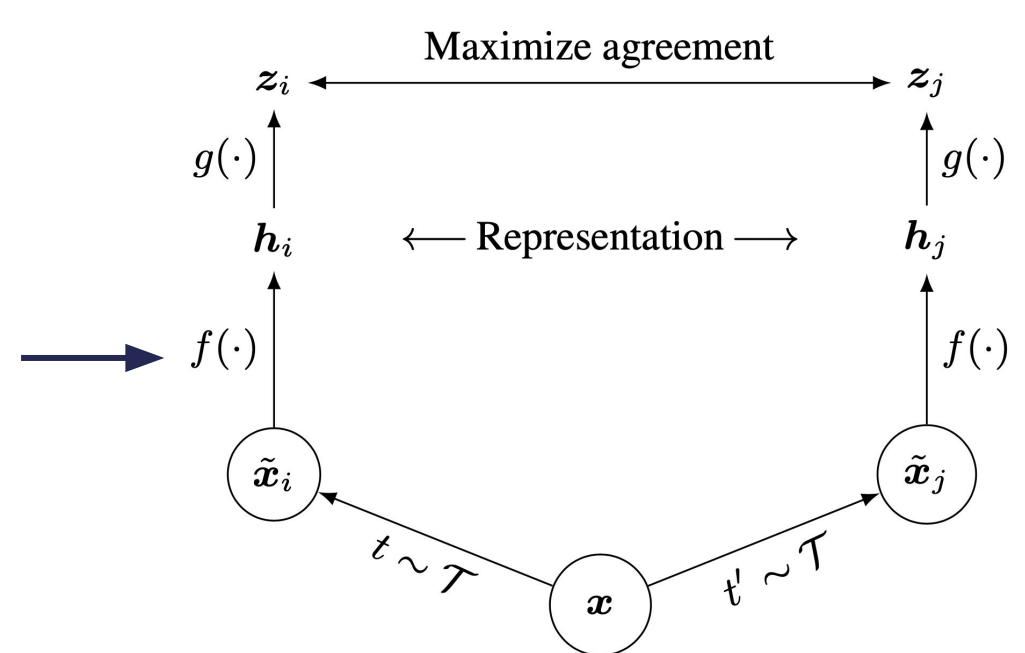
$$\ell(i, j) = -\log \frac{\exp(s_{i,j}/\tau)}{\sum_{k=1}^{2N} \mathbb{1}_{[k \neq i]} \exp(s_{i,k}/\tau)}$$

negatives: sum over all other views (excluding this image!)

$$s_{ij} = \text{cosine similarity}(z_i, z_j)$$



Batch



SimCLR: Training & InfoNCE Loss

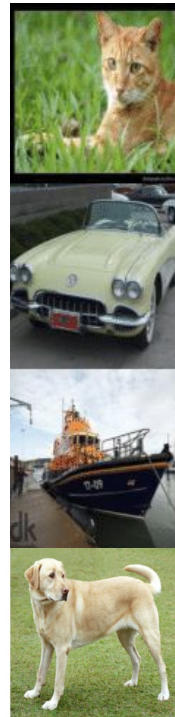
InfoNCE:

$$\ell(i, j) = -\log \frac{\exp(s_{i,j}/\tau)}{\sum_{k=1}^{2N} \mathbb{1}_{[k \neq i]} \exp(s_{i,k}/\tau)}$$

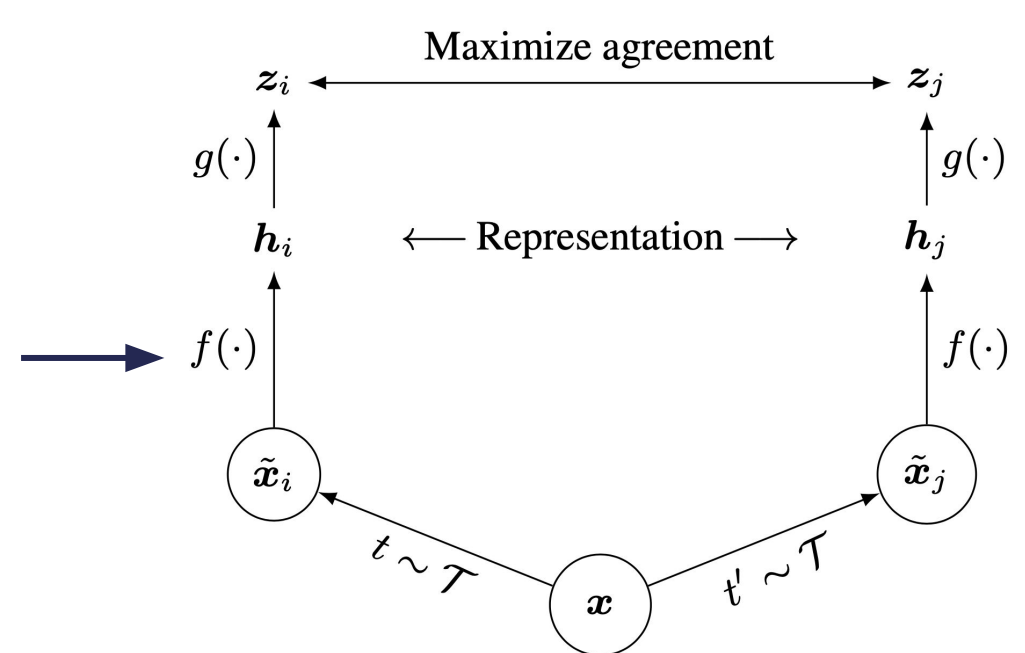
positive (views of same image) temperature (hyperparam)

negatives: sum over all other views (excluding this image!)

$$s_{ij} = \text{cosine similarity}(z_i, z_j)$$



Batch



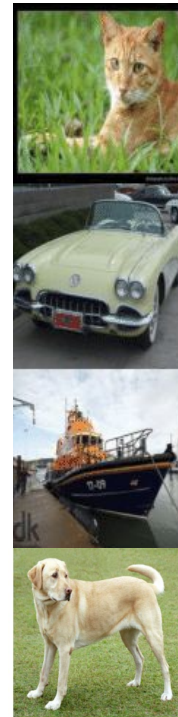
SimCLR: Training & InfoNCE Loss

InfoNCE:

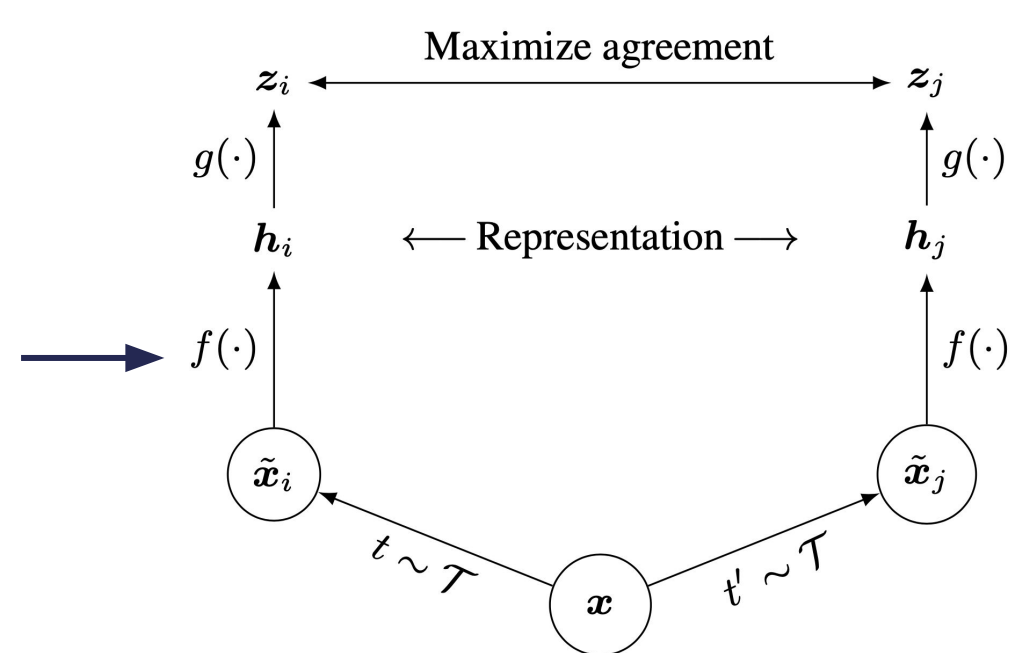
$$\ell(i, j) = -\log \frac{\exp(s_{i,j}/\tau)}{\sum_{k=1}^{2N} \mathbb{1}_{[k \neq i]} \exp(s_{i,k}/\tau)}$$

$$\mathcal{L} = \frac{1}{2N} \sum_{k=1}^N [\ell(2k-1, 2k) + \ell(2k, 2k-1)]$$

Symmetrized loss for all views in batch



Batch



SimCLR: Heavy data augmentation (two random ones applied per image)



(a) Original



(b) Crop and resize



(c) Crop, resize (and flip)



(d) Color distort. (drop)



(e) Color distort. (jitter)



(f) Rotate $\{90^\circ, 180^\circ, 270^\circ\}$



(g) Cutout



(h) Gaussian noise



(i) Gaussian blur



(j) Sobel filtering

SimCLR: Very large batch sizes

- Negative examples are key to contrastive learning
- SimCLR uses batch sizes up to **8192** samples
 - Requires custom optimizer (LARS)
 - Trained on 128 Google TPUs in parallel

ImageNet linear evaluation top-1 accuracy

ResNet-50: 69.3%

ResNet-50-4x: 76.5%

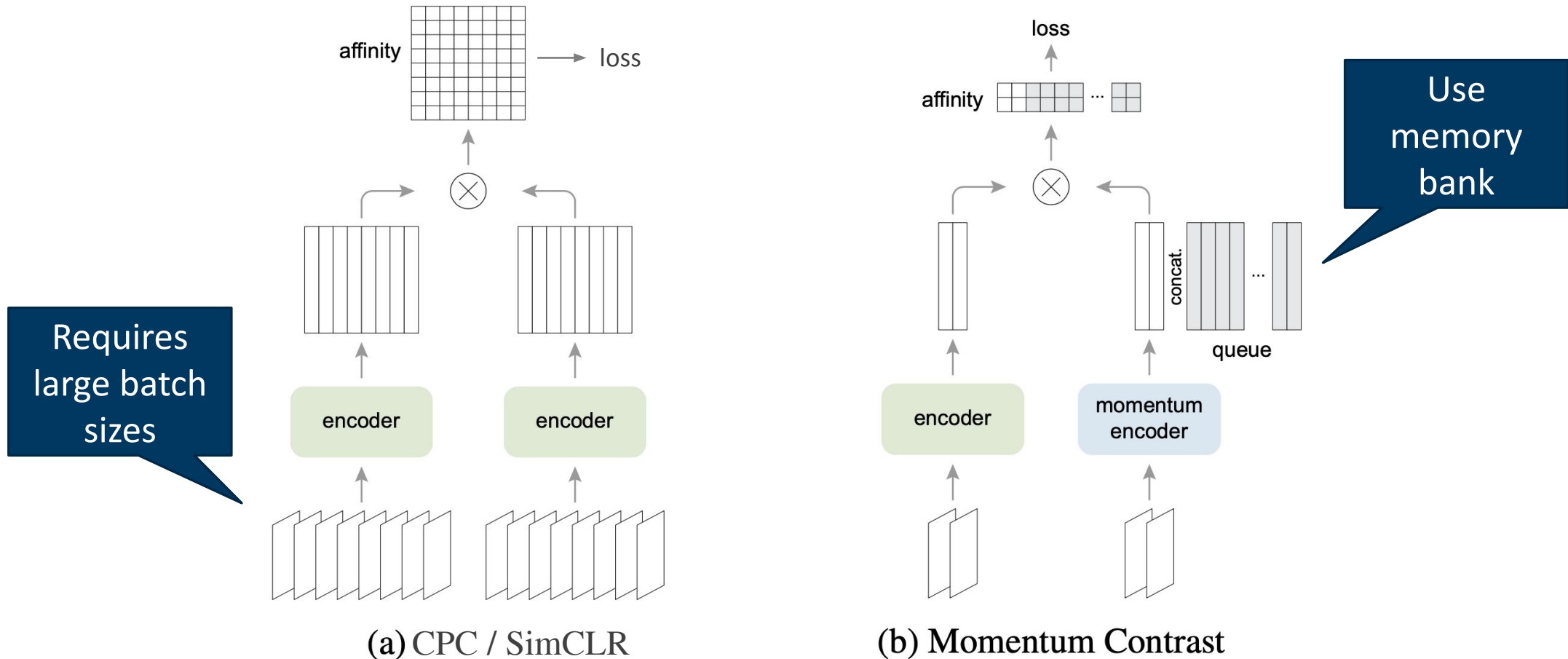
SimCLR: application beyond ImageNet

= Supervised
pretrained on
ImageNet

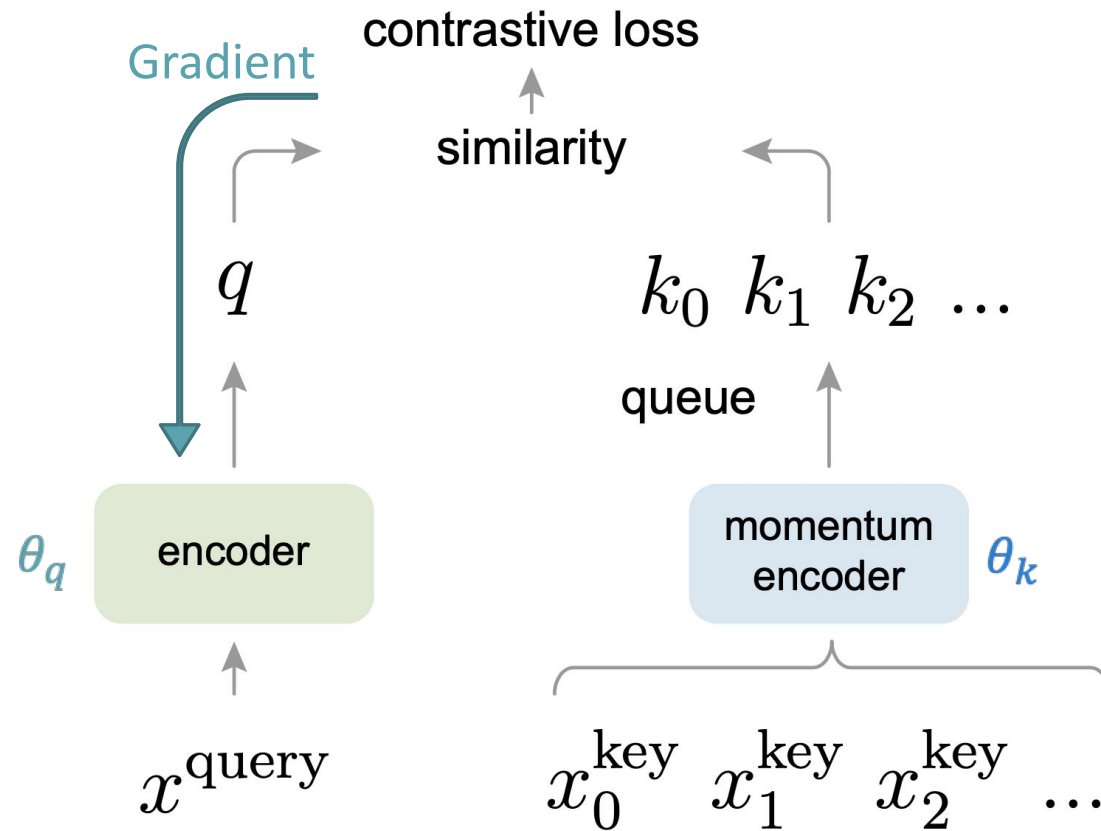
	Food	CIFAR10	CIFAR100	Birdsnap	SUN397	Cars	Aircraft	VOC2007	DTD	Pets	Caltech-101	Flowers
<i>Linear evaluation:</i>												
SimCLR (ours)	76.9	95.3	80.2	48.4	65.9	60.0	61.2	84.2	78.9	89.2	93.9	95.0
Supervised	75.2	95.7	81.2	56.4	64.9	68.8	63.8	83.8	78.7	92.3	94.1	94.2
<i>Fine-tuned:</i>												
SimCLR (ours)	89.4	98.6	89.0	78.2	68.1	92.1	87.0	86.6	77.8	92.1	94.1	97.6
Supervised	88.7	98.3	88.7	77.8	67.0	91.4	88.0	86.5	78.8	93.2	94.2	98.0
Random init	88.3	96.0	81.9	77.0	53.7	91.3	84.8	69.4	64.1	82.7	72.5	92.5

Table 8. Comparison of transfer learning performance of our self-supervised approach with supervised baselines across 12 natural image classification datasets, for ResNet-50 (4×) models pretrained on ImageNet. Results not significantly worse than the best ($p > 0.05$, permutation test) are shown in bold. See Appendix B.8 for experimental details and results with standard ResNet-50.

Hypothesis: Good contrastive learning requires many negative examples



Momentum Contrast (MoCo)



Momentum encoder

$$\theta_k \leftarrow m\theta_k + (1 - m)\theta_q$$

Advantages

- Decouples batch size from number of negative examples
- Can keep a larger set of negative examples

Disadvantages

- Two encoders
- More complex algorithm

MoCo performance

ImageNet linear evaluation top-1 accuracy

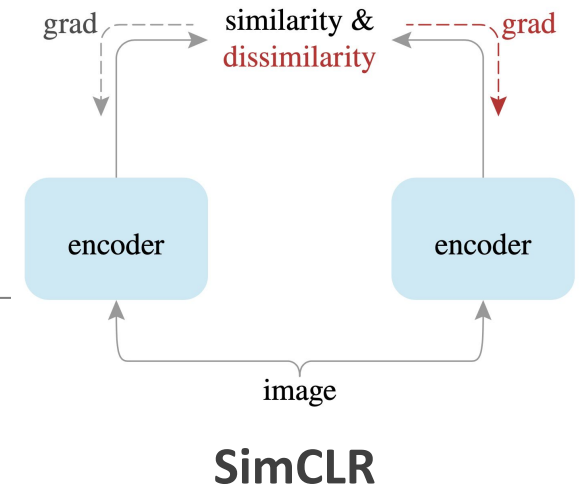
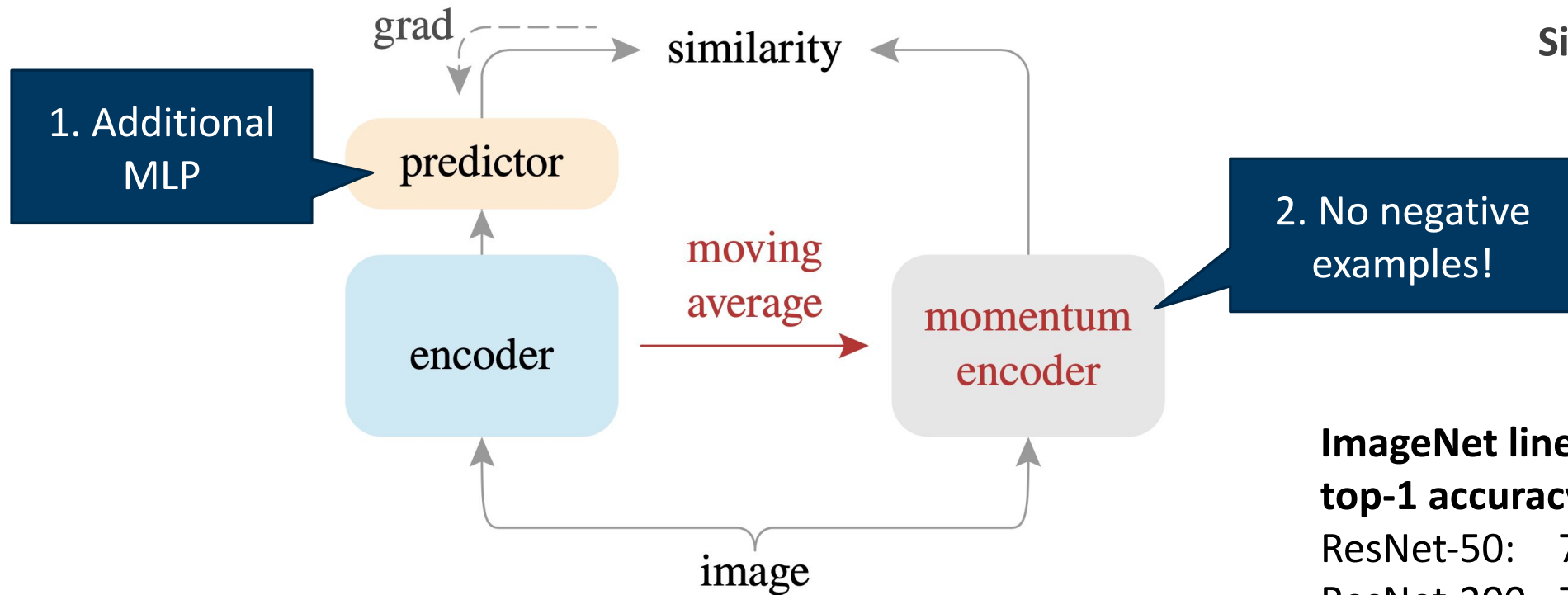
ResNet-50: 60.6%

ResNet-50-4x: 68.8%

After adopting projection head and data augmentation of SimCLR: MoCo-v2

ResNet-50: 71.1%

BYOL: Bootstrap your own latent

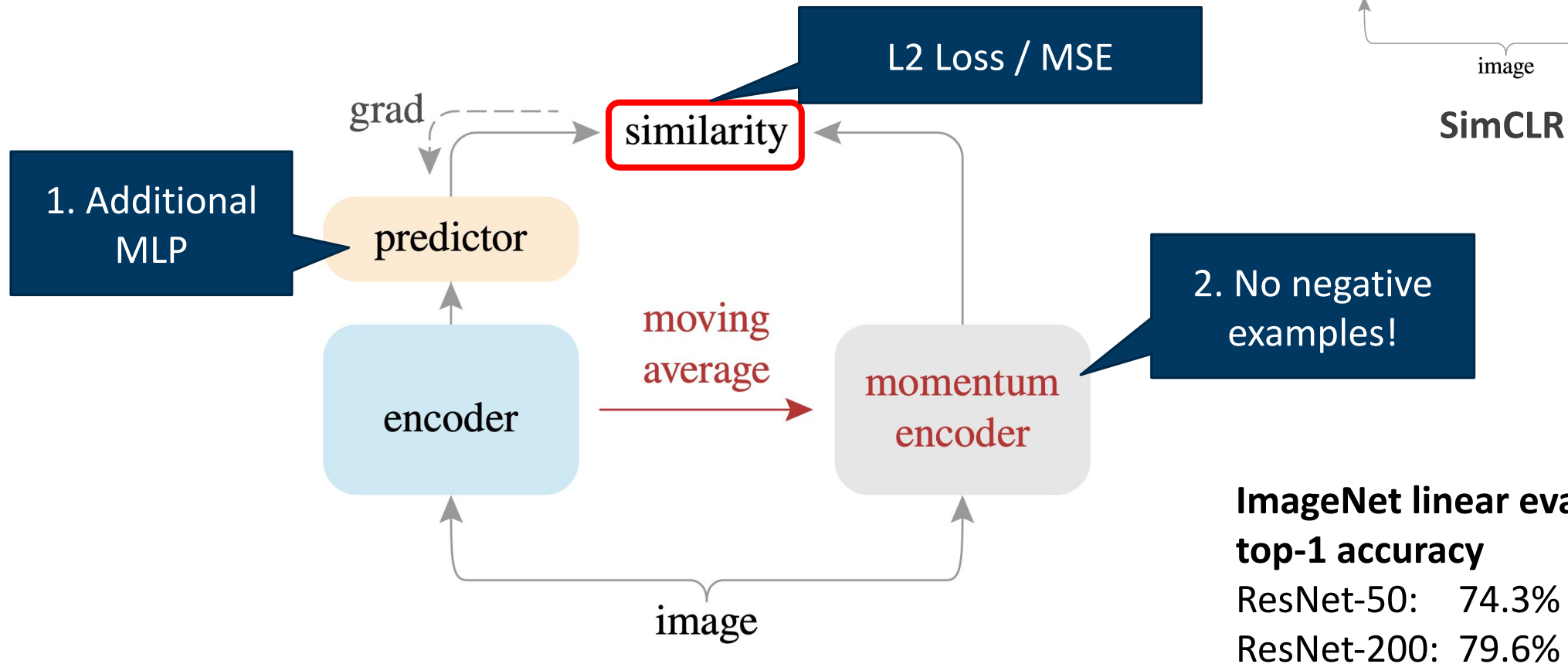


**ImageNet linear evaluation
top-1 accuracy**

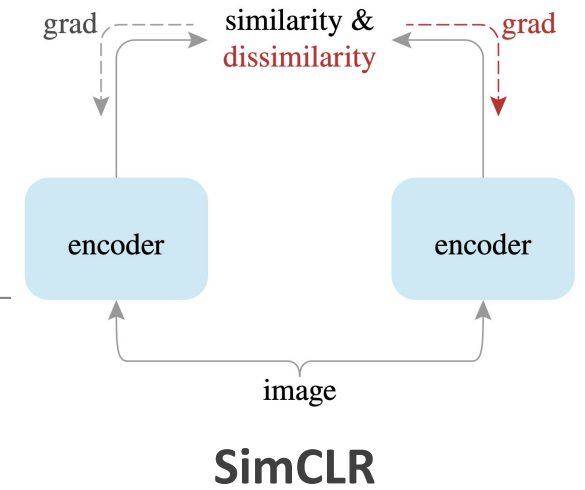
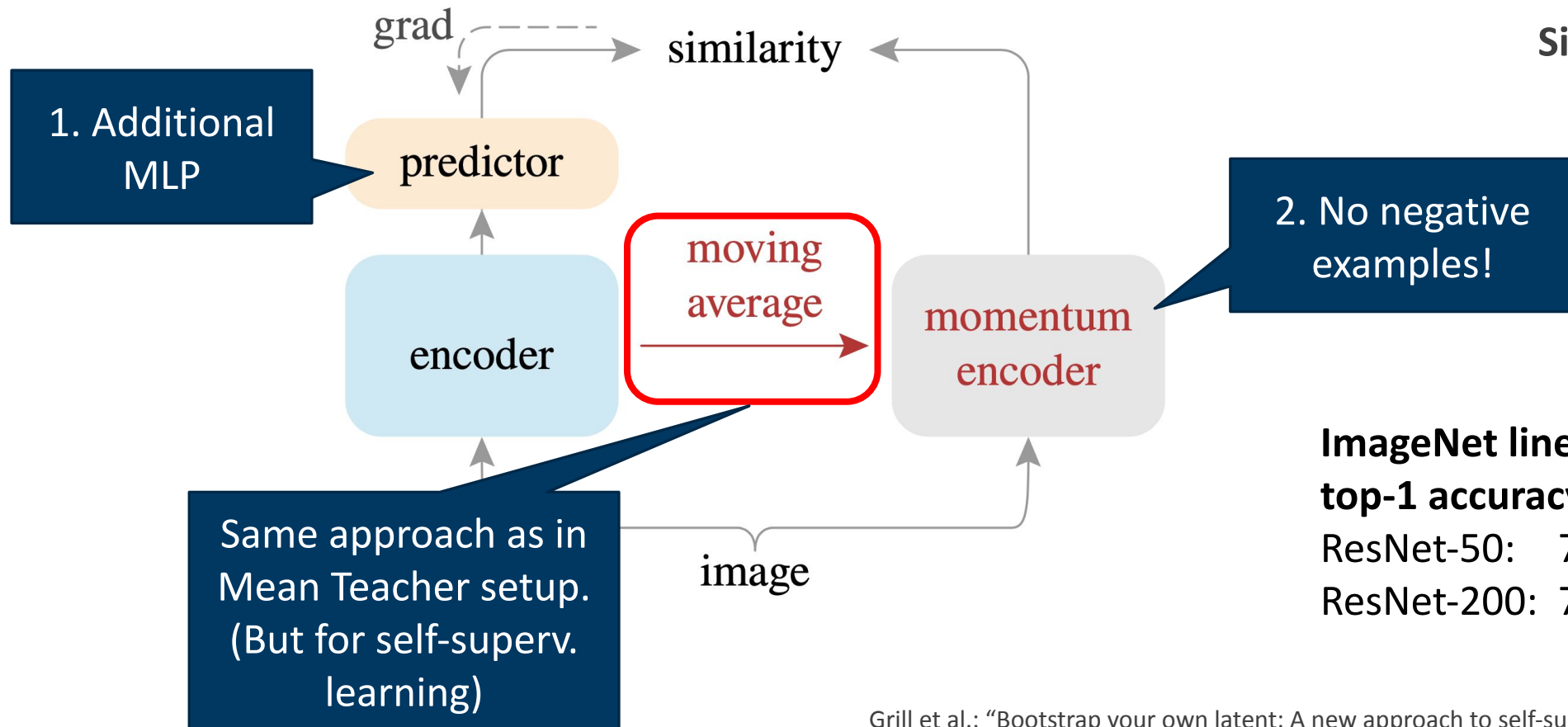
ResNet-50: 74.3%

ResNet-200: 79.6%

BYOL: Bootstrap your own latent



BYOL: Bootstrap your own latent



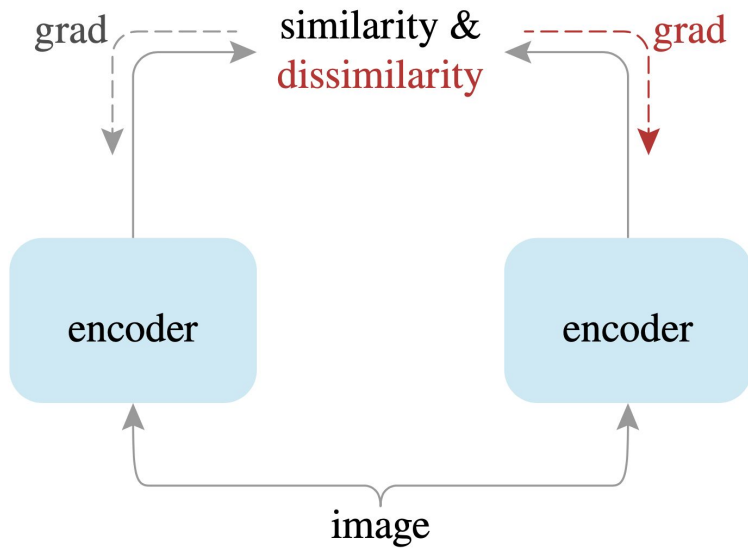
**ImageNet linear evaluation
top-1 accuracy**

ResNet-50: 74.3%

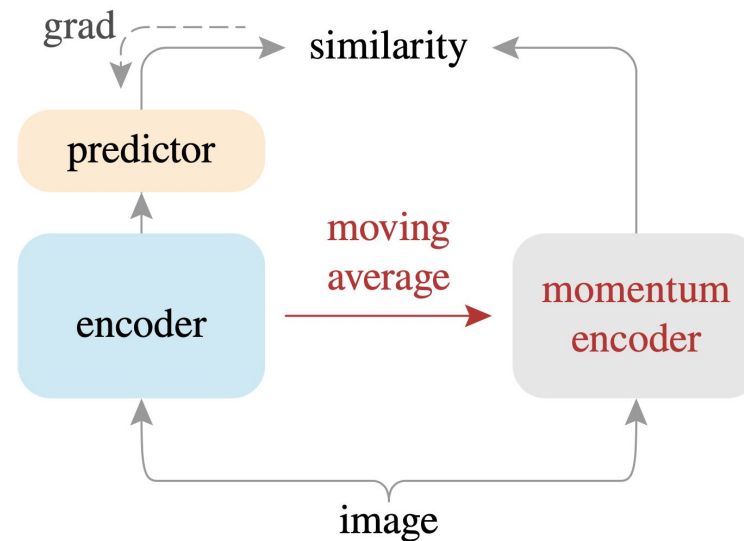
ResNet-200: 79.6%

Simplifying even further: SimSiam

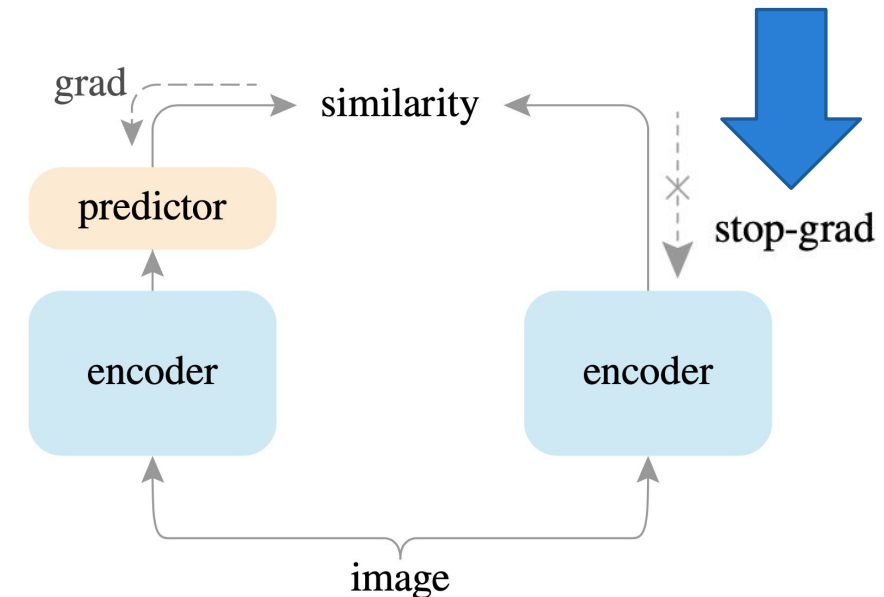
SimCLR



BYOL



SimSiam



No momentum encoder
No negative pairs
No large batch sizes

DINO v1

<https://arxiv.org/abs/2104.14294>

Does the network architecture for SSL make a difference?

- So far: CNN (ResNet)
- DINO v1: vision transformer (ViT)

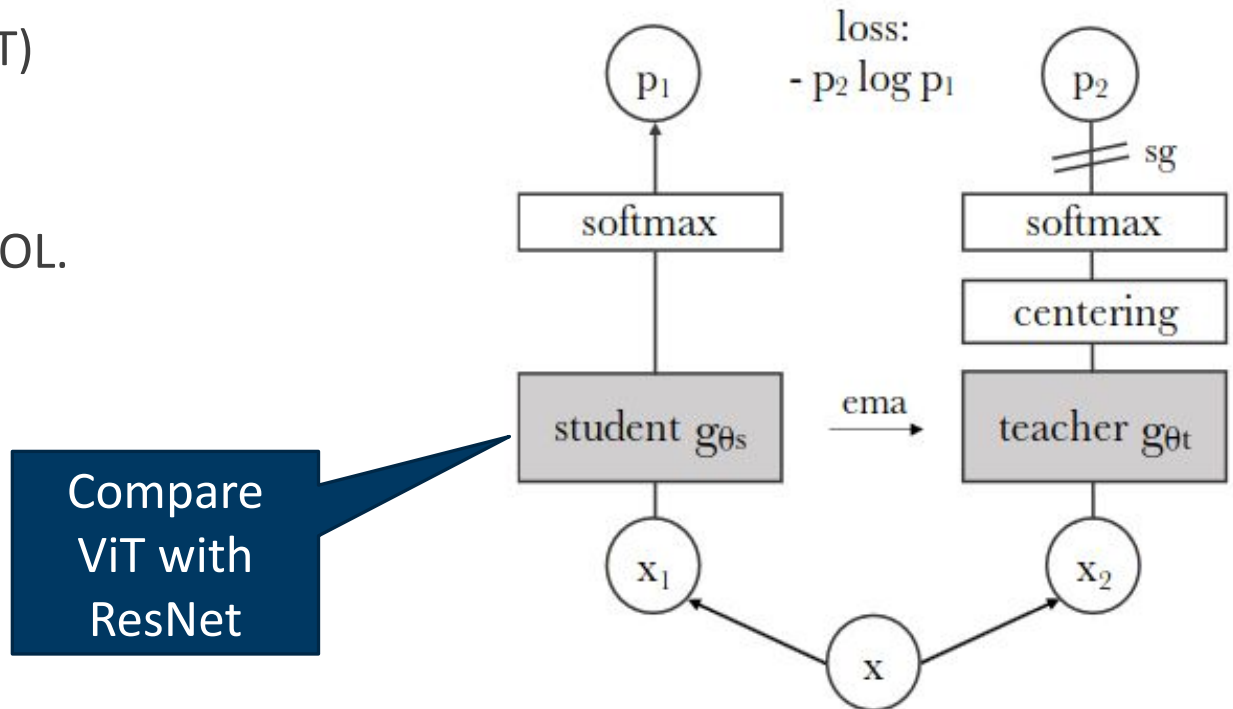
DINO v1

<https://arxiv.org/abs/2104.14294>

Does the network architecture for SSL make a difference?

- So far: CNN (ResNet)
- DINO v1: vision transformer (ViT)

Similar student-teacher set-up to BYOL.



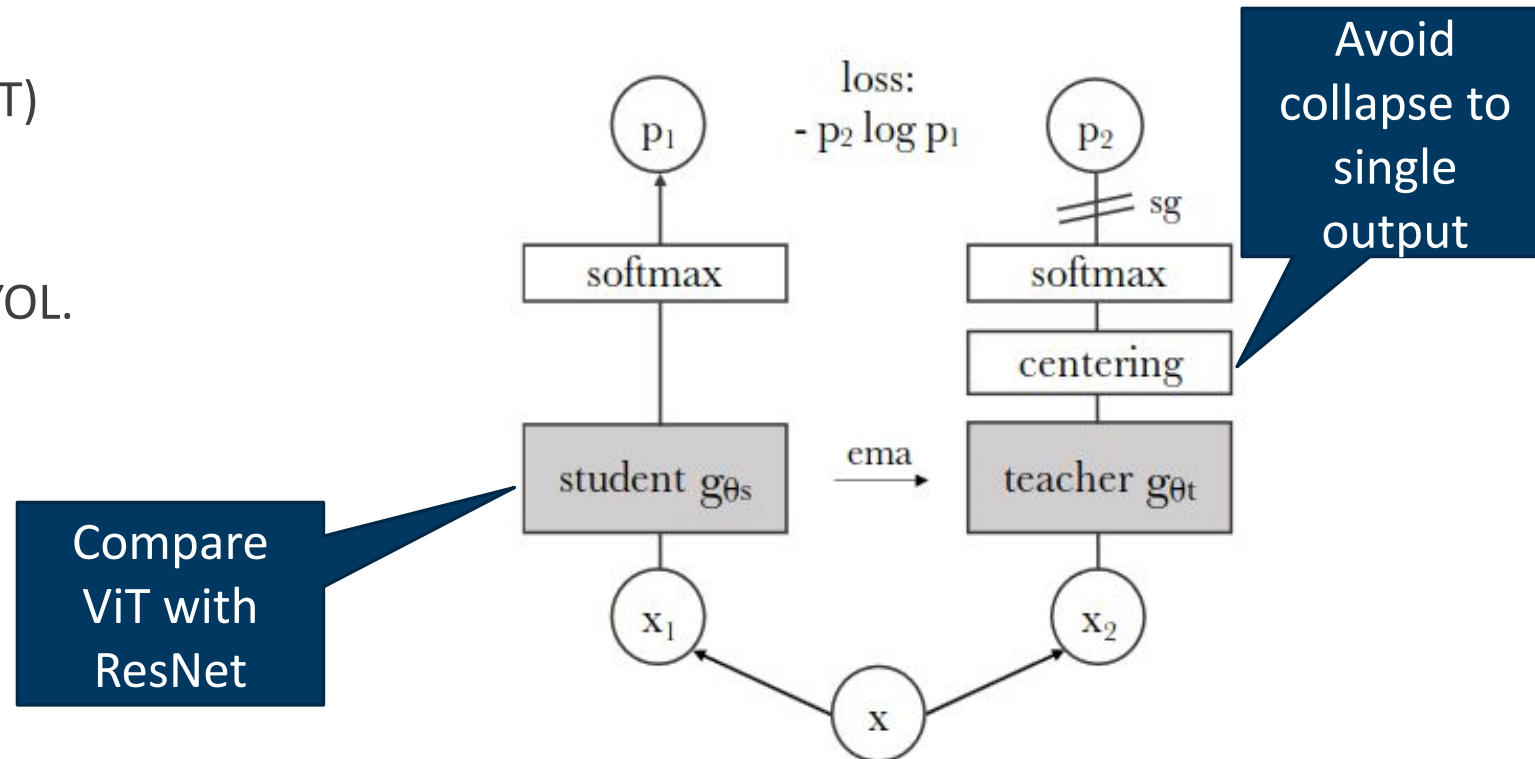
DINO v1

<https://arxiv.org/abs/2104.14294>

Does the network architecture for SSL make a difference?

- So far: CNN (ResNet)
- DINO v1: vision transformer (ViT)

Similar student-teacher set-up to BYOL.



DINO v1

<https://arxiv.org/abs/2104.14294>

Does the network architecture for SSL make a difference?

- So far: CNN (ResNet)
- DINO v1: vision transformer (ViT)

Comparison on ImageNet: ViT better than ResNet.

Method	Arch.	Param.	im/s	Linear	k -NN
DINO	RN50	23	1237	75.3	67.5
SCLR [12]	RN50w4	375	117	76.8	69.3
SwAV [10]	RN50w2	93	384	77.3	67.3
BYOL [30]	RN50w2	93	384	77.4	–
DINO	ViT-B/16	85	312	78.2	76.1
SwAV [10]	RN50w5	586	76	78.5	67.1
BYOL [30]	RN50w4	375	117	78.6	–
BYOL [30]	RN200w2	250	123	79.6	73.9
DINO	ViT-S/8	21	180	79.7	78.3
SCLRv2 [13]	RN152w3+SK	794	46	79.8	73.1
DINO	ViT-B/8	85	63	80.1	77.4

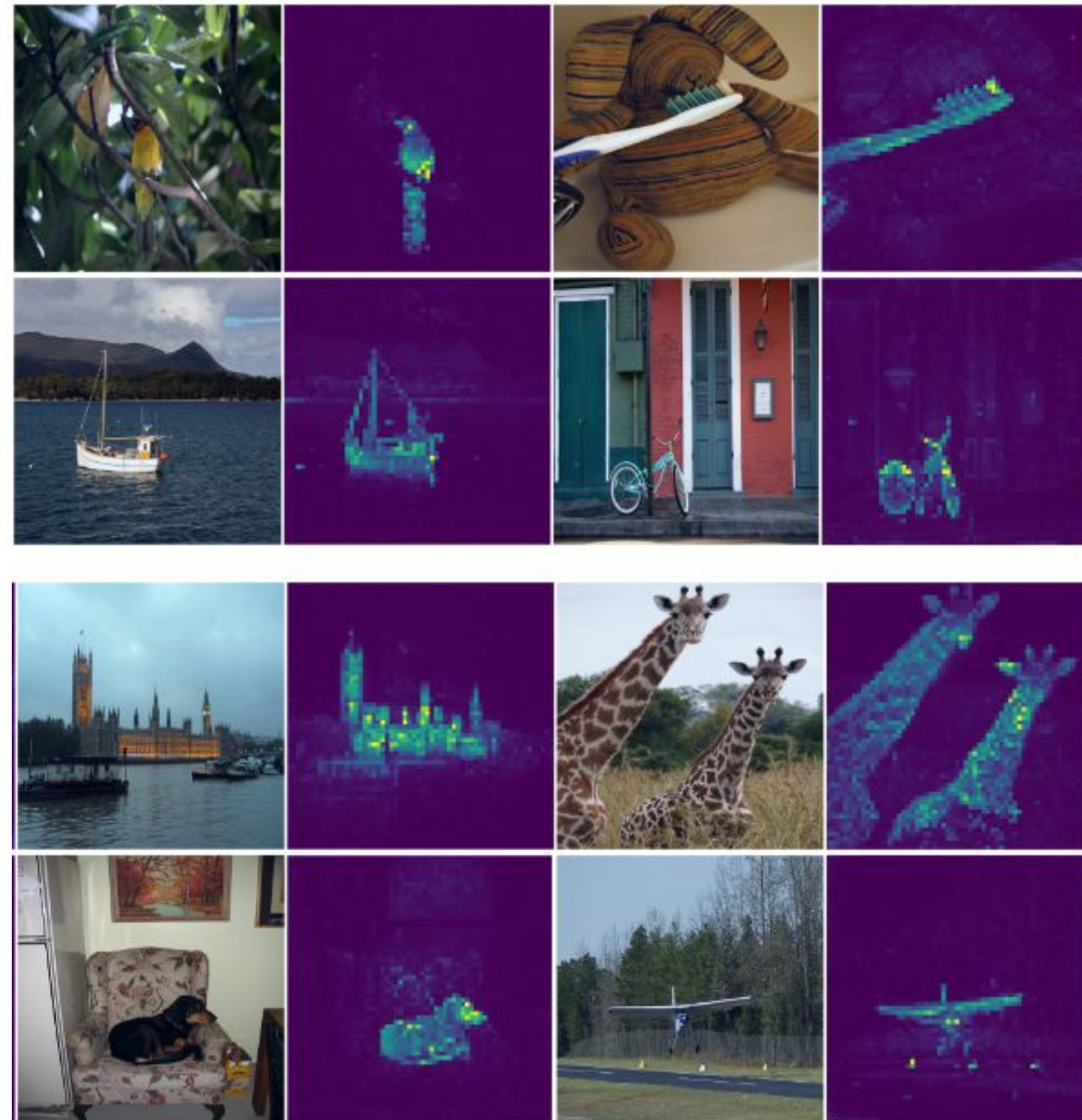
DINO v1

Does the network architecture for SSL make a difference?

- So far: CNN (ResNet)
- DINO v1: vision transformer (ViT)

Attention Maps of last layer:

Emerging semantic segmentation / saliency!



DINO v2

<https://arxiv.org/abs/2304.07193>

Scaling SSL to larger pre-training datasets.

- So far: methods used ImageNet 1k (1M images) or ImageNet 22k (14M images).
- DINO v2: scale to larger dataset.

DINO v2

<https://arxiv.org/abs/2304.07193>

Scaling SSL to larger pre-training datasets.

- So far: methods used ImageNet 1k (1M images) or ImageNet 22k (14M images).
- DINO v2: scale to larger dataset.

Dataset (LVD-142M):

- 1.2B images from the internet
- Remove near duplicates + cluster images to avoid redundancy -> 142M unique images

DINO v2

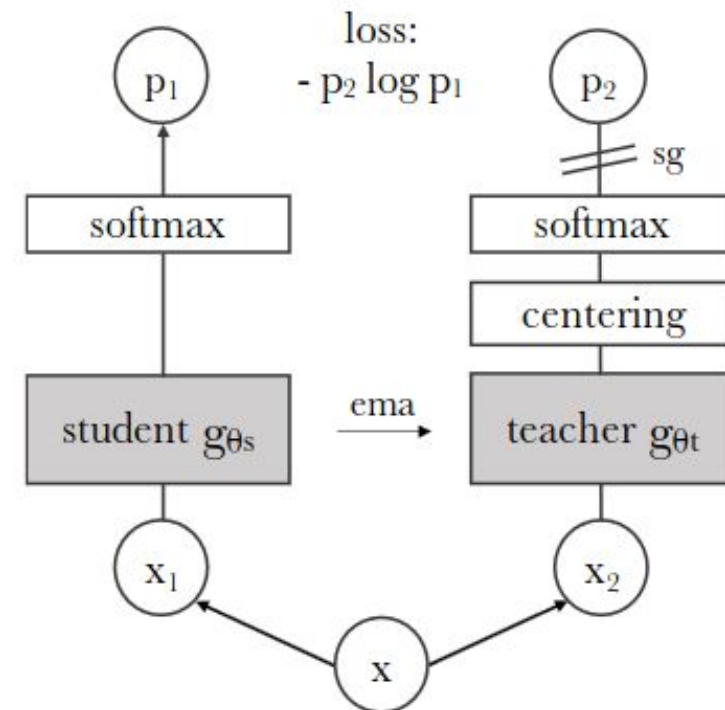
<https://arxiv.org/abs/2304.07193>

Scaling SSL to larger pre-training datasets.

- So far: methods used ImageNet 1k (1M images) or ImageNet 22k (14M images).
- DINO v2: scale to larger dataset.

Training: extension of DINO v1:

- Image-level SSL
 - like DINO v1, different centering
- Patch-level SSL
 - some patches of student are masked, predict the teacher output for these patches
- Feature regularization (uniform span)



DINO v2

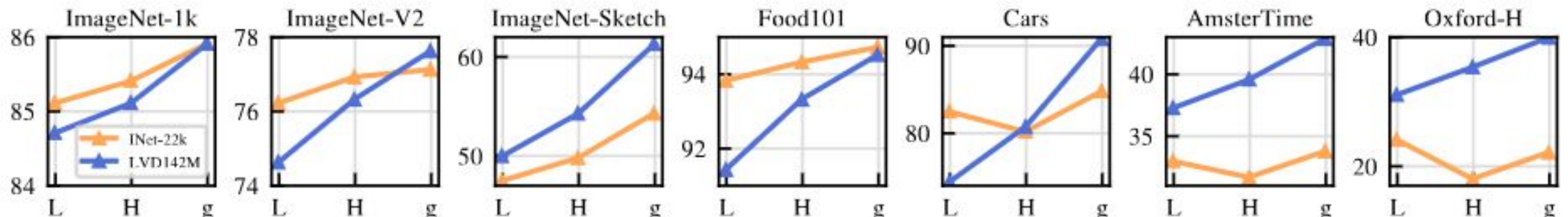
<https://arxiv.org/abs/2304.07193>

Scaling SSL to larger pre-training datasets.

- So far: methods used ImageNet 1k (1M images) or ImageNet 22k (14M images).
- DINO v2: scale to larger dataset.

Scaling behaviour for vision transformers of different size: L (large), H (huge), g (giant):

larger model improves with scale!



DINO v2

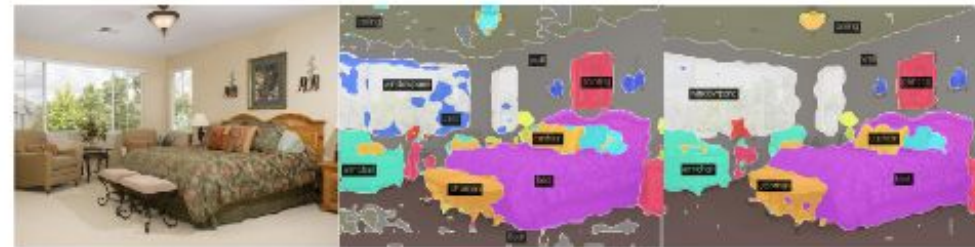
Scaling SSL to larger pre-training datasets.

- So far: methods used ImageNet 1k (1M images)
- DINO v2: scale to larger dataset.

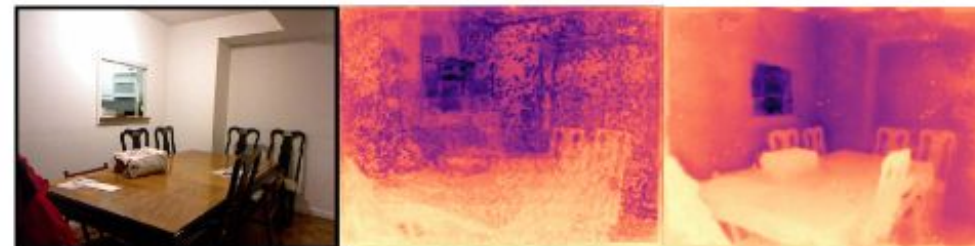
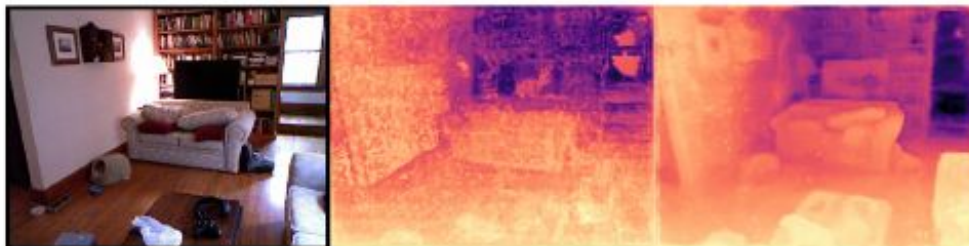
ImageNet: competitive performance to weakly supervised models with text supervision (next lecture)

Method	Arch.	Data	Text sup.	kNN	linear
				val	val
Weakly supervised					
CLIP	ViT-L/14	WIT-400M	✓	79.8	84.3
CLIP	ViT-L/14 ₃₃₆	WIT-400M	✓	80.5	85.3
SWAG	ViT-H/14	IG3.6B	✓	82.6	85.7
OpenCLIP	ViT-H/14	LAION	✓	81.7	84.4
OpenCLIP	ViT-G/14	LAION	✓	83.2	86.2
EVA-CLIP	ViT-g/14	custom*	✓	83.5	86.4
Self-supervised					
MAE	ViT-H/14	INet-1k	×	49.4	76.6
DINO	ViT-S/8	INet-1k	×	78.6	79.2
SEERv2	RG10B	IG2B	×	—	79.8
MSN	ViT-L/7	INet-1k	×	79.2	80.7
EsViT	Swin-B/W=14	INet-1k	×	79.4	81.3
Mugs	ViT-L/16	INet-1k	×	80.2	82.1
iBOT	ViT-L/16	INet-22k	×	72.9	82.3
DINOv2	ViT-S/14	LVD-142M	×	79.0	81.1
	ViT-B/14	LVD-142M	×	82.1	84.5
	ViT-L/14	LVD-142M	×	83.5	86.3
	ViT-g/14	LVD-142M	×	83.5	86.5

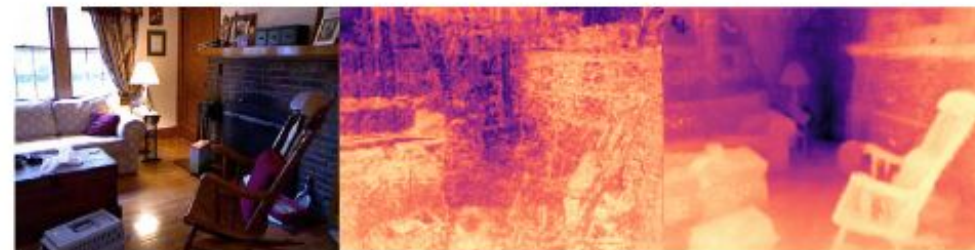
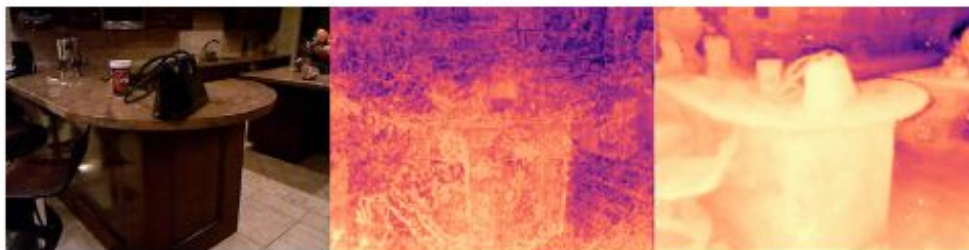
ADE20K



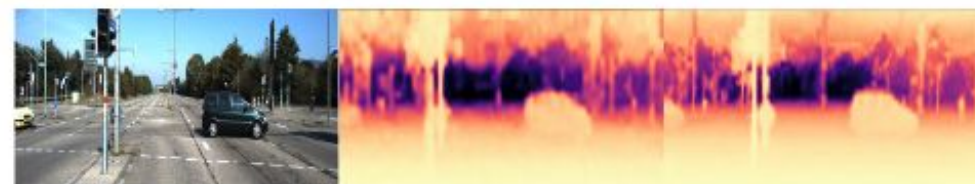
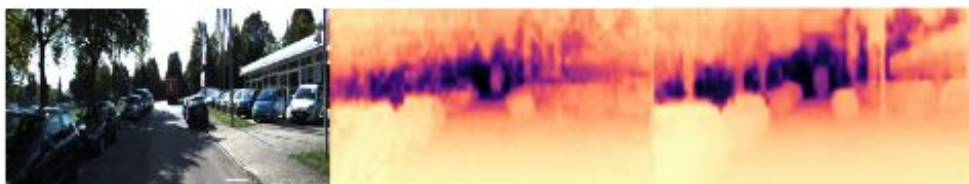
NYUd



SUN-RGBd



KITTI



Input

OpenCLIP-G

DINOv2-g

Input

OpenCLIP-G

DINOv2-g

DINO v2: good features for downstream tasks

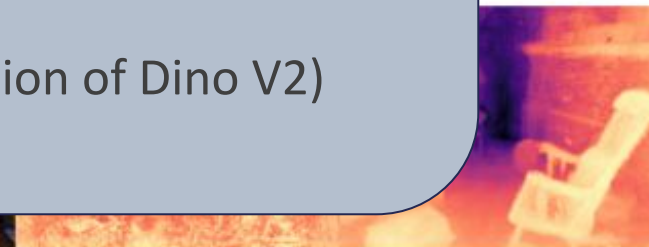
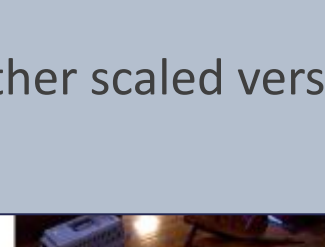
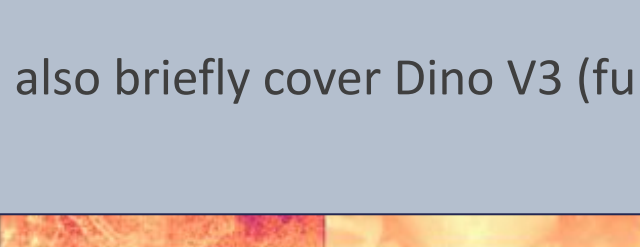
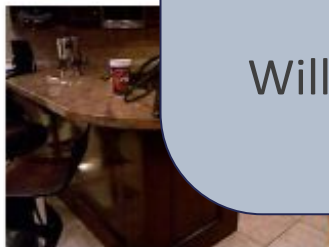
ADE20K



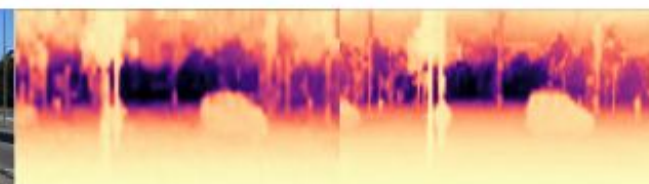
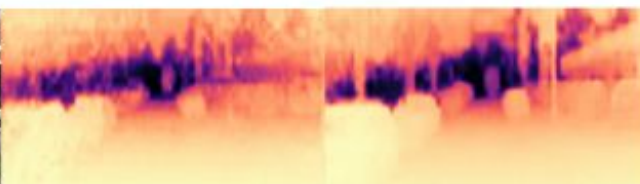
NYUd



SUN-RGBd



KITTI



Input

OpenCLIP-G

DINOv2-g

Input

OpenCLIP-G

DINOv2-g

DINO v2: features can be used for different downstream tasks (without costly fine-tuning).

Towards *vision foundation models* (analogous to LLMs)
More on this in the next lecture.

Will also briefly cover Dino V3 (further scaled version of Dino V2)

DINO v2: good features for downstream tasks

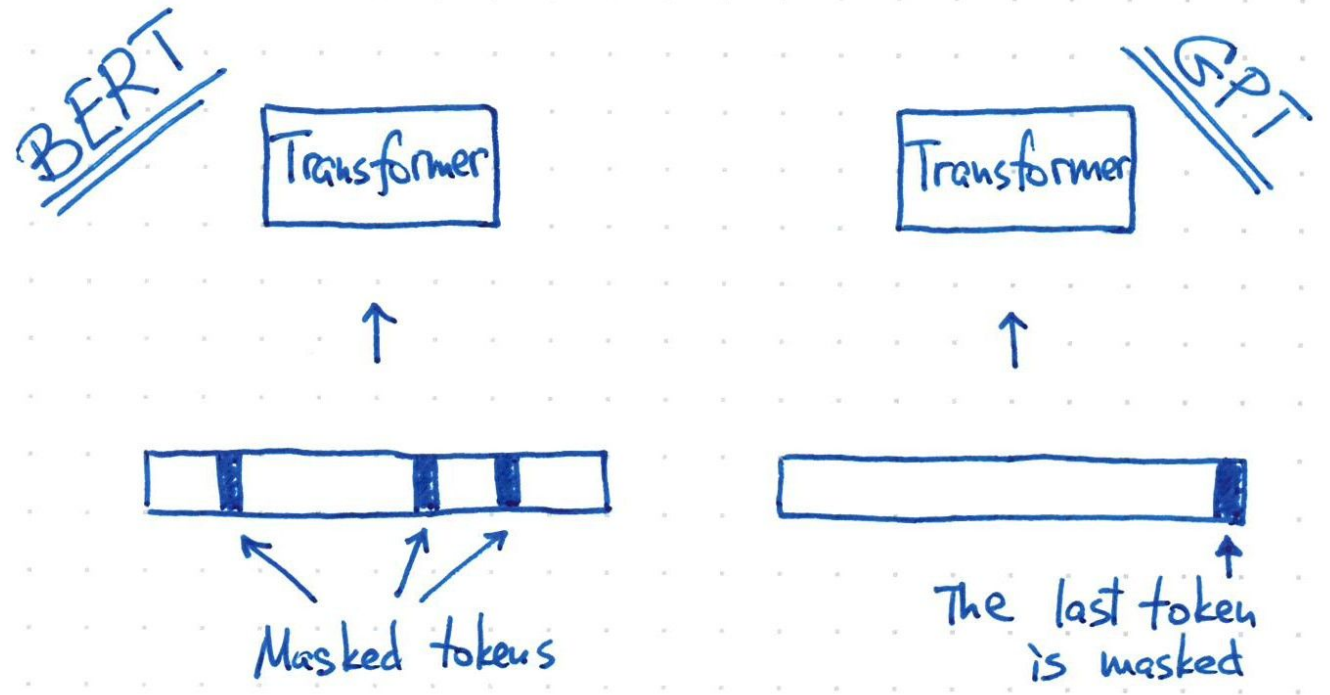
Masked Auto Encoder

<https://arxiv.org/abs/2111.06377>

Remember motivation from LLMs:
predict masked tokens from context.

CPC is quite similar to this, but not
state-of-the art.

State-of-the-art methods so far: rely
on loss across different images.

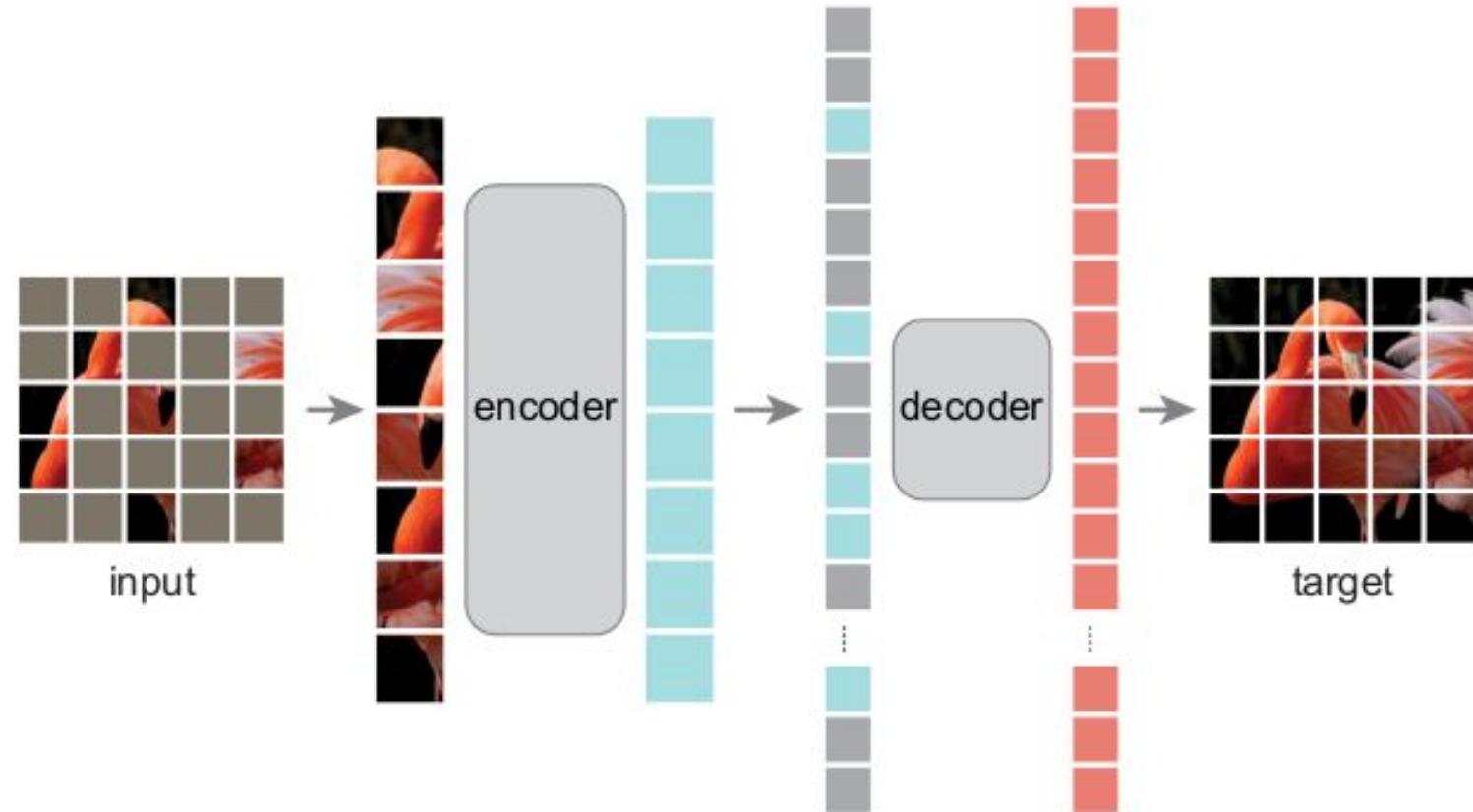


Masked Auto Encoder

<https://arxiv.org/abs/2111.06377>

Masked Auto Encoder (MAE):

- Pretrain ViT with auto encoder based loss
- Mask input tokens (patches), reconstruct mask from context

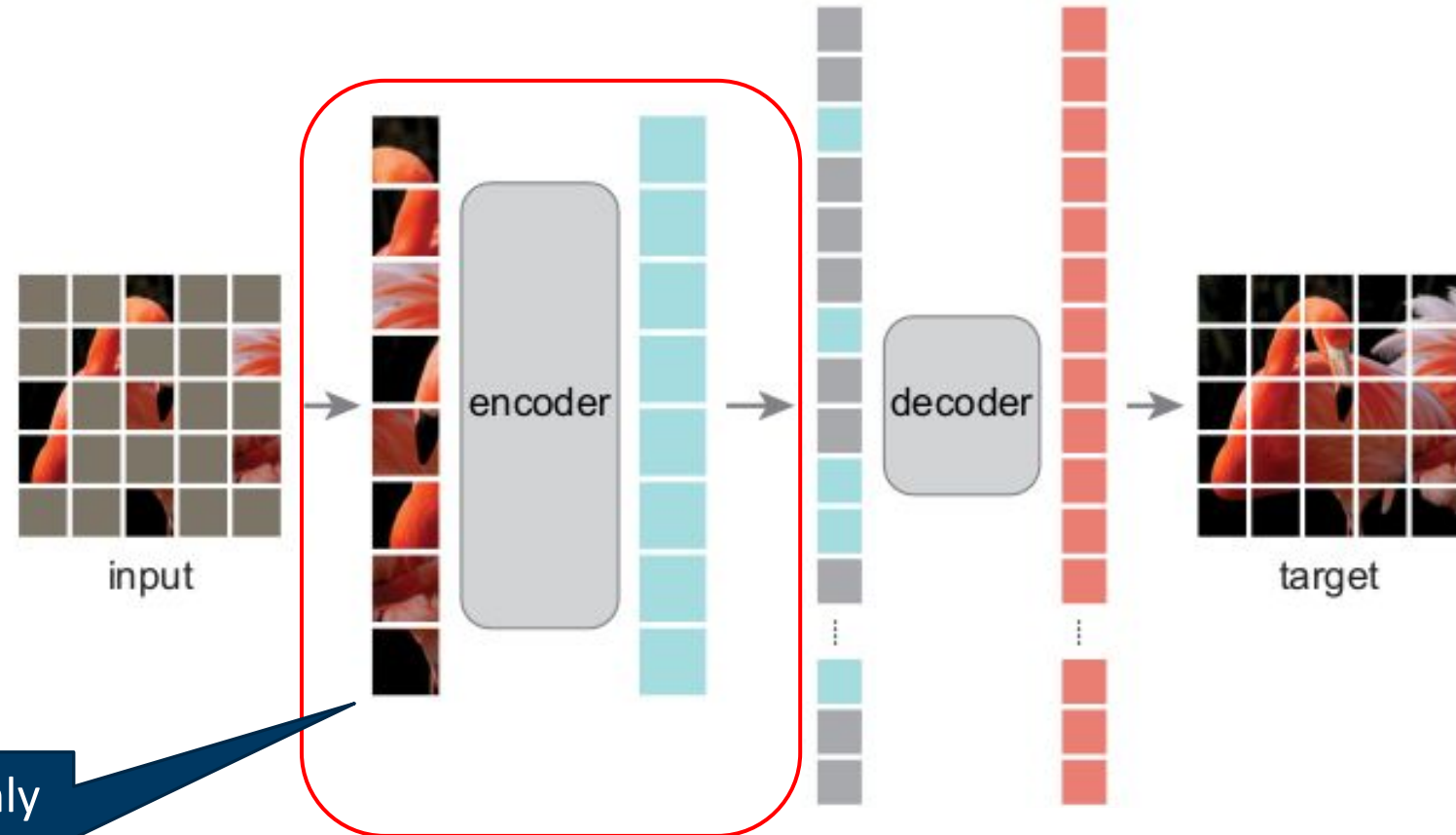


Masked Auto Encoder

<https://arxiv.org/abs/2111.06377>

Masked Auto Encoder (MAE):

- Pretrain ViT with auto encoder based loss
- Mask input tokens (patches), reconstruct mask from context



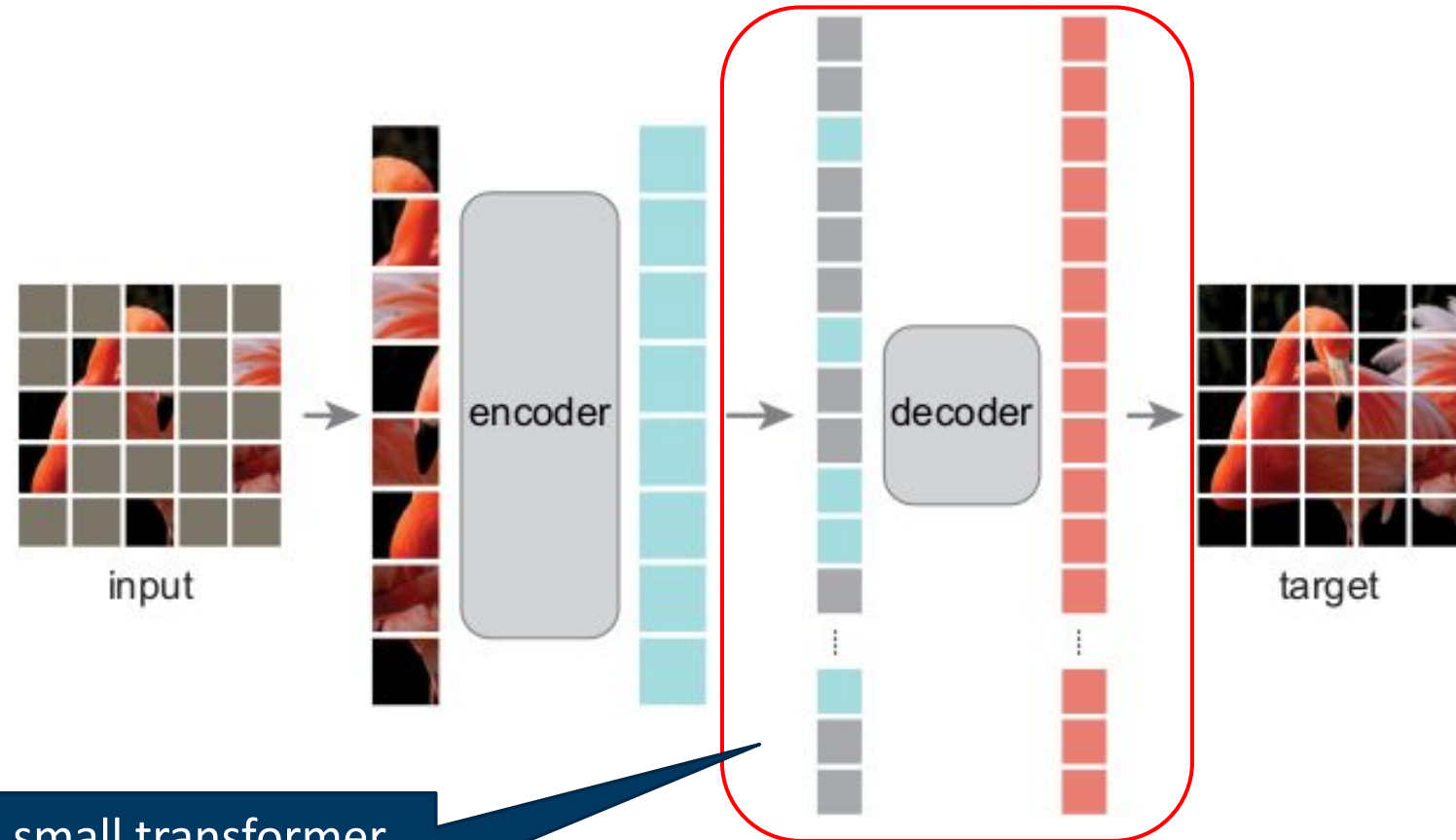
Encoder processes only
tokens with content
(more efficient!)

Masked Auto Encoder

<https://arxiv.org/abs/2111.06377>

Masked Auto Encoder (MAE):

- Pretrain ViT with auto encoder based loss
- Mask input tokens (patches), reconstruct mask from context



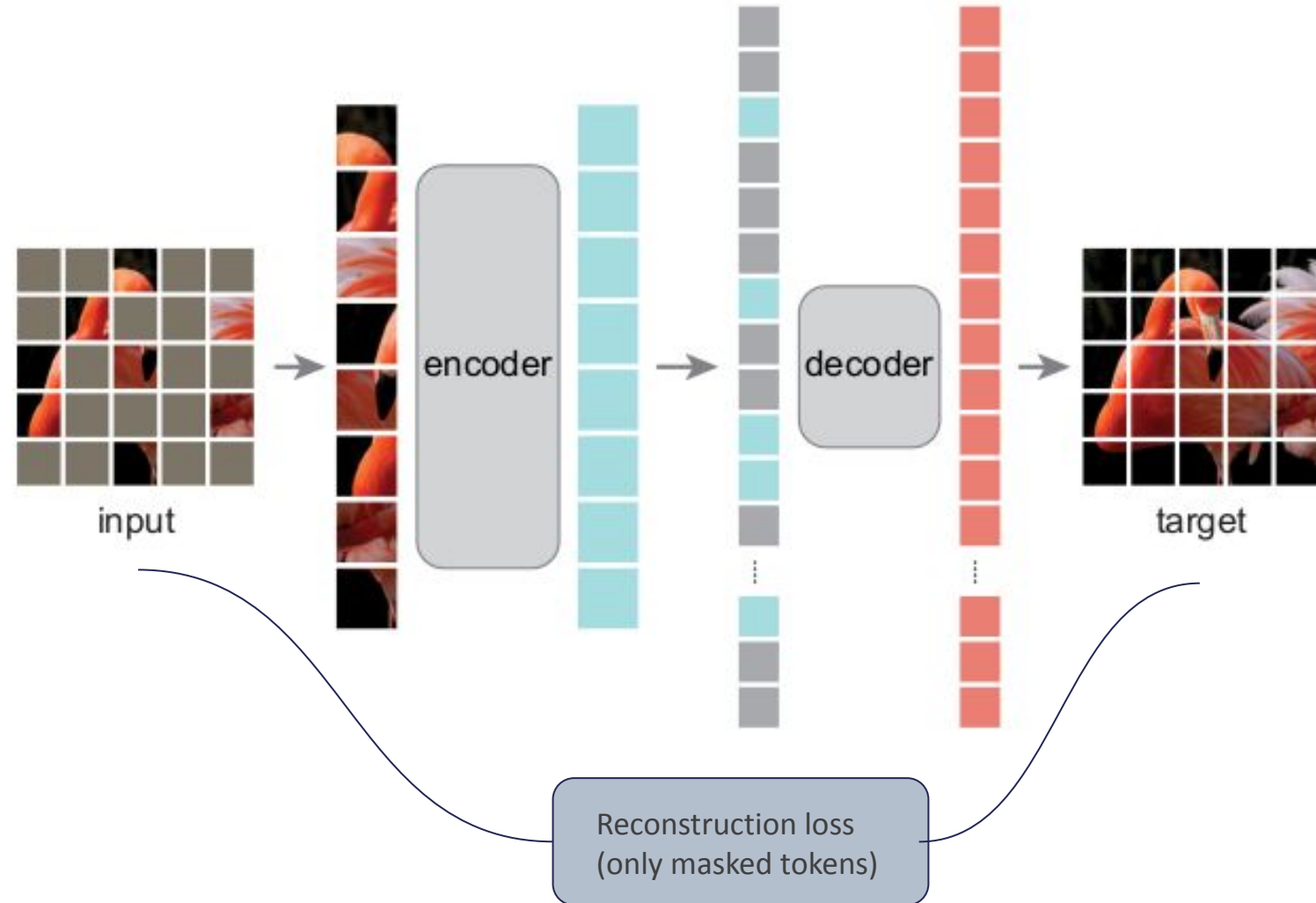
Decoder: small transformer,
processes all tokens,
predicts output image

Masked Auto Encoder

<https://arxiv.org/abs/2111.06377>

Masked Auto Encoder (MAE):

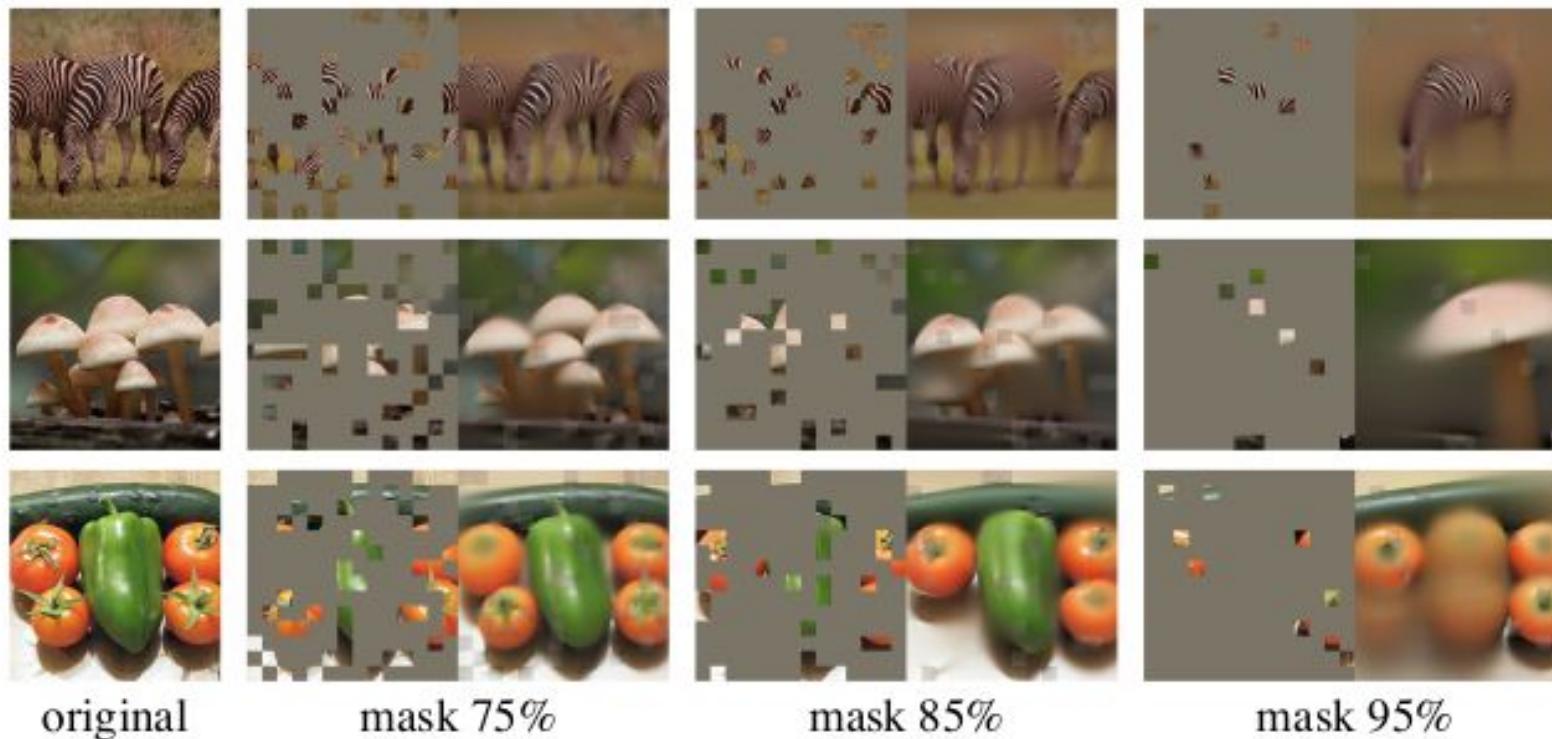
- Pretrain ViT with auto encoder based loss
- Mask input tokens (patches), reconstruct mask from context



Masked Auto Encoder

<https://arxiv.org/abs/2111.06377>

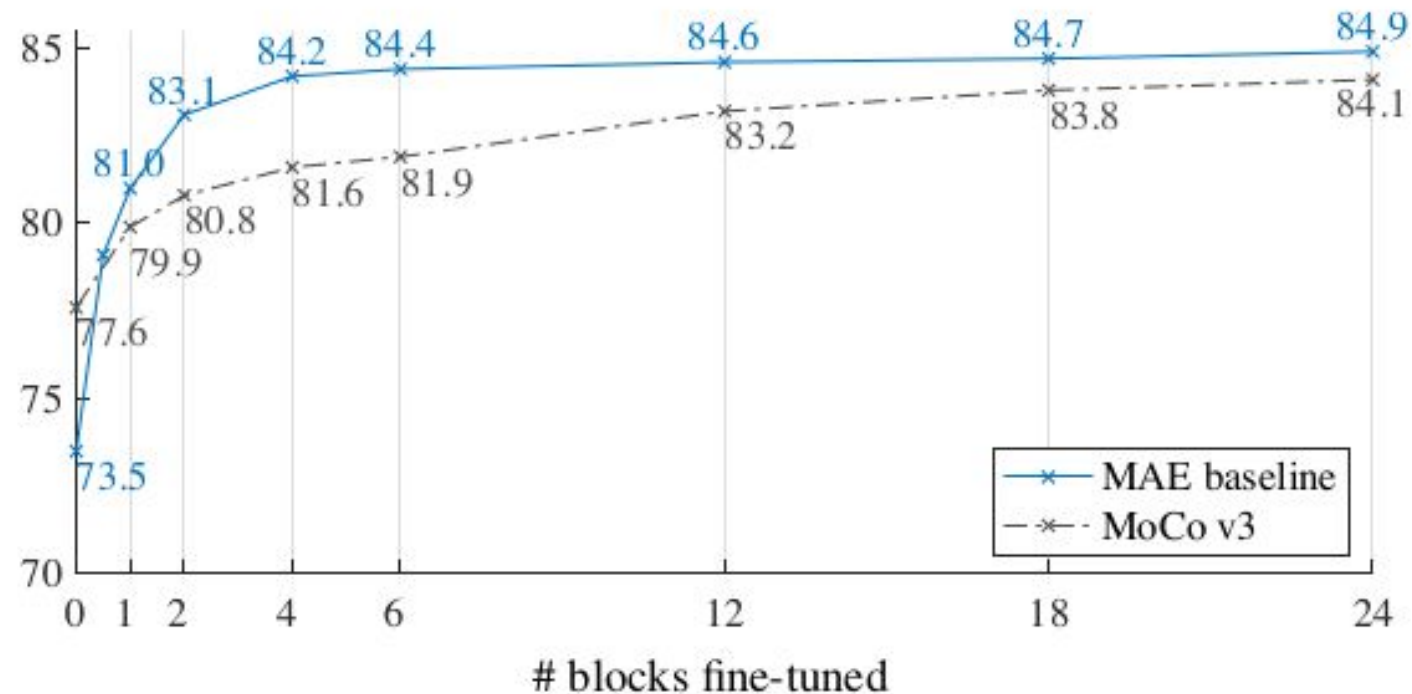
MAE uses high masking ratio (75%):
more difficult reconstruction task, leads to better pretraining



Masked Auto Encoder

<https://arxiv.org/abs/2111.06377>

MAE performance is worse for doing linear adaptation, significantly better when fine-tuning two or more layers.



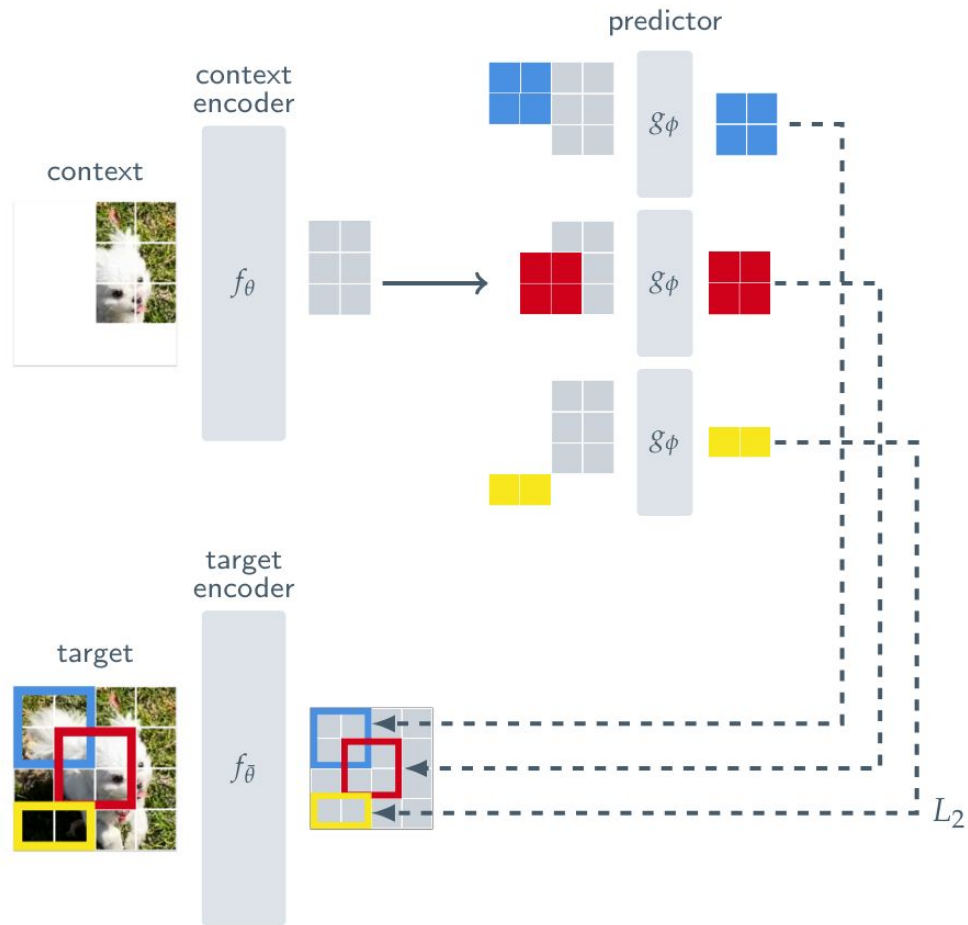
I-JEPA

- SIMCLR, MoCo, BYOL, SimSiam: Compare different views across images.
 - Either with negatives (SIMCLR, MoCo) or without (BYOL, SimSiam, Dino)
 - Does not explicitly rely on context within images.
- MAE: Prediction of image content from context.
 - Single image level.
 - Directly within image / pixel space -> needs decoder + noisy.

I-JEPA

- SIMCLR, MoCo, BYOL, SimSiam: Compare different views across images.
 - Either with negatives (SIMCLR, MoCo) or without (BYOL, SimSiam, Dino)
 - Does not explicitly rely on context within images.
- MAE: Prediction of image content from context.
 - Single image level.
 - Directly within image / pixel space -> needs decoder + noisy.
- (I-)JEPA: Joint embedding predictive architecture
 - Predict (image) content from context in embedding space.
 - General purpose approach for self supervised learning:
<https://openreview.net/pdf?id=BZ5a1r-kVsf>

I-JEPA



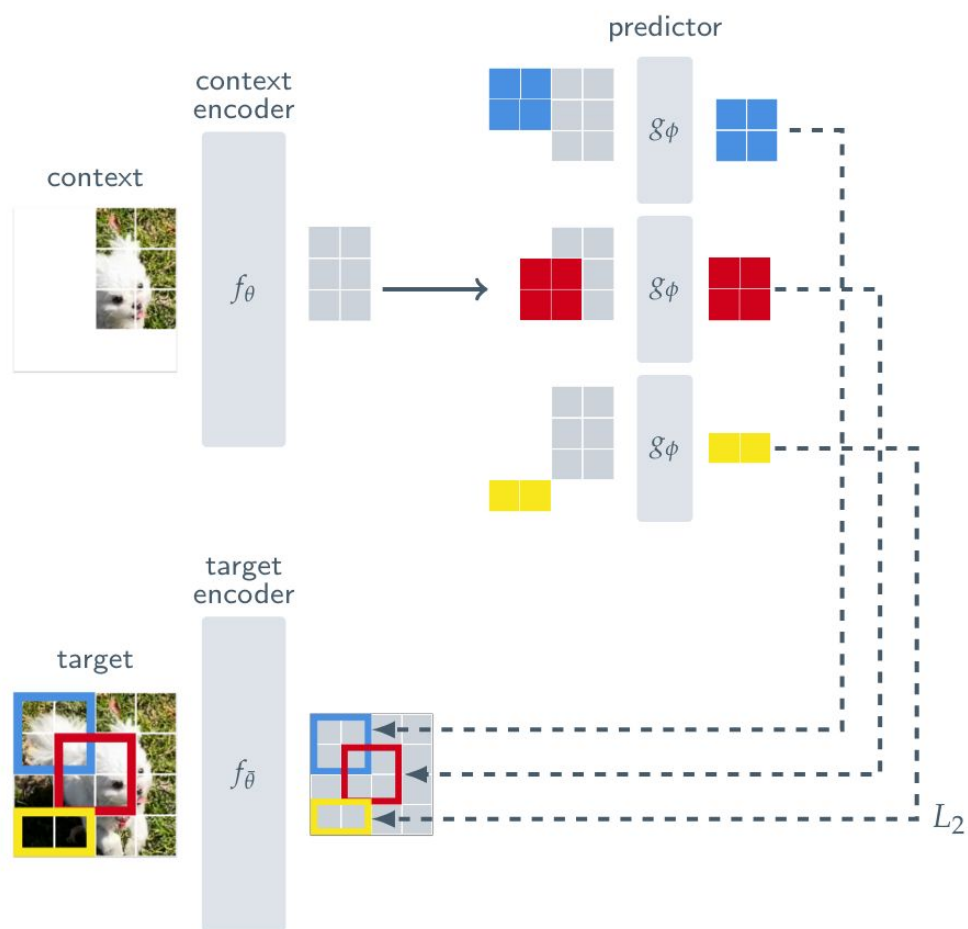
Target encoder is EMA of context encoder

Predictor is small ViT

Loss: L2 distance between all targets

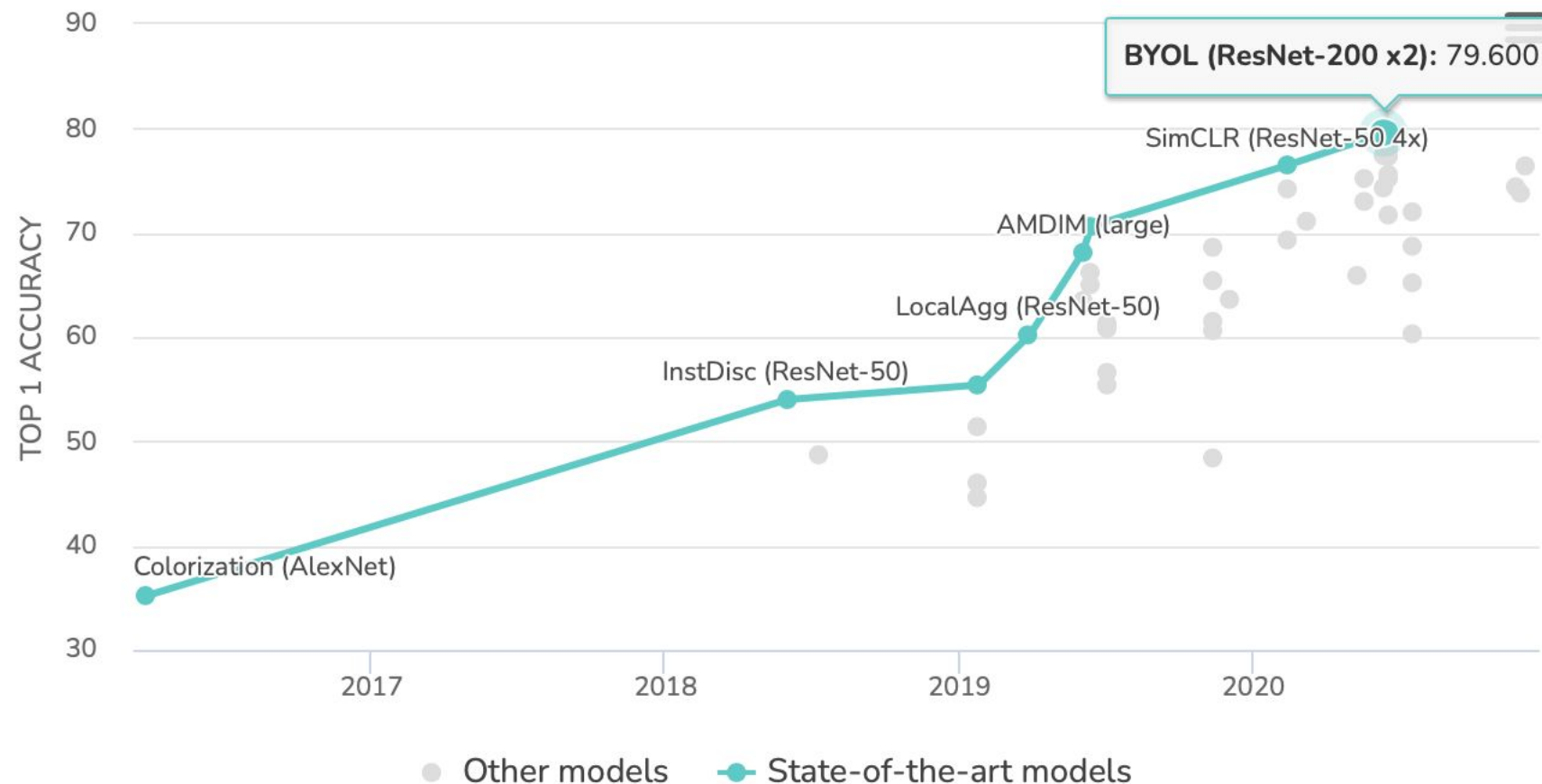
$$\frac{1}{M} \sum_{i=1}^M D(\hat{\mathbf{s}}_y(i), \mathbf{s}_y(i)) = \frac{1}{M} \sum_{i=1}^M \sum_{j \in B_i} \|\hat{\mathbf{s}}_{y_j} - \mathbf{s}_{y_j}\|_2^2$$

I-JEPA



Method	Arch.	Epochs	Top-1
<i>Methods without view data augmentations</i>			
data2vec [8]	ViT-L/16	1600	77.3
	ViT-B/16	1600	68.0
MAE [36]	ViT-L/16	1600	76.0
	ViT-H/14	1600	77.2
CAE [22]	ViT-B/16	1600	70.4
	ViT-L/16	1600	78.1
I-JEPA	ViT-B/16	600	72.9
	ViT-L/16	600	77.5
	ViT-H/14	300	79.3
	ViT-H/16 ₄₄₈	300	81.1
<i>Methods using extra view data augmentations</i>			
SimCLR v2 [21]	RN152 (2×)	800	79.1
DINO [18]	ViT-B/8	300	80.1
iBOT [79]	ViT-L/16	250	81.0

Self-supervised methods over time



DINO v1: 80.1 %

DINO v2: 83.5 %
(larger dataset)

MAE: 81+ %
(2+ layer
fine-tuning)

I-JEPA: 81%

Models evaluated via linear probe

Next Week: Vision Foundation Models

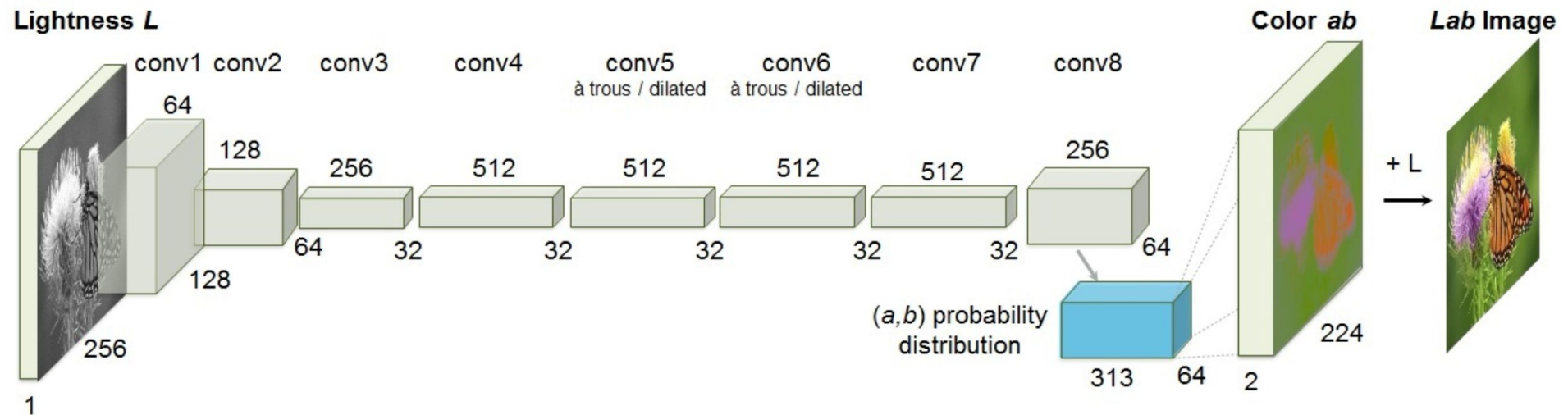
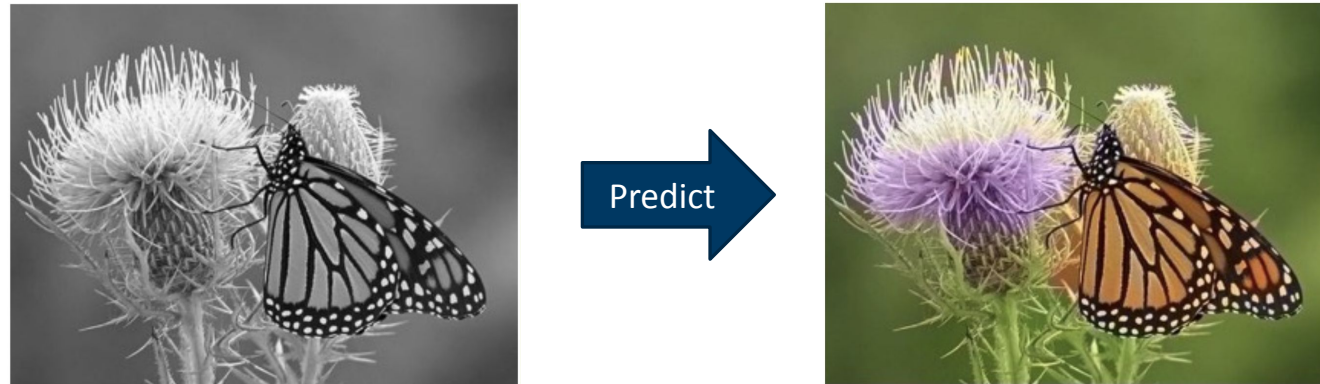
Powerful deep learning models for vision tasks; mostly trained via self-supervised or weakly-supervised (vision + text) learning; trained on very large datasets

- More powerful backbones for transfer learning; require less new data
- Can solve (some) tasks in context from prompts (without re-training)
- Integrated into Multi-Modal-Language Models / Vision Language Models

Questions!

Old slide pretext tasks

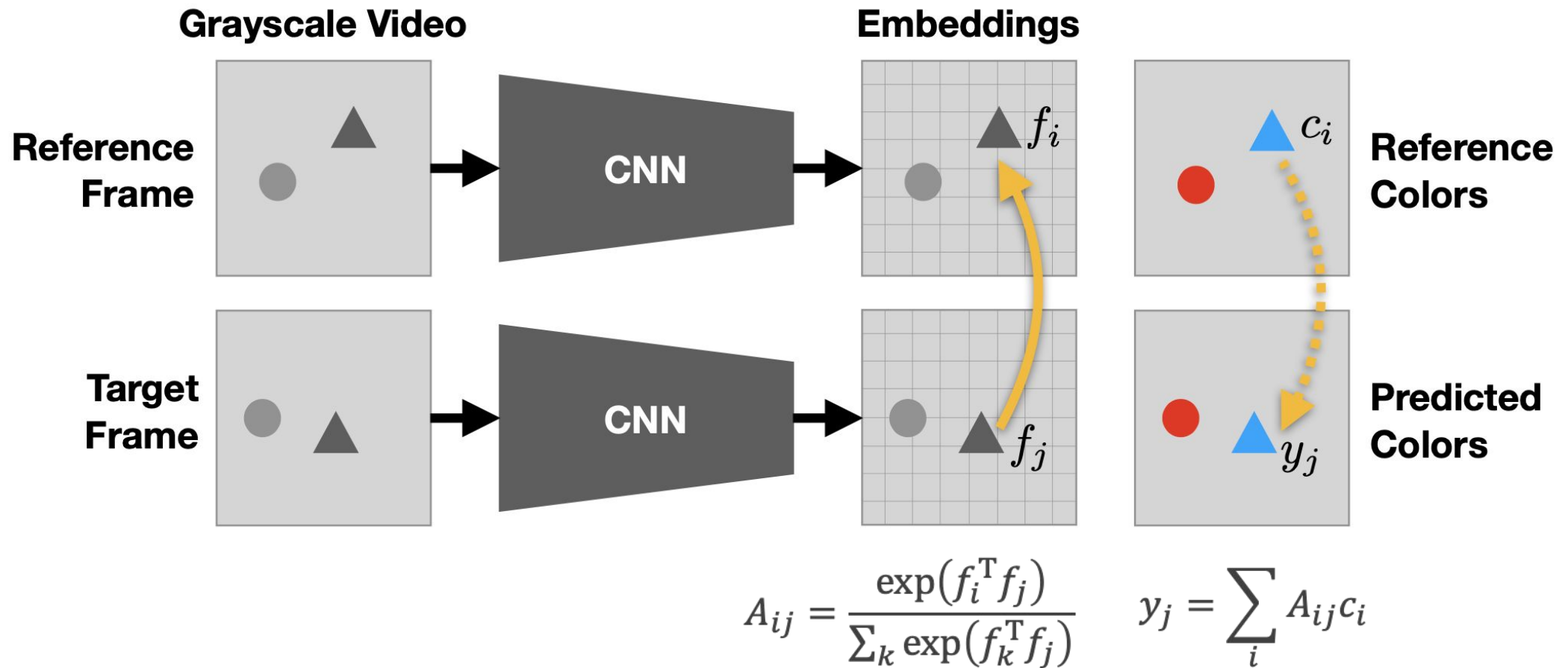
Colorization: Images



Colorization: Videos

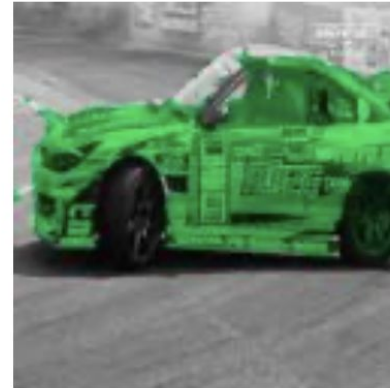
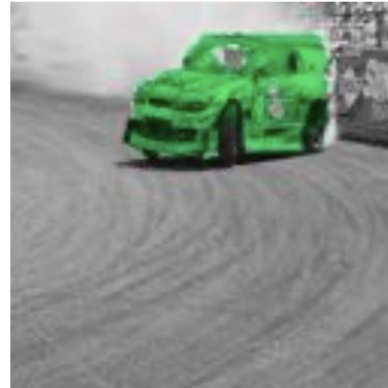


Colorization: Videos

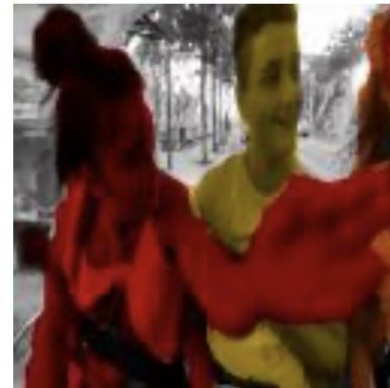
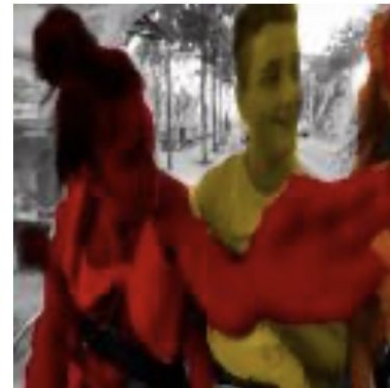


Transfer to video segmentation: examples

Inputs

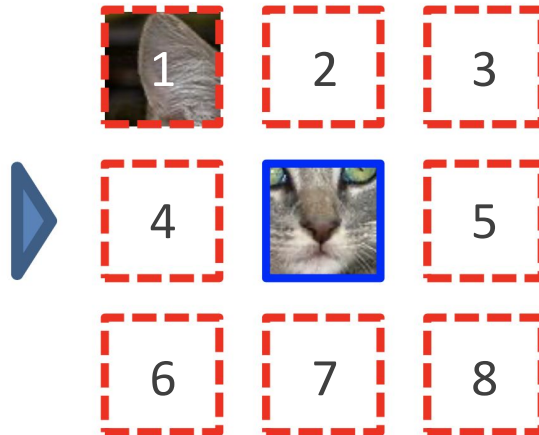


Predicted Segmentations



Context prediction

Example:



Question 1:



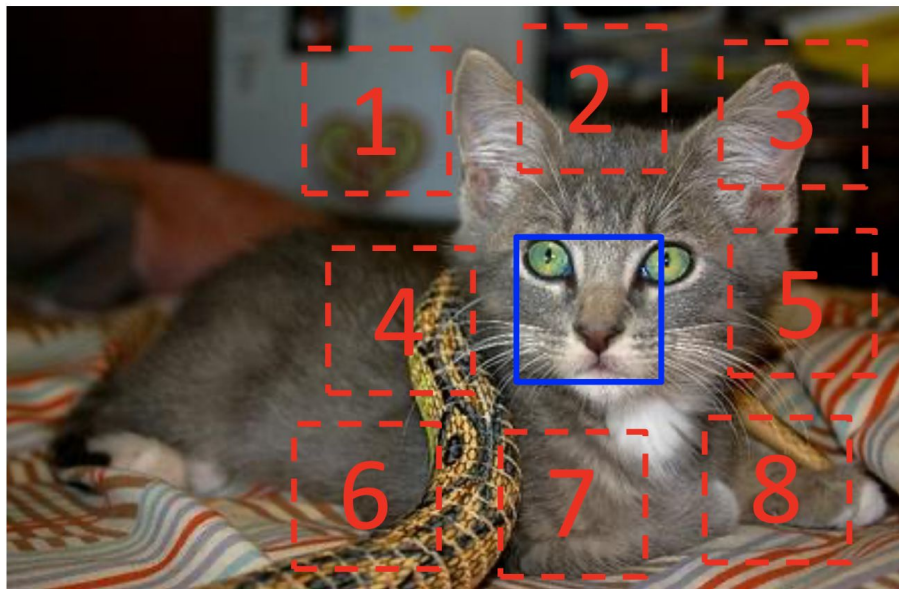
?
bottom right

Question 2:

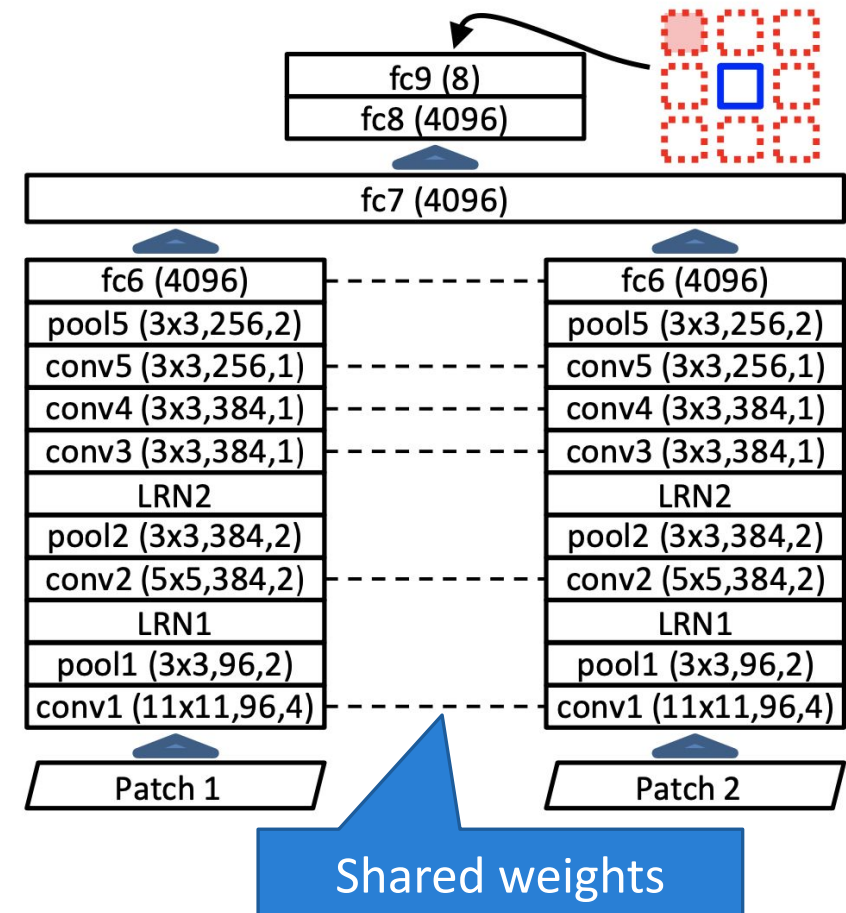


?
top center

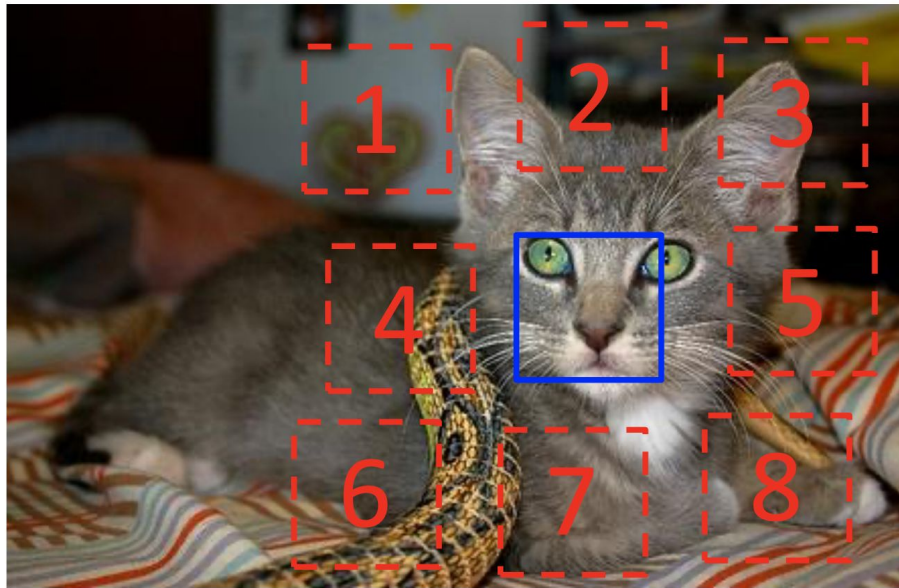
Context prediction



$$X = \left(\begin{array}{c} \text{[Kitten Face Patch]} \\ \text{[Kitten Ear Patch]} \end{array} \right); Y = 3$$



Context prediction: trivial solutions & shortcuts



$$X = (\text{patch 4}, \text{patch 7}); Y = 3$$



Boundary patterns or textures continuing between patches

- Gap between patches
- Jitter location of patches slightly

Chromatic aberration!

- Network learned to predict absolute location of patches